Khulna University of Engineering & Technology
Department of Computer Science and Engineering

Computer Graphics and Image Processing Laboratory
Course code: CSE 4102

Jungle Rest Stop

An interactive 3D scene built only from basic shapes with OpenGL 3.3
with vending machine, ATM, coffee machine & library



Submitted by	**Submitted to**
Md. Nayeem Roll: 2107050 Department of Computer Science and Engineering Khulna University of Engineering & Technology	Md Tajmilur Rahman Lecturer, Dept. of CSE, KUET Md. Mubtashim Abrar Nihal Lecturer, Dept. of CSE, KUET

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1 Introduction

Jungle Rest Stop is a real-time 3D scene developed with OpenGL 3.3. A small rest stop stands in a clearing of a dense jungle, next to a parking lot and a road. It has three connected rooms: a main lounge with three vending machines, a coffee machine and a television, an ATM room, and a library with an aquarium. The user explores the scene on foot and operates the machines, while visitors arrive by car, use the same machines and leave again.

Every object is built only from basic shapes (cube, cylinder, sphere, cone, torus, wedge and a few more) that are placed with translation, rotation and scaling. No 3D model files and no image files are loaded: all meshes and all textures are generated by the program. The scene runs at 60 frames per second on an integrated Intel Iris Xe GPU and offers a *Basic* and an *Enhanced* rendering mode.

2 Overview of the Scene

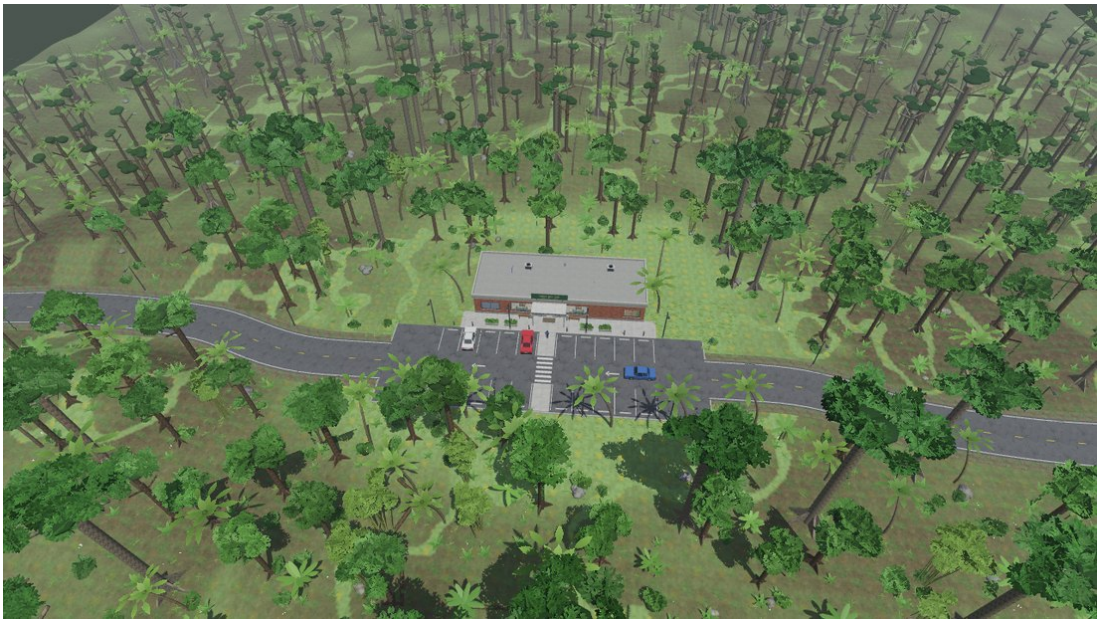


Figure 1: The rest stop in its jungle clearing.



Figure 2: The rest stop by day: building, sidewalk, parking lot and street lamps.



Figure 3: The main lounge: sofas, TV wall, coffee counter and the three vending machines.



Figure 4: The three vending machines: cold drinks, crispy snacks and sweet treats.



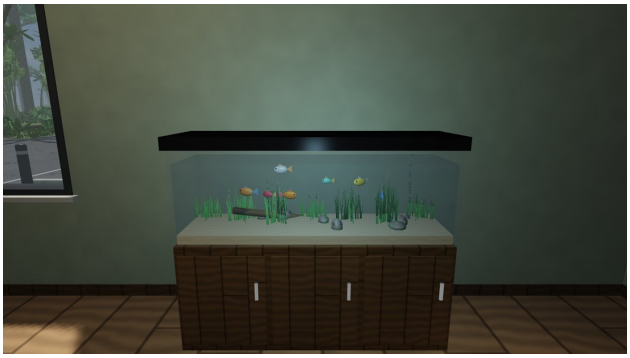
(a) ATM room



(b) Coffee machine (brewing)



(c) Library



(d) Aquarium in the library

Figure 5: The ATM room, the coffee counter and the library.



(a) Sunrise 6:18



(b) Day 10:30



(c) Sunset 17:36



(d) Night 21:00

Figure 6: The same view at four times of day.

3 Tools and Environment

Item	Used
Language	C++17 (MinGW-w64 / MSYS2 `g++`)
Graphics API	OpenGL 3.3 Core profile, GLSL 330
Window / input	GLFW 3
OpenGL loader	GLAD 2
Text	Windows GDI renders one font atlas at start-up (Consolas)
Sound	Windows `waveOut` (winmm); 14 short CC0 (public domain) recordings
Assets	no models, no images: all meshes and textures are generated in code; only the sounds are files
Test PC	Intel Core i5-1135G7, Intel Iris Xe (integrated GPU), 1280×720 window, 4×MSAA

4 Controls

The scene is controlled with the keyboard and the mouse. Pressing **H** shows the same list inside the program (Figure 7).

Table 1: Keyboard and mouse controls

Key	Action	Key	Action
W A S D	walk	Shift	run
Mouse	look around	Space	jump
E / left click	use the part under the crosshair	V	camera: first person, third person, free
Mouse wheel, + / –	change the field of view	Alt	free the mouse to click the on-screen buttons
B	Basic / Enhanced rendering	T, G, F	textures, shadows, fog on / off
N	switch day / night	[ ]	slower / faster time
M	pause the clock	R	roof cut-away view
F1 – F4	lounge, spot, room and outdoor lights	K	call a visitor car
L	frame cap 30 / 60 FPS	O	sound on / off
F12	save a screenshot	H, Esc	help screen, quit



Figure 7: The help screen (H), with the status panel (top left), the wallet and the bag (top right) and the clickable buttons (bottom right).

5 System Design

5.1 Modules

The code is divided into small modules, each with a single responsibility (Figure 8).

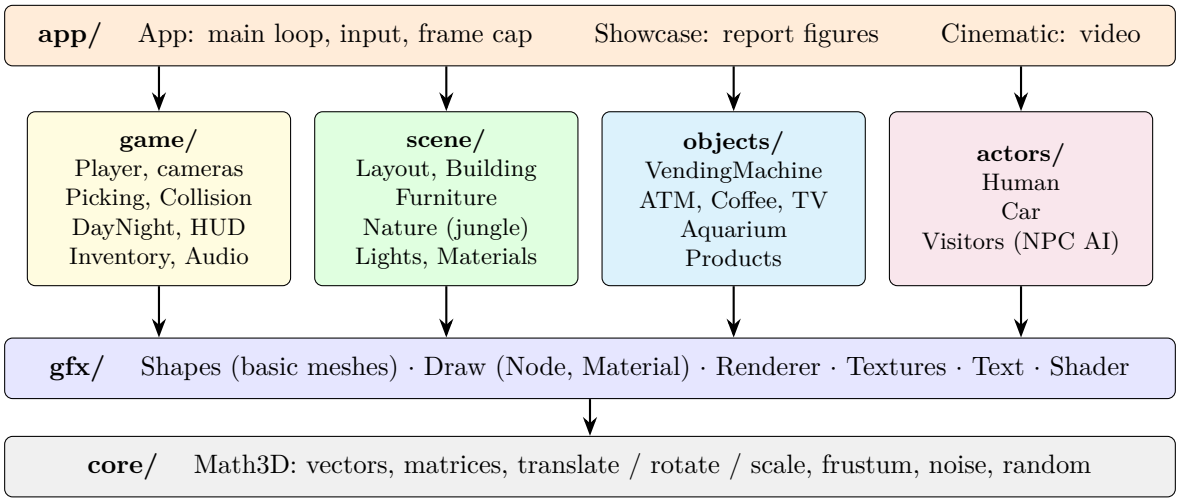


Figure 8: Module overview. Arrows mean "uses".

5.2 Frame loop

Each frame runs the same sequence (Figure 9). All movement uses the measured frame time Δt, so every animation has the same speed at 30 and at 60 frames per second.

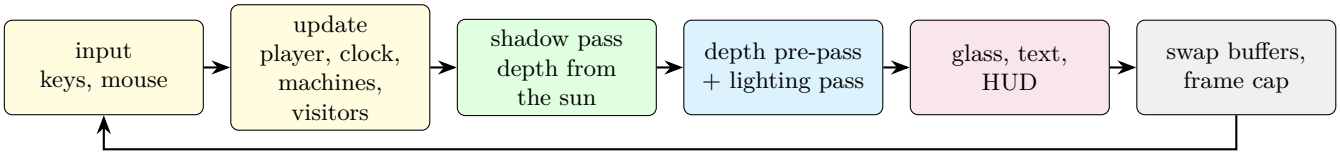


Figure 9: One frame of the program.

5.3 Scene layout

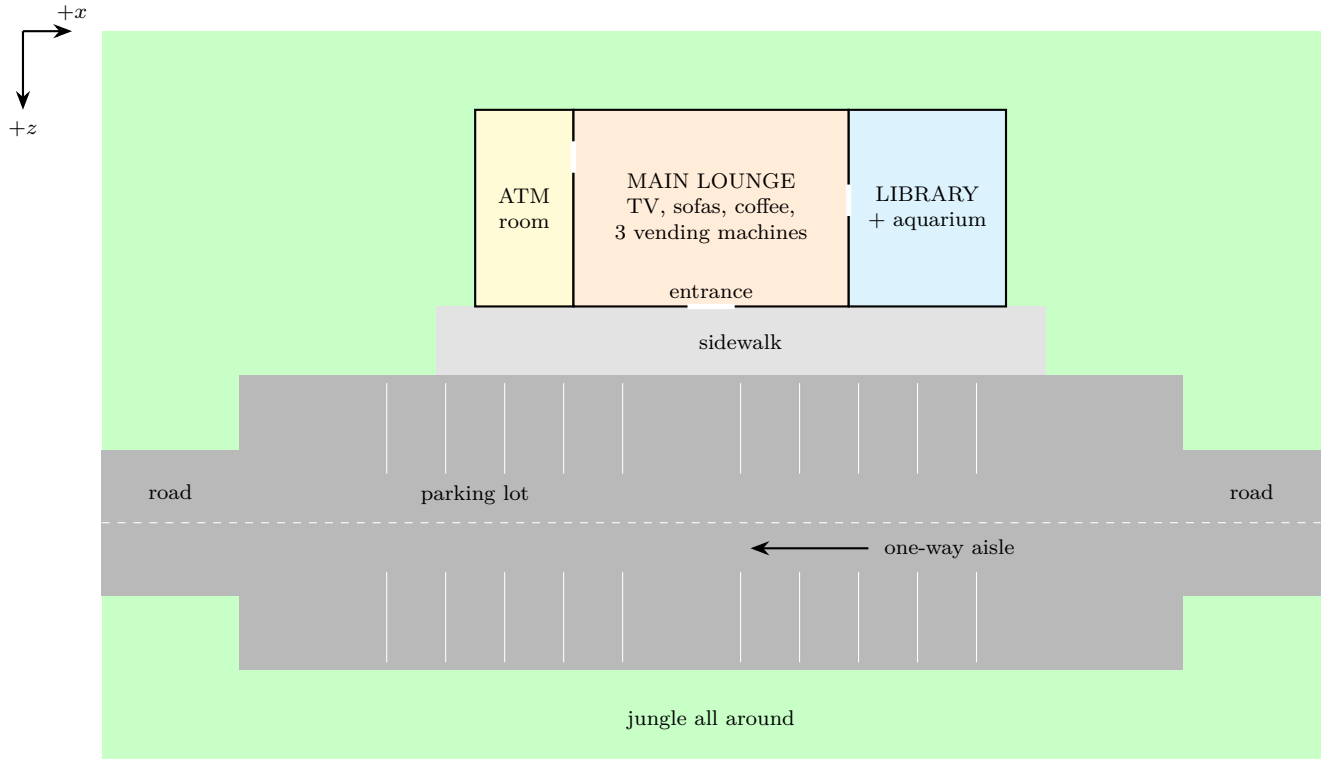


Figure 10: Top view of the layout (1 grid unit = 1 m). The building faces $+z$, towards the parking lot.

6 The Basic Shapes

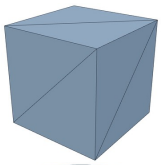
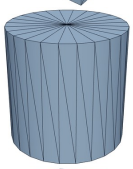
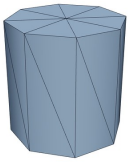
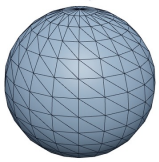
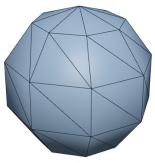
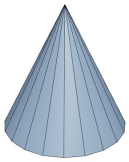
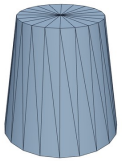
All objects are built from the fifteen *unit* meshes listed in Table 2. Each mesh is centred at the origin and has a size of about one unit, so a part of any size is obtained by scaling it. The meshes are generated once at start-up (`gfx/Shapes.cpp`) and stay on the GPU; every part of every object is only a reference to one of these meshes plus a 4×4 matrix. The pictures show each mesh with the edges of its triangles drawn on top.

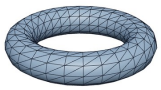
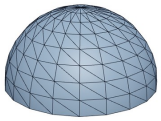
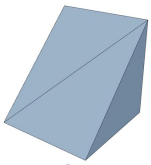
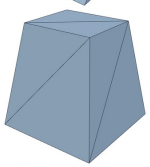
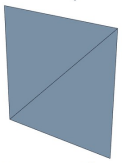
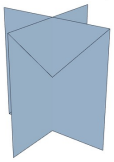
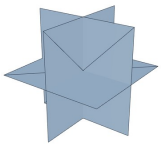
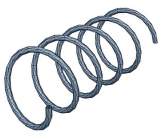
Generation rules. Flat shapes are written as lists of corner points. Every face has its own vertices, so that the face has one flat normal and the edges stay sharp; a quad face (a, b, c, d) is split into the triangles (a, b, c) and (a, c, d). Round shapes are generated by loops over angles. For a ring of n steps the angle is $\theta_i = 2\pi i/n$, and the side of a cylinder with bottom radius r_b and top radius r_t uses

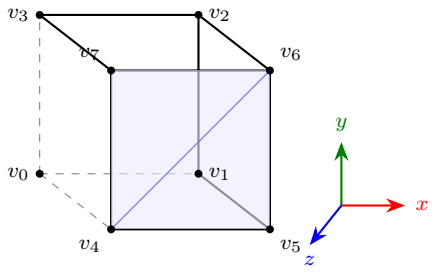
$$\mathbf{p}_i^{top} = (r_t \cos\theta_i,\ 0.5,\ r_t \sin\theta_i), \quad \mathbf{p}_i^{bot} = (r_b \cos\theta_i,\ -0.5,\ r_b \sin\theta_i), \quad \mathbf{n}_i = \text{normalize}(\cos\theta_i,\ r_b - r_t,\ \sin\theta_i).$$

Neighbouring ring points form one side quad $(\mathbf{p}_i^{bot},\ \mathbf{p}_{i+1}^{bot},\ \mathbf{p}_{i+1}^{top},\ \mathbf{p}_i^{top})$, and a cap is a fan of triangles around the centre vertex $(0, \pm 0.5, 0)$. Round shapes have one normal per vertex pointing straight out of the surface, so they look smooth even with few sides.

Table 2: The basic shapes, how their vertices are generated, and their vertex / triangle counts

Shape	Name	Vertices	V / T
	Cube	8 corners $(\pm.5, \pm.5, \pm.5)$ (Figure 11); each of the 6 faces has its own 4 vertices and the face normal	24 / 12
	Cylinder	ring loop with $r_b = r_t = 0.5$, $n = 24$, two caps	102 / 96
	Cylinder (low)	the same with $n = 8$; used for tree trunks and bamboo	38 / 32
	Sphere	$\mathbf{p}_{ij} = 0.5(\sin\phi_i \cos\theta_j,\ \cos\phi_i,\ \sin\phi_i \sin\theta_j)$, $\phi_i = \pi i/14$, $\theta_j = 2\pi j/20$, normal $\mathbf{n} = 2\mathbf{p}$	315 / 560
	Sphere (low)	the same with 5 rings and 8 sectors; foliage blobs, rocks	54 / 80
	Cone	ring loop with $r_t = 0$ (all top vertices meet at the tip $(0, 0.5, 0)$), $n = 20$, bottom cap	64 / 60
	Truncated cone	ring loop with $r_t = 0.35$, $n = 20$, two caps; cups, pots, limbs	86 / 80

Shape	Name	Vertices	V / T
	Torus	$\mathbf{p}(u, w) =$ $(0.4 + 0.1\cos w)(\cos u, 0, \sin u) + (0, 0.1\sin w, 0)$, 28×10 steps	319 / 560
	Hemisphere	sphere formula with $\phi \in [0, \pi/2]$ (8 rings) plus a flat disc at $y = 0$	211 / 340
	Wedge	left side $w_0(-.5, -.5, .5)$, $w_1(-.5, -.5, -.5)$, $w_2(-.5, .5, -.5)$; right side w_3, w_4, w_5 = the same with $x = +.5$; 2 triangles and 3 quads	18 / 8
	Tapered box	bottom corners $(\pm.5, -.5, \pm.5)$, top corners $(\pm.35, .5, \pm.3)$; side normal $\mathbf{n} = \text{normalize}((\mathbf{b} - \mathbf{a}) \times (\mathbf{d} - \mathbf{a}))$	24 / 12
	Quad	$(-.5, -.5, 0)$, $(.5, -.5, 0)$, $(.5, .5, 0)$, $(-.5, .5, 0)$, normal $(0, 0, 1)$; fronds, vines	4 / 2
	Cross card	two crossed quads with $y \in [0, 1]$ (base on the ground); grass, ferns, flowers	8 / 4
	Leaf card	three quads through the centre, one in each plane (xy, yz, xz); leaf clusters	12 / 6
	Coil	a 6-sided ring around each of 101 points of the helix $\mathbf{c}(t) = (R\cos 2\pi kt,\ R\sin 2\pi kt,\ L(t - 0.5))$, $k = 5$ turns	606 / 1200



Corner	(x, y, z)	Corner	(x, y, z)
v_0	$(-.5, -.5, -.5)$	v_4	$(-.5, -.5, .5)$
v_1	$(.5, -.5, -.5)$	v_5	$(.5, -.5, .5)$
v_2	$(.5, .5, -.5)$	v_6	$(.5, .5, .5)$
v_3	$(-.5, .5, -.5)$	v_7	$(-.5, .5, .5)$

Faces: front $(v_4v_5v_6v_7)$, back $(v_1v_0v_3v_2)$, top $(v_7v_6v_2v_3)$, bottom $(v_0v_1v_5v_4)$, right $(v_5v_1v_2v_6)$, left $(v_0v_4v_7v_3)$. The front face is split into $(v_4v_5v_6)$ and $(v_4v_6v_7)$.

Figure 11: The unit cube: 8 corners, 6 faces, 12 triangles. The front face (shaded) looks towards $+z$.

The ground is one more generated mesh: a 150×150 grid whose vertices are lifted to the terrain height $h(x, z)$ (Section 8.5.8).

7 Transformations and Hierarchical Modelling

7.1 Homogeneous coordinates and the basic transformations

A point is written as $\mathbf{p} = (x, y, z, 1)^T$, so that translation, rotation and scaling are all 4×4 matrix products. The matrices are stored column-major and sent to OpenGL unchanged.

$$T(\mathbf{t}) = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad S(\mathbf{s}) = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A rotation given in degrees $(\theta_x, \theta_y, \theta_z)$ is applied as $R = R_y(\theta_y)\, R_x(\theta_x)\, R_z(\theta_z)$. Rotation about an arbitrary unit axis $\mathbf{a}$ uses Rodrigues' formula

$$R(\mathbf{a}, \theta) = \cos\theta\, I + (1 - \cos\theta)\, \mathbf{a}\mathbf{a}^T + \sin\theta\, [\mathbf{a}]_\times, \qquad [\mathbf{a}]_\times = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}.$$

A negative scale factor mirrors a part (used for the fruit bowl and for mirrored book rows).

7.2 Composite transformation of a part

Every part of every object is placed with

$$M_{part} = M_{parent} \cdot T(\text{position}) \cdot R(\text{rotation}) \cdot S(\text{size}), \qquad \mathbf{p}_{world} = M_{part}\, \mathbf{p}_{unit}.$$

Read from right to left, the unit shape is first **scaled** to its size, then **rotated**, then **translated** to its place inside the parent, and finally carried into the world by the parent's matrix. The order matters: scaling after a rotation would shear the shape. Normals are transformed with the normal matrix

$$N = \left(M_{3\times3}^{-1}\right)^T, \qquad \mathbf{n}_{world} = \text{normalize}(N\, \mathbf{n}),$$

which keeps them perpendicular to the surface even when a shape is stretched unevenly.

7.3 Hierarchical modelling with nodes

Objects are written as trees of `Node`s. A node stores the matrix of its coordinate frame; a child node is created with an extra translation and rotation relative to its parent, so moving or turning a parent moves all of its children:

```
struct Node {
    DrawSink* sink; Mat4 M; // M = this node's world matrix
    Node at(const Vec3& p, const Vec3& rotDeg) const // child: M * T(p) * R(rot)
        { return Node(*sink, rotateYXZ(translate(M, p), rotDeg)); }
    void part(MeshId mesh, Vec3 pos, Vec3 size, const Material& m, Vec3 rotDeg = {}) const
        { sink->add(mesh, scale(rotateYXZ(translate(M, pos), rotDeg), size), m); }
};
// a sofa arm: a box and a cylinder lying along z (rotated 90 degrees about x)
n.box({x, 0.42f, 0.0f}, {0.16f, 0.44f, 0.88f}, fabric);
n.part(M_CYLINDER, {x, 0.64f, 0.0f}, {0.19f, 0.88f, 0.19f}, fabric, {90, 0, 0});
```

The same object code can send its shapes to different *sinks*: the static world, the per-frame list, a counter, or a collector that records every part with its name (used for the exploded views and part tables in Section 8).

A matrix built from corner points. Some parts are not rectangles (car windows, door glass). For a parallelogram with corners $\mathbf{a}$, $\mathbf{b}$, $\mathbf{d}$ and $\mathbf{b}+\mathbf{d}-\mathbf{a}$ the model matrix is built directly from the edge vectors:

$$M = \begin{bmatrix} \mathbf{b}-\mathbf{a} & \mathbf{d}-\mathbf{a} & t\,\hat{\mathbf{n}} & \mathbf{a}+\frac{1}{2}(\mathbf{b}-\mathbf{a})+\frac{1}{2}(\mathbf{d}-\mathbf{a}) \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \hat{\mathbf{n}} = \frac{(\mathbf{b}-\mathbf{a})\times(\mathbf{d}-\mathbf{a})}{|(\mathbf{b}-\mathbf{a})\times(\mathbf{d}-\mathbf{a})|},$$

where t is the thickness. The unit cube corner $(-.5,-.5,0)$ lands exactly on $\mathbf{a}$, $(.5,-.5,0)$ on $\mathbf{b}$ and $(-.5,.5,0)$ on $\mathbf{d}$. A triangle is made the same way from the wedge. A *beam* between two points $\mathbf{a}$ and $\mathbf{b}$ (pillars, branches) turns the shape's y axis onto $\mathbf{d}=\mathbf{b}-\mathbf{a}$ with the axis $\mathbf{y}\times\hat{\mathbf{d}}$ and the angle $\operatorname{atan2}(|\mathbf{y}\times\hat{\mathbf{d}}|,\ \mathbf{y}\cdot\hat{\mathbf{d}})$, and scales it to the length $|\mathbf{d}|$.

7.4 Viewing and projection

The camera matrix is built with `lookAt` from the eye $\mathbf{e}$, the target $\mathbf{c}$ and the up vector $\mathbf{u}$:

$$\mathbf{f} = \frac{\mathbf{c}-\mathbf{e}}{|\mathbf{c}-\mathbf{e}|}, \quad \mathbf{s} = \frac{\mathbf{f}\times\mathbf{u}}{|\mathbf{f}\times\mathbf{u}|}, \quad \mathbf{u}' = \mathbf{s}\times\mathbf{f}, \qquad V = \begin{bmatrix} s_x & s_y & s_z & -\mathbf{s}\cdot\mathbf{e} \\ u'_x & u'_y & u'_z & -\mathbf{u}'\cdot\mathbf{e} \\ -f_x & -f_y & -f_z & \mathbf{f}\cdot\mathbf{e} \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The main view uses a perspective projection with vertical field of view f, aspect ratio a and the near and far planes z_n, z_f. The sun's shadow camera and the 2D text overlay use orthographic projections:

$$P_{persp} = \begin{bmatrix} \frac{1}{a\tan(f/2)} & 0 & 0 & 0 \\ 0 & \frac{1}{\tan(f/2)} & 0 & 0 \\ 0 & 0 & -\frac{z_f+z_n}{z_f-z_n} & -\frac{2z_fz_n}{z_f-z_n} \\ 0 & 0 & -1 & 0 \end{bmatrix} \qquad P_{ortho} = \begin{bmatrix} \frac{2}{r-l} & 0 & 0 & -\frac{r+l}{r-l} \\ 0 & \frac{2}{t-b} & 0 & -\frac{t+b}{t-b} \\ 0 & 0 & \frac{-2}{z_f-z_n} & -\frac{z_f+z_n}{z_f-z_n} \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

A vertex therefore goes through $\mathbf{p}_{clip} = P\,V\,M\,\mathbf{p}$, the perspective division $\mathbf{p}_{ndc} = \mathbf{p}_{clip}/w_{clip}$, and the viewport transformation to pixels,

$$x_{win} = \tfrac{W}{2}\,(x_{ndc}+1), \qquad y_{win} = \tfrac{H}{2}\,(y_{ndc}+1), \qquad z_{win} = \tfrac{1}{2}(z_{ndc}+1).$$

The field of view f is changed with the mouse wheel between 25° and 110° (Section 15.1), and the aspect ratio $a = W/H$ follows the window size when the window is resized.

8 The Objects

Each object is shown complete and with its parts pulled apart, followed by the basic shapes it uses and the transformation of every part. For the main objects, the world coordinates of their key parts are also listed. All tables are produced by the program from the drawing code of the scene. Table 4 summarises the number of basic shapes of each object. The world-coordinate tables give, for each key part, two vertices A and B that really exist in the mesh of its shape (Table 3); their world position is $\mathbf{p}_{world} = M_{place}\,M_{part}\,\mathbf{v}$.

Table 3: The two mesh vertices A and B listed for each shape (unit-shape coordinates, Table 2)

Shape	Vertex A	Vertex B	Which vertices
cube	$(-.5,-.5,-.5)$	$(.5,.5,.5)$	opposite corners v_0 and v_6 (Figure 11)
cylinder	$(.5,-.5,0)$	$(.5,.5,0)$	first vertex of the bottom ring and of the top ring ($\theta_0=0$)
truncated cone	$(.5,-.5,0)$	$(.35,.5,0)$	first vertex of the bottom ring and of the top ring
cone	$(.5,-.5,0)$	$(0,.5,0)$	first vertex of the bottom ring, and the tip
sphere	$(0,-.5,0)$	$(0,.5,0)$	the two poles ($\phi=\pi$ and $\phi=0$)
hemisphere	$(.5,0,0)$	$(0,.5,0)$	first vertex of the rim, and the pole
torus	$(.5,0,0)$	$(.3,0,0)$	outer and inner edge of the tube at $u=0$
wedge	$(-.5,-.5,.5)$	$(.5,.5,-.5)$	corners w_0 and w_5
tapered box	$(-.5,-.5,-.5)$	$(.35,.5,.3)$	a bottom corner and a top corner

Table 4: Number of basic shapes per object (Cyl-L / Sph-L = low-poly versions, TCone = truncated cone, TBox = tapered box)

Object	Total	Cube	Cyl	Cyl-L	Sph	Sph-L	Cone	TCone	Torus	Hemi	Wedge	TBox	Quad	Cross	Leaf	Coil
Vending machine	**287**	79	177	–	3	–	–	16	–	–	–	–	–	–	–	12
Products (6)	**23**	9	12	–	1	–	–	1	–	–	–	–	–	–	–	–
ATM	**49**	48	–	–	–	–	–	–	–	1	–	–	–	–	–	–
Coffee machine	**32**	16	9	–	1	–	–	2	4	–	–	–	–	–	–	–
Television	**5**	4	–	–	1	–	–	–	–	–	–	–	–	–	–	–
Aquarium	**76**	12	–	1	35	6	7	1	–	–	7	–	–	7	–	–
Sofa (3 seats)	**21**	12	9	–	–	–	–	–	–	–	–	–	–	–	–	–
Armchair	**13**	6	7	–	–	–	–	–	–	–	–	–	–	–	–	–
Coffee table + rug	**14**	10	–	–	3	–	–	–	–	1	–	–	–	–	–	–
Bookshelf	**31**	21	–	–	–	–	–	–	–	–	–	–	10	–	–	–
Reading table + chair	**10**	7	3	–	–	–	–	–	–	–	–	–	–	–	–	–
Wall clock	**17**	14	2	–	–	–	–	–	1	–	–	–	–	–	–	–
Pendant lamp	**5**	–	3	–	1	–	–	1	–	–	–	–	–	–	–	–
Table + floor lamp	**10**	–	6	–	2	–	–	2	–	–	–	–	–	–	–	–
Potted plants (3)	**27**	–	4	–	–	1	7	3	–	–	–	–	6	1	5	–
Counter + wall cabinet	**12**	12	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Bench	**10**	10	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Waste bin	**4**	1	2	–	–	–	–	–	–	1	–	–	–	–	–	–
Light switch	**3**	2	–	–	1	–	–	–	–	–	–	–	–	–	–	–
Security camera	**5**	3	1	–	1	–	–	–	–	–	–	–	–	–	–	–
Paintings (3)	**18**	10	2	–	–	–	5	–	–	1	–	–	–	–	–	–
Sliding doors	**10**	10	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Ceiling + roof fixtures	**11**	3	5	–	1	–	–	–	2	–	–	–	–	–	–	–
Street lamp + bollard	**8**	1	4	–	–	–	–	1	–	1	–	1	–	–	–	–
Man	**49**	8	6	–	24	–	–	8	1	1	–	1	–	–	–	–
Woman	**50**	7	6	–	24	–	–	10	1	1	–	1	–	–	–	–
Car	**144**	76	23	–	24	–	–	8	2	1	9	1	–	–	–	–
Rainforest tree	**30**	–	–	5	–	3	–	–	–	–	4	–	–	–	18	–
Coconut palm	**37**	–	–	14	–	5	–	–	–	–	–	–	18	–	–	–
Emergent tree	**35**	–	–	5	–	3	–	–	–	–	5	–	–	–	22	–
Bamboo clump	**130**	–	–	100	–	–	–	–	–	–	–	–	–	–	30	–
Banana plant	**28**	–	–	6	–	22	–	–	–	–	–	–	–	–	–	–
Undergrowth	**23**	–	–	4	–	5	–	–	–	3	–	–	5	1	5	–
Bird	**4**	2	–	–	–	1	1	–	–	–	–	–	–	–	–	–

8.1 Machines

8.1.1 Vending machine

The vending machine is $1.0 \times 1.95 \times 0.8$ m. Its origin is on the floor in the middle and it faces $+z$. Three machines stand in the lounge: cold drinks (red), crispy snacks (orange) and sweet treats (blue).

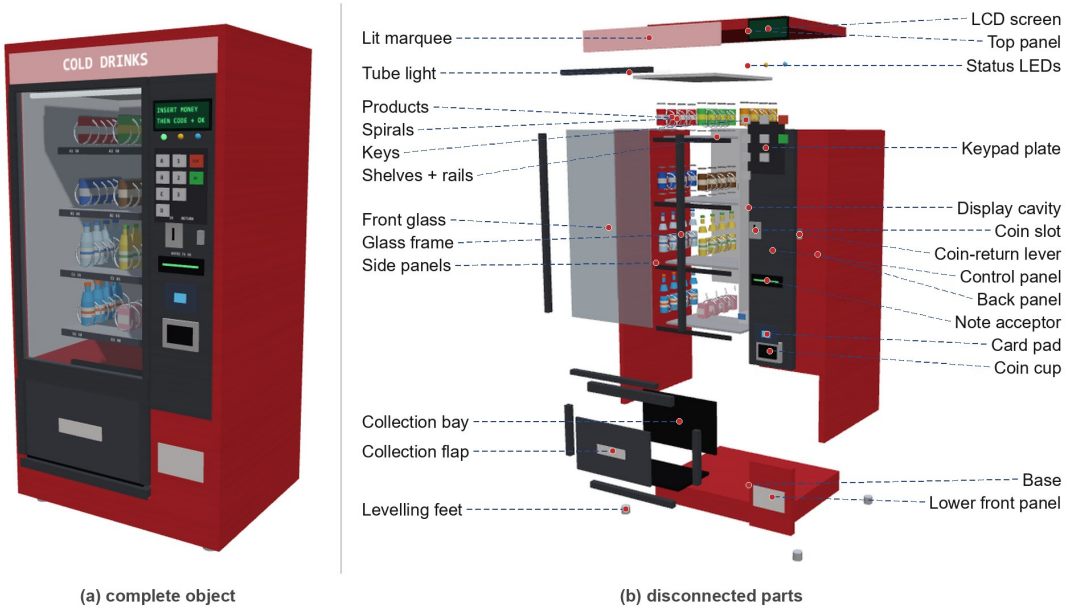


Figure 12: Vending machine: (a) complete, (b) disconnected parts with their names.

Table 5: Vending machine: the basic shapes it uses (287 shapes in total)

Shape	Name	No.	Used for
	cylinder	177	Levelling feet, Tube light, Products
	cube	79	Back panel, Side panels, Top panel, Base, Lit marquee, Glass frame, Display cavity, Shelves + rails, Front glass, Control panel, Lower front panel, LCD screen, Keypad plate, Coin slot, Note acceptor, Card pad, Coin cup, Collection bay, Products, Keys, Coin-return lever, Collection flap
	truncated cone	16	Products
	coil	12	Spirals
	sphere	3	Status LEDs

Table 6: Vending machine: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Back panel	cube	1	(1, 1.86, .04)	(0, 1, -.38)	–	
Side panels	cube	2	(.04, 1.86, .8)	($\pm$.48, 1, 0)	–	
Top panel	cube	1	(1, .06, .8)	(0, 1.93, 0)	–	
Base	cube	1	(1, .14, .8)	(0, .11, 0)	–	
Levelling feet	cylinder	4	(.06, .05, .06)	($\pm$.42, .025, $\pm$.32)	–	
Lit marquee	cube	1	(.94, .12, .02)	(0, 1.86, .4)	–	glow
Glass frame	cube	4	(.03/.66, .03/1.11, .04)	(-.47/-.15/.17, .65/1.22/1.78, .38)	–	
Display cavity	cube	4	(.01/.64, .02/1.14, .02/.7)	(-.46/-.15/.16, 1.22/1.78, -.35/0)	–	
Tube light	cylinder	1	(.022, .6, .022)	(-.15, 1.76, .3)	(0, 0, 90)	glow
Shelves + rails	cube	8	(.62, .015/.03, .01/.38)	(-.15, .72. . .1.54, -.06/.13)	–	
Front glass	cube	1	(.62, 1.14, .008)	(-.15, 1.22, .39)	–	glass
Control panel	cube	1	(.27, 1.27, .04)	(.33, 1.19, .38)	–	

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Lower front panel	cube	2	(.2/.27, .12/.38, .004/.04)	(.33, .36, .38/.4)	–	
LCD screen	cube	2	(.22/.24, .09/.11, .002/.004)	(.33, 1.66, .4/.4)	–	glow
Keypad plate	cube	1	(.22, .27, .004)	(.33, 1.41, .4)	–	
Coin slot	cube	2	(.008/.07, .045/.09, .004/.008)	(.3, 1.23/1.24, .4/.41)	–	
Note acceptor	cube	2	(.15/.19, .008/.06, .004/.008)	(.33, 1.12, .4/.41)	–	glow
Card pad	cube	2	(.05/.14, .035/.1, .003/.008)	(.33, 1, .4/.41)	–	glow
Coin cup	cube	2	(.11/.15, .07/.11, .01/.04)	(.33, .86, .39/.41)	–	
Collection bay	cube	6	(.04/.6/.66, .02. . .36, .02/.04/.34)	(-.47/-.15/.17, .19. . .6, .06. . .39)	–	
Spirals	coil	12	(1, 1, 1)	(-.36/-.15/.06, .81. . .1.62, -.06)	–	
Products	cylinder	172	(.008. . .065, .006. . .11, .008. . .065)	(-.36. . .078, .81. . .1.68, -.17. . .052)	(0, 0, 0/12)	+14
Products	cube	24	(.06/.065, .04/.1, .001/.045)	(-.15/.06, .81/1.34, -.17. . .075)	–	+1
Products	truncated cone	16	(.062, .04, .062)	(-.36/-.15/.06, .89/1.16, -.17. . .052)	–	
Keys	cube	9	(.05, .045, .014)	(.26/.33/.4, 1.32. . .1.5, .41)	–	
Coin-return lever	cube	1	(.02, .05, .02)	(.42, 1.23, .42)	–	
Status LEDs	sphere	3	(.022, .022, .022)	(.26/.33/.4, 1.58, .41)	–	glow
Collection flap	cube	2	(.22/.58, .06/.34, .004/.015)	(-.15, .4, .39/.4)	–	

Table 7: Vending machine in the world, placed with $M_{place} = T(2.4, 0, -4.5)$: centre of each key part and the world position $M_{place}\, M_{part}\, \mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Back panel	cube	(2.40, 1.00, −4.88)	(1.90, 0.07, −4.90)	(2.90, 1.93, −4.86)
Side panels (1)	cube	(1.92, 1.00, −4.50)	(1.90, 0.07, −4.90)	(1.94, 1.93, −4.10)
Side panels (2)	cube	(2.88, 1.00, −4.50)	(2.86, 0.07, −4.90)	(2.90, 1.93, −4.10)
Top panel	cube	(2.40, 1.93, −4.50)	(1.90, 1.90, −4.90)	(2.90, 1.96, −4.10)
Base	cube	(2.40, 0.11, −4.50)	(1.90, 0.04, −4.90)	(2.90, 0.18, −4.10)
Front glass	cube	(2.25, 1.22, −4.12)	(1.94, 0.65, −4.12)	(2.56, 1.79, −4.11)
Keys (1)	cube	(2.65, 1.50, −4.09)	(2.63, 1.48, −4.10)	(2.68, 1.52, −4.08)
Keys (2)	cube	(2.65, 1.44, −4.09)	(2.63, 1.42, −4.10)	(2.68, 1.46, −4.08)
Collection flap (1)	cube	(2.25, 0.40, −4.11)	(1.96, 0.23, −4.12)	(2.54, 0.57, −4.10)
Collection flap (2)	cube	(2.25, 0.40, −4.10)	(2.14, 0.37, −4.10)	(2.36, 0.43, −4.10)
Levelling feet (1)	cylinder	(1.98, 0.03, −4.82)	(2.01, 0.00, −4.82)	(2.01, 0.05, −4.82)
Levelling feet (2)	cylinder	(1.98, 0.03, −4.18)	(2.01, 0.00, −4.18)	(2.01, 0.05, −4.18)

Construction. The machine is a small tree of nodes (Figure 13); moving parts are child nodes whose own transformation changes over time.

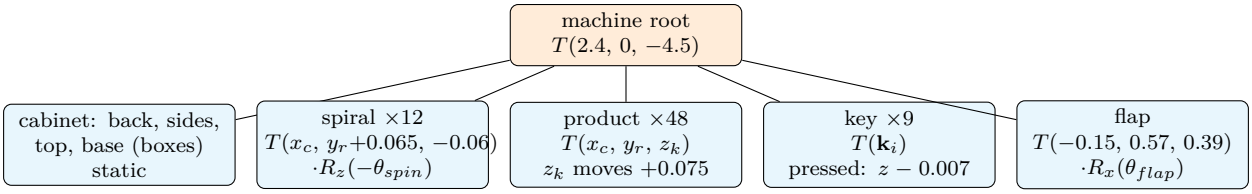


Figure 13: Node tree of the vending machine. Changing θ_{spin} turns a spiral about its own axis; changing θ_{flap} swings the flap about its top edge.

Worked example: one vertex of the right side panel, from the unit cube to the world.

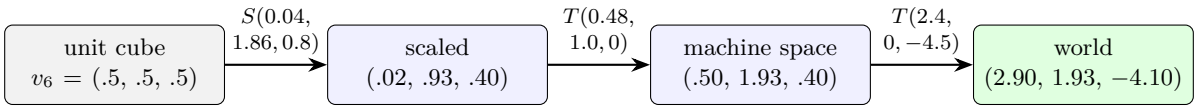


Figure 14: $\mathbf{p}_{world} = T(2.4, 0, -4.5) \cdot T(0.48, 1.0, 0) \cdot S(0.04, 1.86, 0.8) \cdot \mathbf{v}$ for the corner v_6.

Table 8: The 8 corners of the vending machine side panel, step by step

Corner	unit **v**	after S	after T (machine space)	world
v_0	(−.5, −.5, −.5)	(−.02, −.93, −.40)	(.46, .07, −.40)	(2.86, 0.07, −4.90)
v_1	(.5, −.5, −.5)	(.02, −.93, −.40)	(.50, .07, −.40)	(2.90, 0.07, −4.90)
v_2	(.5, .5, −.5)	(.02, .93, −.40)	(.50, 1.93, −.40)	(2.90, 1.93, −4.90)
v_3	(−.5, .5, −.5)	(−.02, .93, −.40)	(.46, 1.93, −.40)	(2.86, 1.93, −4.90)
v_4	(−.5, −.5, .5)	(−.02, −.93, .40)	(.46, .07, .40)	(2.86, 0.07, −4.10)
v_5	(.5, −.5, .5)	(.02, −.93, .40)	(.50, .07, .40)	(2.90, 0.07, −4.10)
v_6	(.5, .5, .5)	(.02, .93, .40)	(.50, 1.93, .40)	(2.90, 1.93, −4.10)
v_7	(−.5, .5, .5)	(−.02, .93, .40)	(.46, 1.93, .40)	(2.86, 1.93, −4.10)

Panel size: 2.86 → 2.90 (4 cm), 0.07 → 1.93 (1.86 m), −4.90 → −4.10 (80 cm). The other two machines use $x = 3.7$ and $x = 5.0$ in the first translation.

Function. Each machine is a state machine (Figure 15). Money is an integer number of Taka. The user inserts a Tk 10 coin or a Tk 50 note, types a code such as `B2` (row A–D, column 1–3) and presses **OK**. The spiral turns 360° in 1.6 s and the products advance by one place (0.075 m). The front product then falls freely with gravity $g = 9.81\,\mathrm{m/s^2}$ and bounces on the tray with a restitution of 0.32:

$$v_y \leftarrow v_y - g\,\Delta t, \qquad \mathbf{p} \leftarrow \mathbf{p} + \mathbf{v}\,\Delta t, \qquad \text{if } p_y < y_{floor}:\ v_y \leftarrow -0.32\,v_y.$$

When the bounce speed drops below 0.35 m/s the product lies still. Pushing the flap puts it into the player's bag, the change falls into the coin cup as small coins, and an empty slot is refilled after 40 s. The LCD, the price tags and the key labels are text drawn on the machine (Section 17.3).

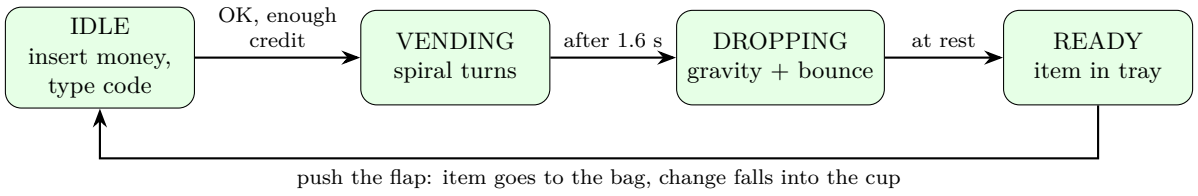


Figure 15: Vending machine state machine.

Colours, textures and sound. Painted body (red, orange or blue), dark metal trims, steel shelves, polished metal coin plate and spirals, transparent front glass ($\alpha = 0.18$), glowing marquee, tube light and LEDs. Sounds: a coin sound when money is inserted, the motor while the spiral turns, a thud when the product hits the tray and small coin sounds when the change lands in the cup.

8.1.2 Products

Every machine holds 12 slots with 4 products each. Six product types exist; their colours come from the product list of each machine (for example Cola red and white, Mango Juice yellow and green).

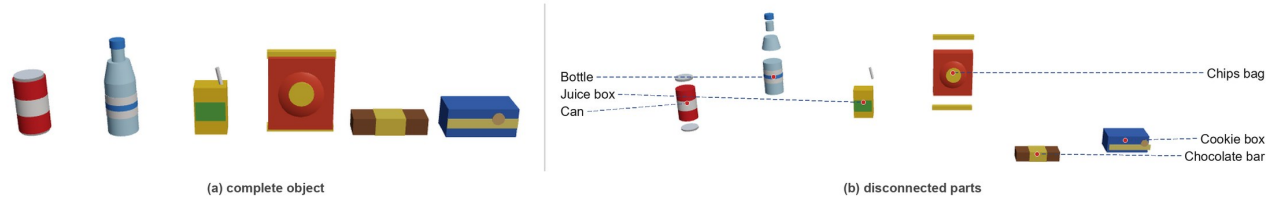


Figure 16: Products: can, bottle, juice box, chips bag, chocolate bar and cookie box.

Table 9: Products (6): the basic shapes it uses (23 shapes in total)

Shape	Name	No.	Used for
	cylinder	12	Can, Bottle, Juice box, Chips bag, Cookie box
	cube	9	Juice box, Chips bag, Chocolate bar, Cookie box
	truncated cone	1	Bottle
	sphere	1	Chips bag

Table 10: Products (6): every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Can	cylinder	4	(.058/.064/.065, .006. . ..11, .058/.064/.065)	(-.45, .003/.06/.12, 0)	–	
Bottle	cylinder	5	(.026. . ..064, .012. . ..11, .026. . ..064)	(-.27, .055. . ..18, 0)	–	
Bottle	truncated cone	1	(.062, .04, .062)	(-.27, .13, 0)	–	
Juice box	cube	2	(.06/.065, .04/.1, .001/.045)	(-.09, .05, 0/.023)	–	
Juice box	cylinder	1	(.008, .04, .008)	(-.072, .12, 0)	(0, 0, 12)	
Chips bag	cube	3	(.13/.14, .014/.15, .016/.04)	(.09, .007/.085/.16, 0)	–	
Chips bag	sphere	1	(.12, .13, .06)	(.09, .085, 0)	–	
Chips bag	cylinder	1	(.05, .004, .05)	(.09, .09, .03)	(90, 0, 0)	
Chocolate bar	cube	2	(.06/.15, .035/.037, .05/.052)	(.27, .018, 0)	–	
Cookie box	cube	2	(.14, .02/.06, .002/.07)	(.45, .03, 0/.036)	–	
Cookie box	cylinder	1	(.025, .002, .025)	(.48, .04, .037)	(90, 0, 0)	

Construction. A *can* is a cylinder body, a coloured band and two metal rims. A *bottle* adds a truncated-cone shoulder, a neck and a cap. A *chips bag* is a box with a flattened sphere ("puff") and two crimp strips; its round logo is a cylinder rotated $R_x(90°)$. A *chocolate bar* is a box with a narrower band, a *cookie box* a box with a stripe and a round picture, and a *juice box* a carton with a label and a straw tilted by $R_z(12°)$.

Function. Products sit on the spirals at $z = 0.0525 - 0.075k$ ($k = 0 \ldots 3$) and move forward while the spiral turns. A dropped product becomes a free body (position, velocity, spin) until it rests in the tray.

Colours, textures and sound. Two colours per product, glossy plastic for bottles, polished

metal for the can rims. Products have no sounds of their own (the drop sound belongs to the machine).

8.1.3 ATM

The ATM is $1.0 \times 2.08 \times 0.72$ m and stands at the back wall of the ATM room.

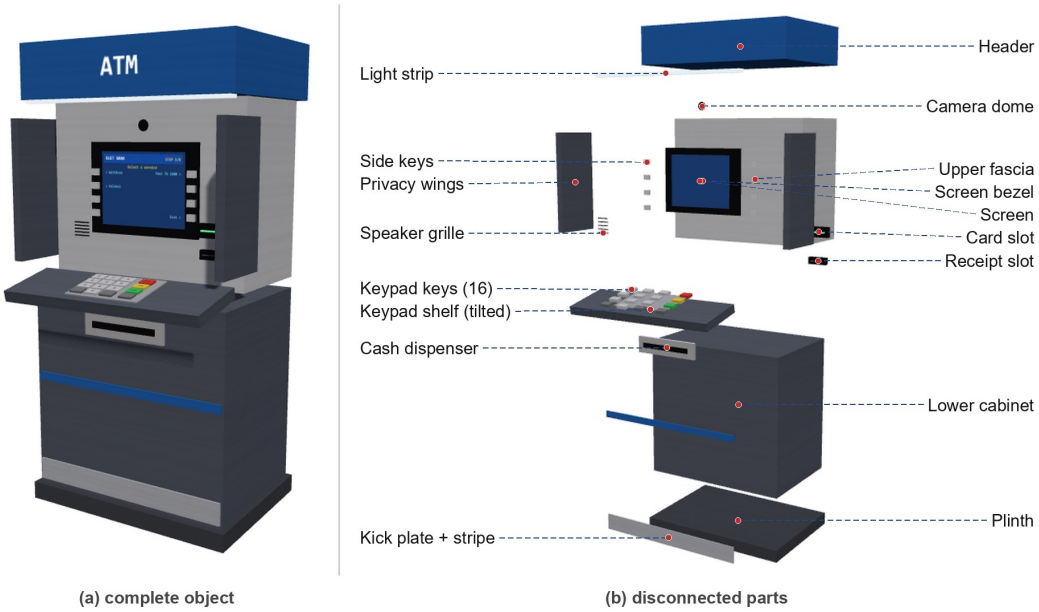


Figure 17: ATM: (a) complete, (b) disconnected parts with their names.

Table 11: ATM: the basic shapes it uses (49 shapes in total)

Shape	Name	No.	Used for
	cube	48	Plinth, Lower cabinet, Upper fascia, Header, Light strip, Privacy wings, Keypad shelf (tilted), Screen bezel, Card slot, Receipt slot, Cash dispenser, Speaker grille, Kick plate + stripe, Screen, Side keys, Keypad keys (16)
	hemisphere	1	Camera dome

Table 12: ATM: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Plinth	cube	1	(1, .08, .72)	(0, .04, 0)	–	
Lower cabinet	cube	1	(.96, .92, .66)	(0, .54, 0)	–	
Upper fascia	cube	1	(.96, .8, .5)	(0, 1.45, -.08)	–	
Header	cube	1	(1.04, .24, .72)	(0, 1.96, 0)	–	
Light strip	cube	1	(1, .02, .05)	(0, 1.83, .33)	–	glow
Privacy wings	cube	2	(.04, .62, .3)	($\pm$.5, 1.45, .28)	–	
Keypad shelf (tilted)	cube	2	(.36/.96, .006/.05, .24/.3)	(0, 1.03/1.06, .4/.4)	(12, 0, 0)	
Screen bezel	cube	1	(.54, .4, .02)	(0, 1.46, .18)	–	
Card slot	cube	2	(.1/.13, .008/.06, .004/.03)	(.34, 1.3, .18/.2)	–	glow
Receipt slot	cube	2	(.08/.11, .004/.04, .004/.03)	(.34, 1.2, .18/.2)	–	
Cash dispenser	cube	2	(.32/.4, .03/.1, .004/.02)	(0, .88, .34/.35)	–	
Camera dome	hemisphere	1	(.05, .025, .05)	(0, 1.73, .18)	(90, 0, 0)	

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Speaker grille	cube	5	(.08, .005, .004)	(-.34, 1.24…1.3, .17)	–	
Kick plate + stripe	cube	2	(.94, .04/.1, .01)	(0, .15/.6, .33/.34)	–	
Screen	cube	1	(.42, .32, .002)	(0, 1.46, .19)	–	glow
Side keys	cube	8	(.04, .03, .016)	(±.24, 1.35…1.53, .19)	–	
Keypad keys (16)	cube	16	(.062, .018, .042)	(-.14…11, 1.05…1.08, .32…48)	(12, 0, 0)	+1

Table 13: ATM in the world, placed with $M_{place} = T(-9.5, 0, -4.55)$: centre of each key part and the world position $M_{place}\, M_{part}\, \mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Plinth	cube	$(-9.50, 0.04, -4.55)$	$(-10.00, 0.00, -4.91)$	$(-9.00, 0.08, -4.19)$
Lower cabinet	cube	$(-9.50, 0.54, -4.55)$	$(-9.98, 0.08, -4.88)$	$(-9.02, 1.00, -4.22)$
Upper fascia	cube	$(-9.50, 1.45, -4.63)$	$(-9.98, 1.05, -4.88)$	$(-9.02, 1.85, -4.38)$
Header	cube	$(-9.50, 1.96, -4.55)$	$(-10.02, 1.84, -4.91)$	$(-8.98, 2.08, -4.19)$
Screen	cube	$(-9.50, 1.46, -4.36)$	$(-9.71, 1.30, -4.36)$	$(-9.29, 1.62, -4.36)$
Keypad shelf (tilted) (1)	cube	$(-9.50, 1.03, -4.15)$	$(-9.98, 1.04, -4.30)$	$(-9.02, 1.03, -4.00)$
Keypad shelf (tilted) (2)	cube	$(-9.50, 1.06, -4.15)$	$(-9.68, 1.09, -4.27)$	$(-9.32, 1.04, -4.04)$
Cash dispenser (1)	cube	$(-9.50, 0.88, -4.21)$	$(-9.70, 0.83, -4.22)$	$(-9.30, 0.93, -4.21)$
Cash dispenser (2)	cube	$(-9.50, 0.88, -4.20)$	$(-9.66, 0.86, -4.21)$	$(-9.34, 0.90, -4.20)$
Card slot (1)	cube	$(-9.16, 1.30, -4.37)$	$(-9.22, 1.27, -4.38)$	$(-9.10, 1.33, -4.36)$
Card slot (2)	cube	$(-9.16, 1.30, -4.35)$	$(-9.21, 1.30, -4.36)$	$(-9.11, 1.30, -4.35)$

Construction. A plinth, a lower cabinet and an upper fascia (boxes) carry the screen bezel, the card slot, the receipt slot, the cash dispenser and two privacy wings. The keypad shelf is a child node tilted towards the user,

$$M_{keypad} = M_{ATM} \cdot T(0,\ 1.035,\ 0.40) \cdot R_x(12^\circ),$$

and the 16 keys are placed in that node on a 4×4 grid, $\mathbf{k}_i = (-0.135 + 0.075\,(i \bmod 4),\ 0.03,\ -0.085 + 0.052\,\lfloor i/4 \rfloor)$, so they follow the slope automatically. A pressed key moves down by 6 mm. The card slides in and out along z, four bank notes slide out of the dispenser, and the receipt is a thin box rotated by $R_x(25^\circ \cdot a)$ while it comes out.

Function. The ATM follows the state machine of Figure 63. The PIN is 1234; amounts of Tk 500, 1,000, 2,000, 5,000 (side keys), fast cash or any other multiple of 100 can be withdrawn. The card is always returned before the cash, and the cash is added to the player's wallet and taken from the bank balance. The screen shows every step with rectangles and text, and a panel on the HUD lists the eight steps (Section 15).

Colours, textures and sound. Dark grey cabinet, light grey fascia, blue header with a glowing light strip, brushed metal keypad plate, coloured CANCEL / CLEAR / ENTER keys, glowing screen. Sounds: a beep for every key and a cash-counting sound when the notes come out.

8.1.4 Coffee machine

The coffee machine ($0.43 \times 0.81 \times 0.40$ m) stands on the counter in the lounge.

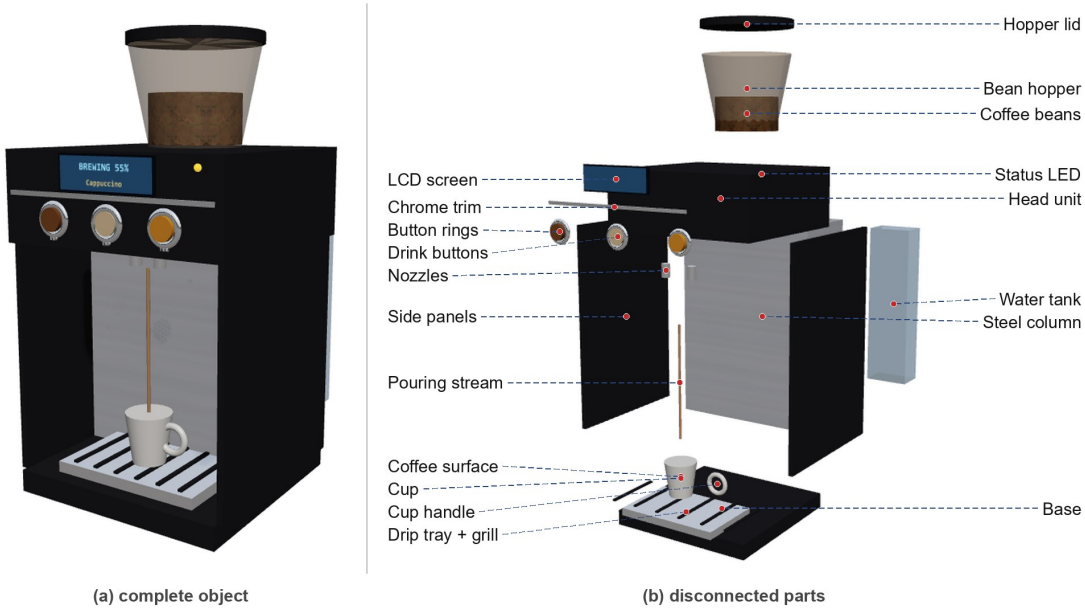


Figure 18: Coffee machine while brewing: (a) complete, (b) disconnected parts with their names.

Table 14: Coffee machine: the basic shapes it uses (32 shapes in total)

Shape	Name	No.	Used for
	cube	16	Base, Steel column, Side panels, Head unit, LCD screen, Drip tray + grill, Water tank, Chrome trim
	cylinder	9	Nozzles, Coffee beans, Hopper lid, Drink buttons, Coffee surface, Pouring stream
	torus	4	Button rings, Cup handle
	truncated cone	2	Bean hopper, Cup
	sphere	1	Status LED

Table 15: Coffee machine: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Base	cube	1	(.4, .04, .4)	(0, .02, 0)	–	
Steel column	cube	1	(.4, .56, .2)	(0, .32, -.1)	–	
Side panels	cube	2	(.012, .56, .38)	($\pm$.2, .32, 0)	–	
Head unit	cube	1	(.43, .16, .4)	(0, .53, 0)	–	
LCD screen	cube	2	(.18/.19, .055/.062, .002/.004)	(0, .57, .2/.2)	–	glow
Nozzles	cylinder	2	(.022, .035, .022)	($\pm$.025, .43, .09)	–	
Drip tray + grill	cube	7	(.008/.28, .004/.02, .16/.18)	(-.11…11, .05/.062, .08)	–	
Bean hopper	truncated cone	1	(.24, .18, .24)	(0, .7, -.06)	(180, 0, 0)	glass
Coffee beans	cylinder	1	(.17, .09, .17)	(0, .66, -.06)	–	
Hopper lid	cylinder	1	(.25, .02, .25)	(0, .8, -.06)	–	
Water tank	cube	1	(.12, .45, .08)	(.12, .33, -.24)	–	glass
Chrome trim	cube	1	(.4, .008, .004)	(0, .54, .2)	–	
Drink buttons	cylinder	3	(.04, .012, .04)	(-.11/0/.11, .5, .2/.2)	(90, 0, 0)	
Button rings	torus	3	(.062, .04, .062)	(-.11/0/.11, .5, .2)	(90, 0, 0)	
Status LED	sphere	1	(.012, .012, .012)	(.17, .59, .2)	–	glow

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Cup	truncated cone	1	(.085, .09, .085)	(0, .11, .09)	(180, 0, 0)	
Cup handle	torus	1	(.06, .06, .06)	(.05, .11, .09)	(90, 0, 0)	
Coffee surface	cylinder	1	(.067, .003, .067)	(0, .11, .09)	–	
Pouring stream	cylinder	1	(.006, .3, .006)	(0, .26, .09)	–	

Table 16: Coffee machine in the world, placed with $M_{place} = T(-5, 0.92, -4.55)$: centre of each key part and the world position $M_{place}\, M_{part}\, \mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Base	cube	$(-5.00, 0.94, -4.55)$	$(-5.20, 0.92, -4.75)$	$(-4.80, 0.96, -4.35)$
Steel column	cube	$(-5.00, 1.24, -4.65)$	$(-5.20, 0.96, -4.75)$	$(-4.80, 1.52, -4.55)$
Head unit	cube	$(-5.00, 1.45, -4.55)$	$(-5.21, 1.37, -4.75)$	$(-4.79, 1.53, -4.35)$
Bean hopper	truncated cone	$(-5.00, 1.62, -4.61)$	$(-4.88, 1.71, -4.61)$	$(-4.92, 1.53, -4.61)$
Drip tray + grill (1)	cube	$(-5.00, 0.97, -4.47)$	$(-5.14, 0.96, -4.56)$	$(-4.86, 0.98, -4.38)$
Drip tray + grill (2)	cube	$(-5.11, 0.98, -4.47)$	$(-5.11, 0.98, -4.55)$	$(-5.11, 0.98, -4.39)$
Cup	truncated cone	$(-5.00, 1.03, -4.46)$	$(-4.96, 1.07, -4.46)$	$(-4.97, 0.98, -4.46)$

Construction. A base, a steel column, two side panels and a head unit form the body. The bean hopper is a glass truncated cone turned upside down with $R_x(180°)$, so it is wider at the top. The three round buttons are cylinders rotated $R_x(90°)$ so that their flat face points forward, each with a torus ring. The cup is also an upside-down truncated cone with a torus handle. The coffee inside the cup is a thin disc whose height and radius follow the cup's taper:

$$h = 0.006 + 0.075\, f, \qquad r = 2\,(0.0298 + (0.0425 - 0.0298)\, h/0.09) - 0.006,$$

where $f \in [0, 1]$ is the fill level. The pouring stream is a thin cylinder from the nozzle down to the coffee surface.

Function. The three buttons make an Espresso, a Cappuccino or a Masala Tea: `IDLE` → `CUP_DROP` (0.5 s, the cup falls) → `GRINDING` (1.6 s) → `BREWING` (4.5 s, the cup fills) → `READY`. Steam puffs rise from the cup (Section 17.2). Taking the cup puts the drink into the bag.

Colours, textures and sound. Black body, brushed steel column, glass hopper with brown bean texture, glowing LCD and status LED (green when idle, orange while busy). Sounds: grinder, pouring and a beep when ready.

8.1.5 Television

The television hangs on the lounge wall at $(0, 1.75, -4.86)$.

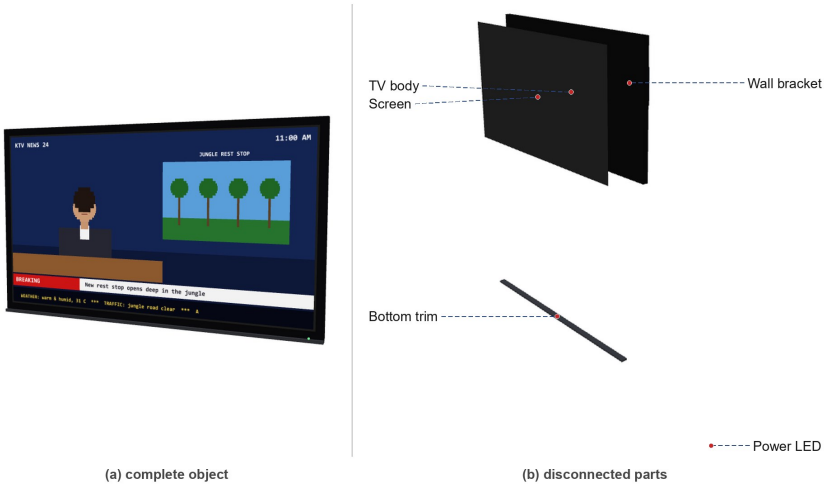


Figure 19: Television: (a) complete with the news channel, (b) disconnected parts.

Table 17: Television: the basic shapes it uses (5 shapes in total)

Shape	Name	No.	Used for
	cube	4	Wall bracket, TV body, Bottom trim, Screen
	sphere	1	Power LED

Table 18: Television: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Wall bracket	cube	1	(.45, .32, .06)	(0, 1, -.06)	–	
TV body	cube	1	(1.86, 1.08, .045)	(0, 1, -.015)	–	
Bottom trim	cube	1	(1.86, .016, .05)	(0, .45, 0)	–	
Screen	cube	1	(1.78, 1, .002)	(0, 1, .009)	–	glow
Power LED	sphere	1	(.012, .012, .012)	(.86, .47, .02)	–	glow

Table 19: Television in the world, placed with $M_{place} = T(0, 1.75, -4.86)$: centre of each key part and the world position $M_{place}\,M_{part}\,\mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Wall bracket	cube	$(0.00, 1.75, -4.92)$	$(-0.23, 1.59, -4.95)$	$(0.23, 1.91, -4.89)$
TV body	cube	$(0.00, 1.75, -4.88)$	$(-0.93, 1.21, -4.90)$	$(0.93, 2.29, -4.85)$
Screen	cube	$(0.00, 1.75, -4.85)$	$(-0.89, 1.25, -4.85)$	$(0.89, 2.25, -4.85)$
Bottom trim	cube	$(0.00, 1.20, -4.86)$	$(-0.93, 1.19, -4.88)$	$(0.93, 1.21, -4.83)$
Power LED	sphere	$(0.86, 1.22, -4.84)$	$(0.86, 1.21, -4.84)$	$(0.86, 1.22, -4.84)$

Construction. A wall bracket, a body box, a bottom trim, the screen and a power LED. The pictures are not textures: each channel is drawn every frame on a *canvas* placed on the screen, using rectangles and text. Circles (heads, the sun, trees, the ball) are drawn as horizontal strips of width $2\sqrt{r^2 - y^2}$, and triangles (mountains) as strips whose width shrinks linearly with the height.

Function. Looking at the screen and pressing E cycles through news (with a moving ticker), nature, live football and a weather map that shows the in-game time, and off. While it is on, the TV adds a flickering coloured point light to the room whose colour follows the picture.

Colours, textures and sound. Black body, dark metal bracket, glowing screen, green or red power LED. A click is heard when the channel changes.

8.1.6 Aquarium and fish

The aquarium ($1.8 \times 1.45 \times 0.5$ m) stands in the library.

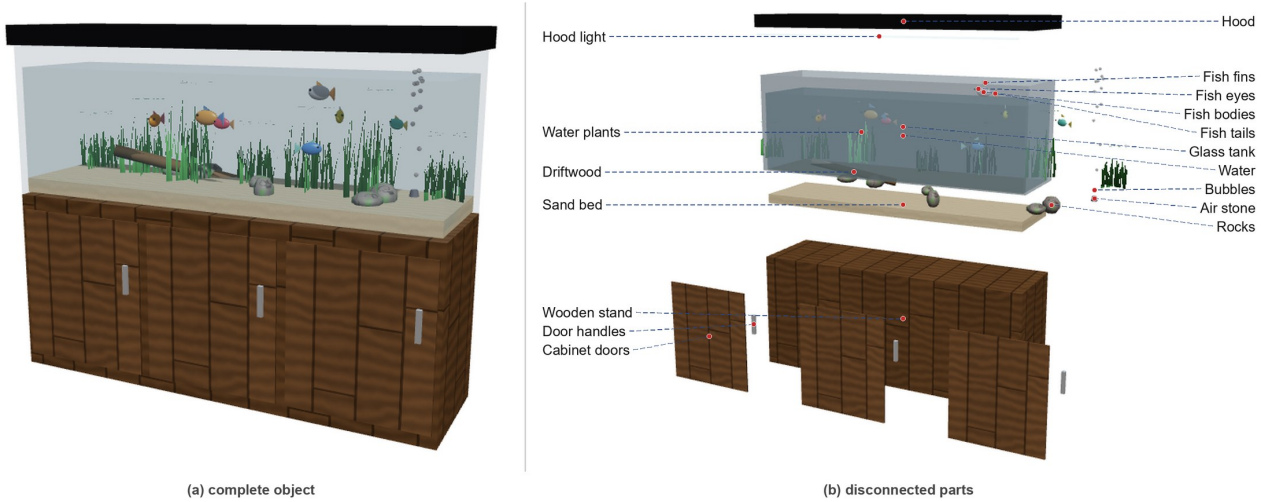


Figure 20: Aquarium: (a) complete, (b) disconnected parts (stand, sand, rocks, plants, glass, water, fish, bubbles).

Table 20: Aquarium: the basic shapes it uses (76 shapes in total)

Shape	Name	No.	Used for
	sphere	35	Fish bodies, Fish eyes, Bubbles
	cube	12	Wooden stand, Cabinet doors, Door handles, Sand bed, Hood, Hood light, Water, Glass tank
	cross card	7	Water plants
	cone	7	Fish tails
	wedge	7	Fish fins
	sphere (low)	6	Rocks
	cylinder (low)	1	Driftwood
	truncated cone	1	Air stone

Table 21: Aquarium: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Wooden stand	cube	1	(1.8, .75, .5)	(0, .38, 0)	–	
Cabinet doors	cube	3	(.56, .64, .01)	(-.6/0/.6, .37, .25)	–	
Door handles	cube	3	(.02, .12, .02)	(-.38/.22/.82, .45, .26)	–	

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Sand bed	cube	1	(1.76, .06, .46)	(0, .79, 0)	–	
Rocks	sphere (low)	6	(.083. . .16, .063. . .12, .082. . .13)	(-.36. . .59, .83, -.14. . .085)	(0, 13. . .114, 0)	
Driftwood	cylinder (low)	1	(.05, .6, .05)	(-.3, .86, -.08)	(0, 30, 75)	
Air stone	truncated cone	1	(.04, .03, .04)	(.68, .83, -.12)	–	
Hood	cube	1	(1.84, .08, .54)	(0, 1.41, 0)	–	
Hood light	cube	1	(1.7, .01, .05)	(0, 1.37, .1)	–	glow
Water plants	cross card	7	(.22, .25/.33/.41, .22)	(-.78. . .78, .81, -.14/-.09/-.04)	(0, 0. . .150, 0)	
Fish bodies	sphere	7	(.024. . .036, .04. . .059, .067. . .099)	(-.39. . .63, 1.01. . .1.19, -.085. . .098)	(0, -118. . .102, 0)	+3
Fish tails	cone	7	(.004/.005/.006, .027. . .04, .038. . .055)	(-.39. . .67, 1.01. . .1.19, -.066. . .1)	(90, -158. . .176, 180)	+3
Fish fins	wedge	7	(.003/.004, .016. . .024, .027. . .04)	(-.39. . .63, 1.03. . .1.23, -.083. . .098)	(0, -173. . .89, 0)	+3
Fish eyes	sphere	14	(.007. . .01, .007. . .01, .007. . .01)	(-.4. . .61, 1.02. . .1.2, -.1. . .11)	(0, -118. . .102, 0)	
Bubbles	sphere	14	(.018, .018, .018)	(.66. . .69, .83. . .1.29, -.12)	–	glass
Water	cube	1	(1.76, .5, .46)	(0, 1.03, 0)	–	glass
Glass tank	cube	1	(1.8, .62, .5)	(0, 1.06, 0)	–	glass

Table 22: Aquarium in the world, placed with $M_{place} = T(9.8, 0, 4.45)\, R_y(180°)$: centre of each key part and the world position $M_{place}\, M_{part}\, \mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Wooden stand	cube	(9.80, 0.38, 4.45)	(10.70, 0.00, 4.70)	(8.90, 0.75, 4.20)
Sand bed	cube	(9.80, 0.79, 4.45)	(10.68, 0.76, 4.68)	(8.92, 0.82, 4.22)
Water	cube	(9.80, 1.03, 4.45)	(10.68, 0.78, 4.68)	(8.92, 1.28, 4.22)
Glass tank	cube	(9.80, 1.06, 4.45)	(10.70, 0.75, 4.70)	(8.90, 1.37, 4.20)
Hood	cube	(9.80, 1.41, 4.45)	(10.72, 1.37, 4.72)	(8.88, 1.45, 4.18)
Fish bodies (1)	sphere	(10.19, 1.06, 4.45)	(10.19, 1.04, 4.45)	(10.19, 1.09, 4.45)
Fish bodies (2)	sphere	(9.39, 1.12, 4.46)	(9.39, 1.10, 4.46)	(9.39, 1.15, 4.46)

Construction. A wooden stand with three doors carries a sand bed, rocks (squashed low spheres), a driftwood branch (low cylinder rotated $R_y(30°)R_z(75°)$), water plants (cross cards), an air stone, the water volume and the glass tank (two transparent boxes) and a hood with a light strip. Each fish is a node $T(x, y, z) \cdot R_y(\psi)$ with a sphere body scaled to $(0.45s, 0.75s, 1.25s)$, a cone tail in a child node that wags with $R_y(25° \sin(9t + 3\varphi))$, a wedge fin and two sphere eyes.

Function. Each of the 7 fish swims on its own ellipse, $x = r_x \cos\varphi$, $z = r_z \sin\varphi$ with $\varphi \mathrel{+}= \omega\, \Delta t$, and faces its direction of motion, $\psi = \operatorname{atan2}(\Delta x, \Delta z)$. "Feed the fish" drops 12 food crumbs that sink slowly; while there is food, every fish swims up to the surface. Bubbles rise from the air stone and the plants sway (Section 17).

Colours, textures and sound. Wood texture on the stand, sand texture, rock texture, bark texture on the driftwood, transparent water ($\alpha = 0.16$) and glass, glowing hood light. A click sounds when the fish are fed.

8.2 Furniture

8.2.1 Sofa

The sofa function builds a sofa for any number of seats n; its width is $W = 0.72n + 0.32$ m. The lounge has two three-seat sofas and four armchairs.

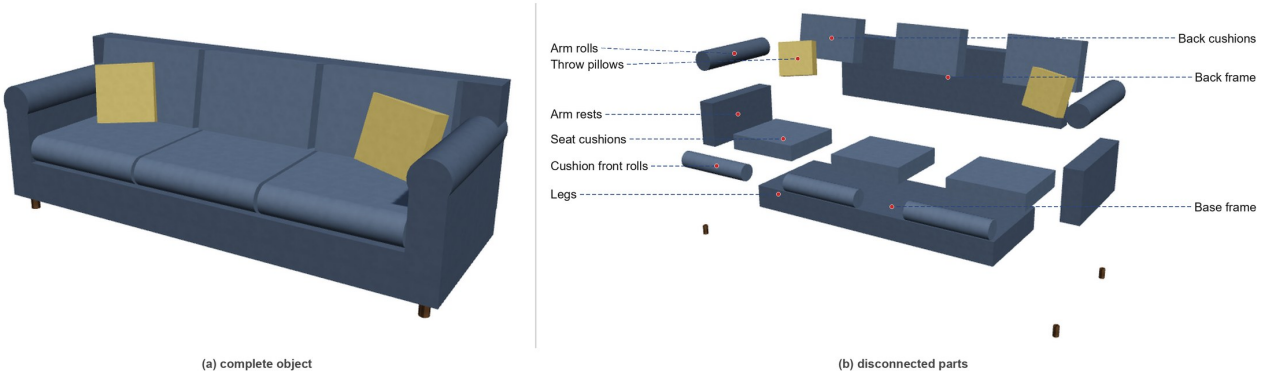


Figure 21: Sofa with three seats: (a) complete, (b) disconnected parts.

Table 23: Sofa (3 seats): the basic shapes it uses (21 shapes in total)

Shape	Name	No.	Used for
	cube	12	Base frame, Back frame, Seat cushions, Back cushions, Arm rests, Throw pillows
	cylinder	9	Cushion front rolls, Arm rolls, Legs

Table 24: Sofa (3 seats): every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Base frame	cube	1	(2.48, .22, .88)	(0, .2, 0)	–	
Back frame	cube	1	(2.48, .66, .16)	(0, .56, -.37)	–	
Seat cushions	cube	3	(.7, .15, .64)	(-.72/0/.72, .38, .05)	–	
Cushion front rolls	cylinder	3	(.15, .7, .15)	(-.72/0/.72, .38, .37)	(0, 0, 90)	
Back cushions	cube	3	(.68, .44, .17)	(-.72/0/.72, .68, -.24)	(-10, 0, 0)	
Arm rests	cube	2	(.16, .44, .88)	($\pm$1.16, .42, 0)	–	
Arm rolls	cylinder	2	(.19, .88, .19)	($\pm$1.16, .64, 0)	(90, 0, 0)	
Legs	cylinder	4	(.05, .09, .05)	($\pm$1.16, .045, $\pm$.36)	–	
Throw pillows	cube	2	(.36, .34, .12)	($\pm$.83, .62, -.12)	(-14, $\pm$15, $\pm$6)	

Table 25: Sofa (3 seats) in the world, placed with $M_{place} = T(0, 0, -0.6)\,R_y(180°)$: centre of each key part and the world position $M_{place}\,M_{part}\,\mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Base frame	cube	$(0.00, 0.20, -0.60)$	$(1.24, 0.09, -0.16)$	$(-1.24, 0.31, -1.04)$
Back frame	cube	$(0.00, 0.56, -0.23)$	$(1.24, 0.23, -0.15)$	$(-1.24, 0.89, -0.31)$
Seat cushions (1)	cube	$(0.72, 0.38, -0.65)$	$(1.07, 0.30, -0.33)$	$(0.37, 0.46, -0.97)$
Seat cushions (2)	cube	$(0.00, 0.38, -0.65)$	$(0.35, 0.30, -0.33)$	$(-0.35, 0.46, -0.97)$
Arm rests (1)	cube	$(1.16, 0.42, -0.60)$	$(1.24, 0.20, -0.16)$	$(1.08, 0.64, -1.04)$
Arm rests (2)	cube	$(-1.16, 0.42, -0.60)$	$(-1.08, 0.20, -0.16)$	$(-1.24, 0.64, -1.04)$
Legs (1)	cylinder	$(1.16, 0.04, -0.24)$	$(1.14, 0.00, -0.24)$	$(1.14, 0.09, -0.24)$
Legs (2)	cylinder	$(1.16, 0.04, -0.96)$	$(1.14, 0.00, -0.96)$	$(1.14, 0.09, -0.96)$

Construction. A base frame and a back frame (boxes); per seat a cushion box, a rounded front roll (cylinder rotated $R_z(90°)$ so that it lies along x) and a back cushion tilted by $R_x(-10°)$; two arm rests made of a box and a rolled cylinder rotated $R_x(90°)$ to lie along z; four wooden legs; and two throw pillows rotated about all three axes.

Function. Visitors can sit on any seat: each seat has a stand point and a sit point. The sitting pose of a person is described in Section 8.3.

Colours, textures and sound. Blue fabric texture (darker on the frame), yellow fabric pillows, wooden legs. No sound.

8.2.2 Armchair

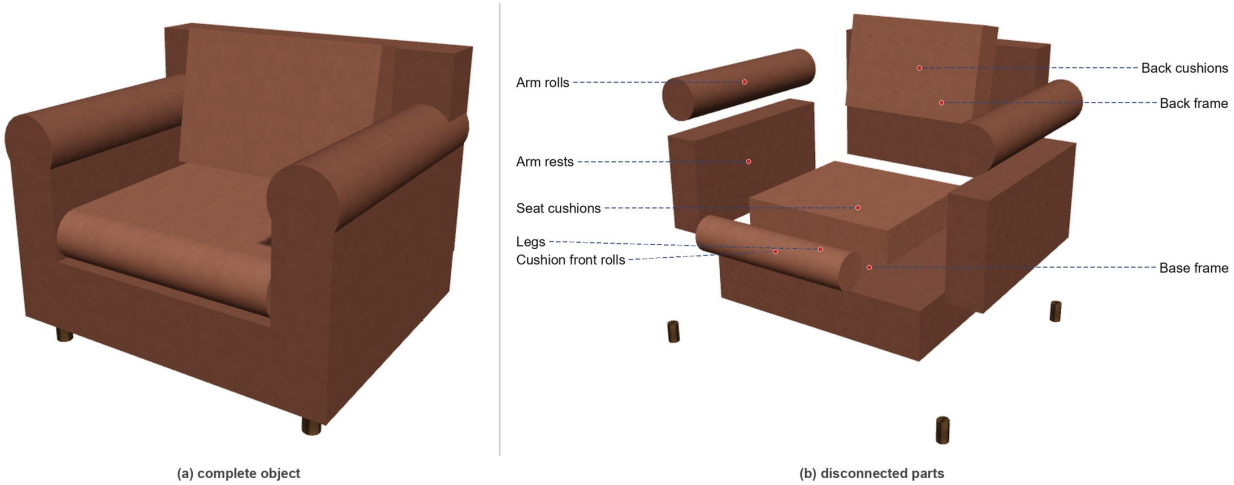


Figure 22: Armchair: (a) complete, (b) disconnected parts.

Table 26: Armchair: the basic shapes it uses (13 shapes in total)

Shape	Name	No.	Used for
	cylinder	7	Cushion front rolls, Arm rolls, Legs
	cube	6	Base frame, Back frame, Seat cushions, Back cushions, Arm rests

Table 27: Armchair: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Base frame	cube	1	(1.04, .22, .88)	(0, .2, 0)	–	
Back frame	cube	1	(1.04, .66, .16)	(0, .56, -.37)	–	
Seat cushions	cube	1	(.7, .15, .64)	(0, .38, .05)	–	
Cushion front rolls	cylinder	1	(.15, .7, .15)	(0, .38, .37)	(0, 0, 90)	
Back cushions	cube	1	(.68, .44, .17)	(0, .68, -.24)	(-10, 0, 0)	
Arm rests	cube	2	(.16, .44, .88)	($\pm$.44, .42, 0)	–	
Arm rolls	cylinder	2	(.19, .88, .19)	($\pm$.44, .64, 0)	(90, 0, 0)	
Legs	cylinder	4	(.05, .09, .05)	($\pm$.44, .045, $\pm$.36)	–	

Construction and function. The armchair is the sofa function called with $n = 1$ (no throw pillows), which shows how one parametric model produces different objects. Armchairs stand in the lounge and in the library, where visitors sit and read.

Colours, textures and sound. Brown fabric texture, wooden legs. No sound.

8.2.3 Coffee table and rug

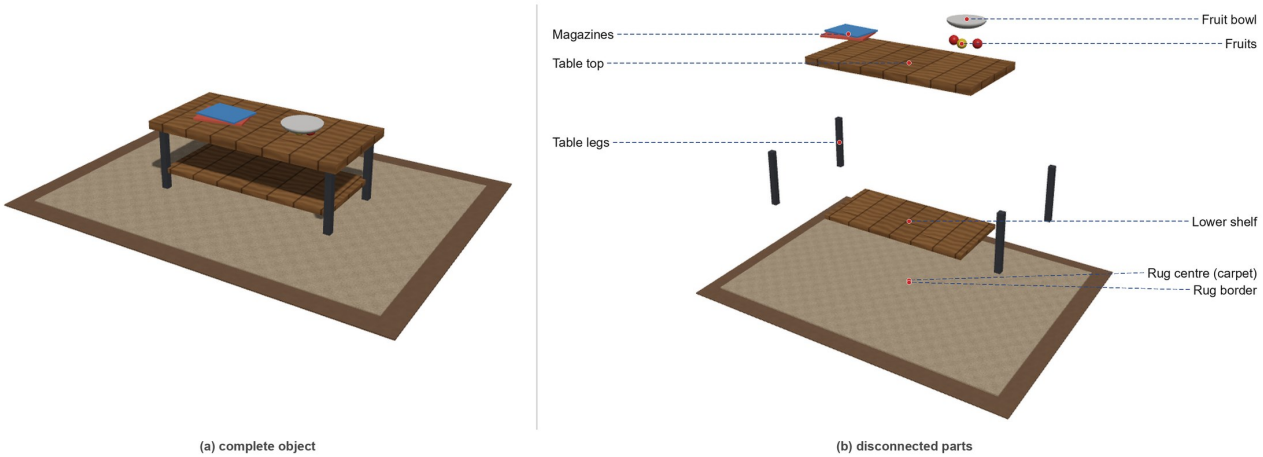


Figure 23: Coffee table on its rug: (a) complete, (b) disconnected parts.

Table 28: Coffee table + rug: the basic shapes it uses (14 shapes in total)

Shape	Name	No.	Used for
	cube	10	Rug border, Rug centre (carpet), Table top, Lower shelf, Table legs, Magazines
	sphere	3	Fruits
	hemisphere	1	Fruit bowl

Table 29: Coffee table + rug: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Rug border	cube	1	(2.4, .008, 1.8)	(0, .004, 0)	–	
Rug centre (carpet)	cube	1	(2.16, .01, 1.56)	(0, .009, 0)	–	
Table top	cube	1	(1.2, .05, .6)	(0, .42, 0)	–	
Lower shelf	cube	1	(1.08, .03, .5)	(0, .13, 0)	–	
Table legs	cube	4	(.04, .4, .04)	($\pm$.54, .2, $\pm$.24)	–	
Magazines	cube	2	(.28/.3, .01/.012, .21/.22)	(-.26, .45/.46, 0/.02)	(0, -5/12, 0)	
Fruit bowl	hemisphere	1	(.24, -.1, .24)	(.24, .51, 0)	–	
Fruits	sphere	3	(.06, .06, .06)	(.19/.24/.29, .47, 0/.03)	–	

Construction. A wooden top and a lower shelf (boxes), four metal legs placed at $(\pm(w/2 - 0.06),\ 0.2,\ \pm(d/2 - 0.06))$, two magazines rotated about y, a fruit bowl and three fruits. The bowl is a hemisphere scaled with a *negative* y factor, $S = (0.24, -0.10, 0.24)$, which mirrors it so that it opens upwards. The rug is two thin boxes: a border and a carpet-textured centre.

Colours, textures and sound. Wood texture, dark metal legs, carpet texture on the rug. No sound.

8.2.4 Bookshelf

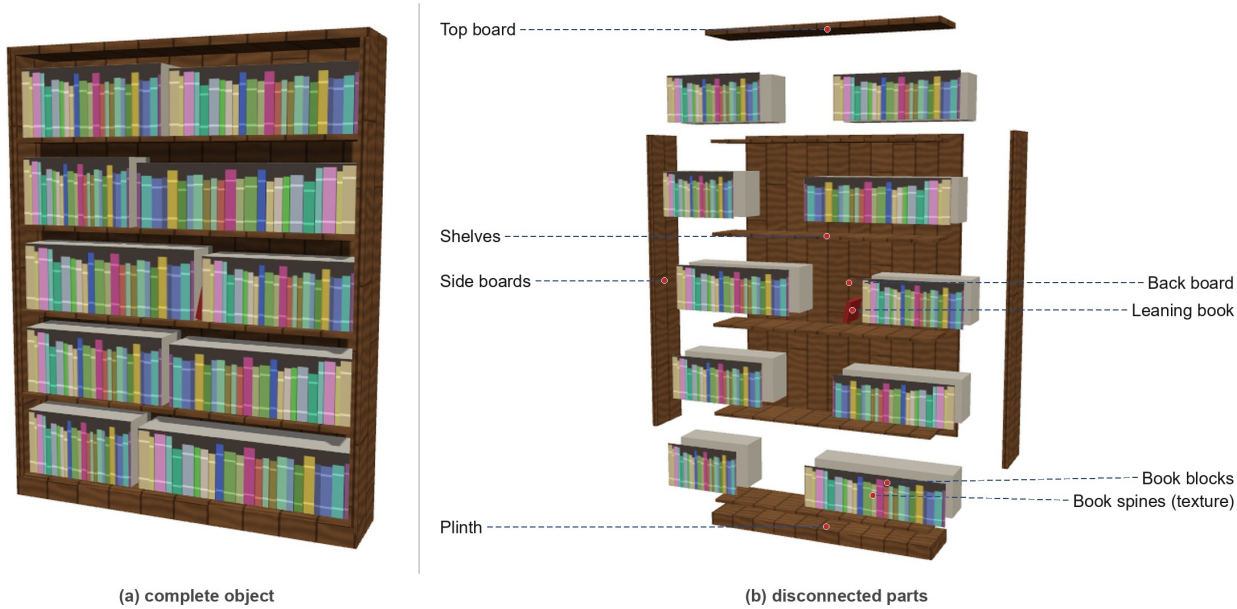


Figure 24: Bookshelf: (a) complete, (b) disconnected parts.

Table 30: Bookshelf: the basic shapes it uses (31 shapes in total)

Shape	Name	No.	Used for
	cube	21	Side boards, Top board, Back board, Plinth, Shelves, Book blocks, Leaning book
	quad	10	Book spines (texture)

Table 31: Bookshelf: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Side boards	cube	2	(.04, 2.3, .38)	($\pm$.88, 1.15, 0)	–	
Top board	cube	1	(1.8, .04, .38)	(0, 2.28, 0)	–	
Back board	cube	1	(1.8, 2.3, .02)	(0, 1.15, -.18)	–	
Plinth	cube	1	(1.8, .1, .38)	(0, .05, 0)	–	
Shelves	cube	5	(1.74, .025, .36)	(0, .1…1.84, 0)	–	
Book blocks	cube	10	(.59…1.06, .31/.34, .24)	(-.55….49, .27…2.03, .02)	–	
Book spines (texture)	quad	10	(.59…1.06, -.31/.31/.34, 1)	(-.55….49, .27…2.03, .14)	(0, 0, 0/180)	
Leaning book	cube	1	(.035, .16, .22)	(.12, 1.07, .02)	(0, 0, -12)	

Table 32: Bookshelf in the world, placed with $M_{place} = T(9, 0, -4.69)$: centre of each key part and the world position $M_{place}\, M_{part}\, \mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Side boards (1)	cube	$(8.12, 1.15, -4.69)$	$(8.10, 0.00, -4.88)$	$(8.14, 2.30, -4.50)$
Side boards (2)	cube	$(9.88, 1.15, -4.69)$	$(9.86, 0.00, -4.88)$	$(9.90, 2.30, -4.50)$
Top board	cube	$(9.00, 2.28, -4.69)$	$(8.10, 2.26, -4.88)$	$(9.90, 2.30, -4.50)$
Back board	cube	$(9.00, 1.15, -4.87)$	$(8.10, 0.00, -4.88)$	$(9.90, 2.30, -4.86)$
Shelves (1)	cube	$(9.00, 0.10, -4.69)$	$(8.13, 0.09, -4.87)$	$(9.87, 0.11, -4.51)$
Shelves (2)	cube	$(9.00, 0.54, -4.69)$	$(8.13, 0.52, -4.87)$	$(9.87, 0.55, -4.51)$
Plinth	cube	$(9.00, 0.05, -4.69)$	$(8.10, 0.00, -4.88)$	$(9.90, 0.10, -4.50)$

Construction. Two side boards, a top, a back board, a plinth and five shelves. The books are not modelled one by one: each row is a plain box (the colour of the pages) with a quad in front that carries the generated *books* texture. Each shelf has two rows of different widths, and on every second shelf one quad is mirrored with a negative x scale, so the pattern does not repeat. One book leans by $R_z(-12°)$.

Function. Visitors stand in front of the shelves and browse.

Colours, textures and sound. Wood texture, books texture, plain page colour. No sound.

8.2.5 Reading table and chair

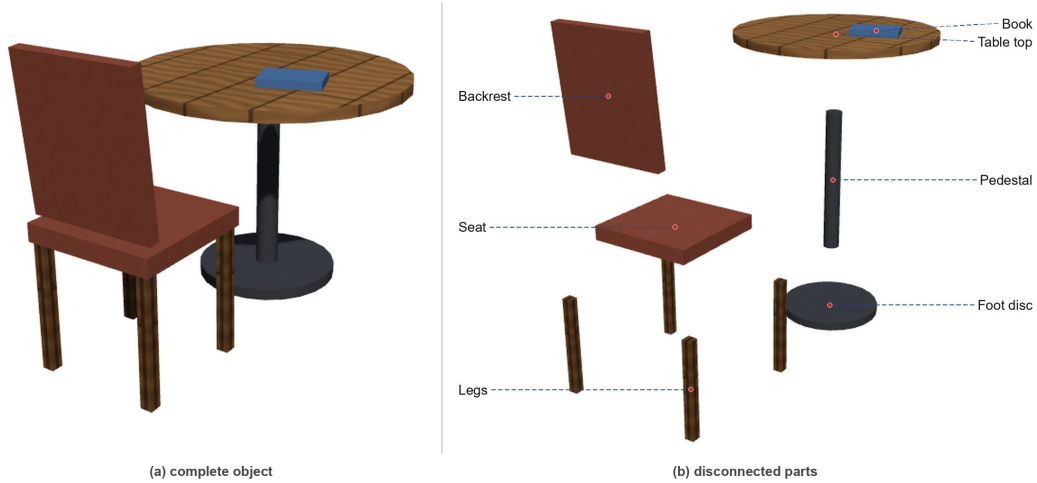


Figure 25: Reading table and chair: (a) complete, (b) disconnected parts.

Table 33: Reading table + chair: the basic shapes it uses (10 shapes in total)

Shape	Name	No.	Used for
	cube	7	Book, Seat, Backrest, Legs
	cylinder	3	Table top, Pedestal, Foot disc

Table 34: Reading table + chair: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Table top	cylinder	1	(1, .04, 1)	(0, .74, 0)	–	
Pedestal	cylinder	1	(.08, .72, .08)	(0, .37, 0)	–	
Foot disc	cylinder	1	(.5, .04, .5)	(0, .02, 0)	–	
Book	cube	1	(.22, .03, .16)	(.15, .78, .1)	(0, 20, 0)	
Seat	cube	1	(.44, .06, .44)	(0, .45, .75)	(0, -180, 0)	
Backrest	cube	1	(.42, .5, .04)	(0, .75, .95)	(-6, -180, 0)	
Legs	cube	4	(.04, .42, .04)	($\pm$.19, .21, .56/.94)	(0, -180, 0)	

Construction. The table is a round top, a pedestal and a foot disc (cylinders) with a book box rotated by $R_y(20°)$. The chair is a seat box, a backrest tilted by $R_x(-6°)$ and four legs at ($\pm$0.19, 0.21, $\pm$0.19); in the figure the chair node is turned by $R_y(180°)$ to face the table.

Colours, textures and sound. Wood texture, dark metal pedestal, red fabric seat. No sound.

8.2.6 Wall clock

Three clocks hang in the building; their hands show the in-game time.

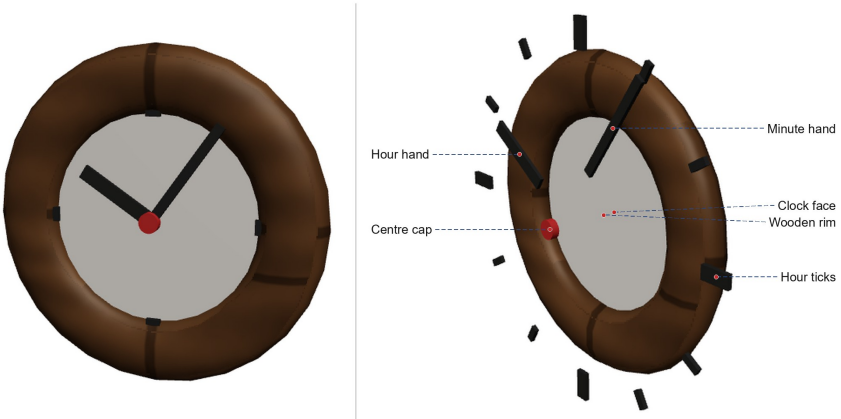


Figure 26: Wall clock: (a) complete, (b) disconnected parts.

Table 35: Wall clock: the basic shapes it uses (17 shapes in total)

Shape	Name	No.	Used for
	cube	14	Hour ticks, Hour hand, Minute hand
	cylinder	2	Clock face, Centre cap
	torus	1	Wooden rim

Table 36: Wall clock: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Wooden rim	torus	1	(.42, .25, .42)	(0, 0, .02)	(90, 0, 0)	

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Clock face	cylinder	1	(.36, .02, .36)	(0, 0, .015)	(90, 0, 0)	
Hour ticks	cube	12	(.01/.018, .025/.04, .006)	(-.14. . .14, -.14. . .14, .03)	(0, 0, -150. . .180)	
Hour hand	cube	1	(.018, .11, .005)	(-.042, .027, .036)	(0, 0, 57)	
Minute hand	cube	1	(.012, .15, .005)	(.041, .057, .04)	(0, 0, -36)	
Centre cap	cylinder	1	(.025, .01, .025)	(0, 0, .044)	(90, 0, 0)	

Table 37: Wall clock in the world, placed with $M_{place} = T(0, 3, -4.87)$: centre of each key part and the world position $M_{place}\, M_{part}\, \mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Wooden rim	torus	$(0.00, 3.00, -4.85)$	$(0.21, 3.00, -4.85)$	$(0.13, 3.00, -4.85)$
Clock face	cylinder	$(0.00, 3.00, -4.86)$	$(0.18, 3.00, -4.87)$	$(0.18, 3.00, -4.84)$
Hour hand	cube	$(-0.04, 3.03, -4.83)$	$(0.00, 2.99, -4.84)$	$(-0.08, 3.06, -4.83)$
Minute hand	cube	$(0.04, 3.06, -4.83)$	$(-0.01, 3.00, -4.83)$	$(0.09, 3.11, -4.83)$
Centre cap	cylinder	$(0.00, 3.00, -4.83)$	$(0.01, 3.00, -4.83)$	$(0.01, 3.00, -4.82)$

Construction. A torus rim and a face disc, both rotated $R_x(90°)$ to face forward. Each of the 12 ticks is a box in its own node rotated by $R_z(-30° \, i)$, so the same box is repeated around the face. The hands are boxes in nodes rotated by

$$\theta_{hour} = -30° \cdot (h \bmod 12), \qquad \theta_{minute} = -6° \cdot m,$$

where h is the in-game hour (with fraction) and m the minutes.

Colours, textures and sound. Wooden rim, light face, dark ticks and hands, red centre cap. No sound.

8.2.7 Pendant lamp



Figure 27: Pendant lamp: (a) complete, (b) disconnected parts.

Table 38: Pendant lamp: the basic shapes it uses (5 shapes in total)

Shape	Name	No.	Used for
	cylinder	3	Ceiling disc, Cord, Diffuser
	truncated cone	1	Shade
	sphere	1	Bulb

Table 39: Pendant lamp: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Ceiling disc	cylinder	1	(.14, .03, .14)	(0, -.015, 0)	–	■ ■
Cord	cylinder	1	(.012, .6, .012)	(0, -.3, 0)	–	■
Shade	truncated cone	1	(.46, .26, .46)	(0, -.73, 0)	–	■ ■
Diffuser	cylinder	1	(.42, .01, .42)	(0, -.86, 0)	–	■ glow
Bulb	sphere	1	(.11, .11, .11)	(0, -.82, 0)	–	■ glow

Construction. A ceiling disc, a cord (thin cylinder), a shade (truncated cone), a diffuser disc and a bulb sphere. The lamp node is placed at the ceiling point and every part hangs below it, so the cord length is one parameter.

Function. Each lamp belongs to a light group. When the group is on, the bulb and the diffuser glow and a point light is placed at the bulb (Section 9).

Colours, textures and sound. Dark (lounge) or bronze (library) shade, glowing bulb. No sound.

8.2.8 Table lamp and floor lamp



Figure 28: Table lamp on a side table, and a floor lamp: (a) complete, (b) disconnected parts.

Table 40: Table + floor lamp: the basic shapes it uses (10 shapes in total)

Shape	Name	No.	Used for
	cylinder	6	Side-table top, Side-table stem, Side-table foot, Table-lamp stem, Floor-lamp base, Floor-lamp pole
	sphere	2	Table-lamp base, Floor-lamp bulb
	truncated cone	2	Table-lamp shade, Floor-lamp shade

Table 41: Table + floor lamp: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Side-table top	cylinder	1	(.5, .04, .5)	(-.55, .55, 0)	–	
Side-table stem	cylinder	1	(.05, .52, .05)	(-.55, .28, 0)	–	
Side-table foot	cylinder	1	(.34, .03, .34)	(-.55, .015, 0)	–	
Table-lamp base	sphere	1	(.16, .18, .16)	(-.55, .66, 0)	–	
Table-lamp stem	cylinder	1	(.02, .2, .02)	(-.55, .82, 0)	–	
Table-lamp shade	truncated cone	1	(.34, .24, .34)	(-.55, .99, 0)	–	glow
Floor-lamp base	cylinder	1	(.32, .03, .32)	(.55, .015, 0)	–	
Floor-lamp pole	cylinder	1	(.03, 1.6, .03)	(.55, .8, 0)	–	
Floor-lamp shade	truncated cone	1	(.44, .3, .44)	(.55, 1.62, 0)	–	glow
Floor-lamp bulb	sphere	1	(.08, .08, .08)	(.55, 1.55, 0)	–	glow

Construction. The table lamp is a ceramic base (sphere), a stem and a fabric shade (truncated cone); the side table is a round top, a stem and a foot disc. The floor lamp is a base disc, a 1.6 m pole, a shade and a bulb.

Function. The table lamps belong to the lounge light group; the floor lamp is the reading lamp of the library and belongs to the spot-light group. When on, the shades glow warm white.

Colours, textures and sound. Fabric shades, dark metal, polished metal stem. No sound.

8.2.9 Potted plants

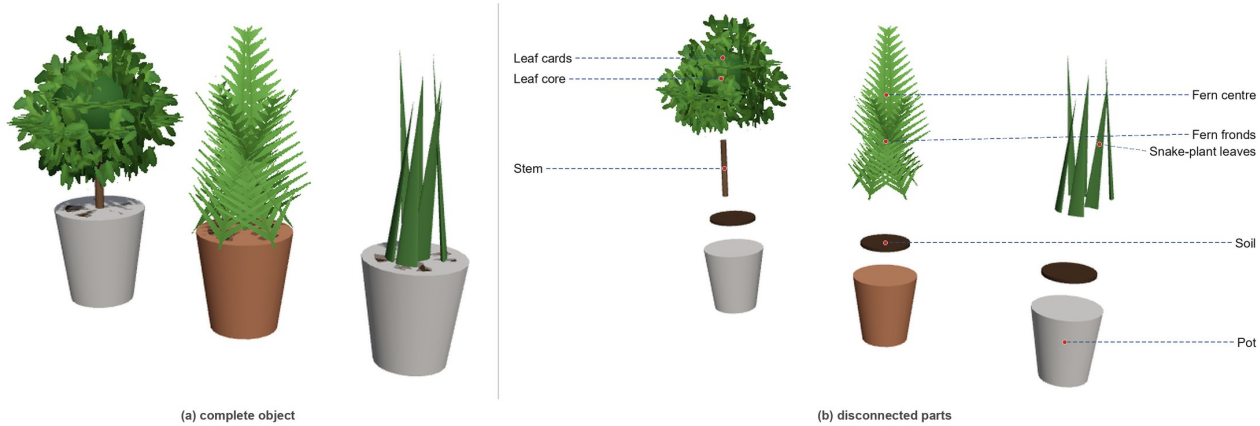


Figure 29: The three potted plants: (a) complete, (b) disconnected parts.

Table 42: Potted plants (3): the basic shapes it uses (27 shapes in total)

Shape	Name	No.	Used for
	cone	7	Snake-plant leaves
	quad	6	Fern fronds
	leaf card	5	Leaf cards
	cylinder	4	Soil, Stem
	truncated cone	3	Pot
	sphere (low)	1	Leaf core
	cross card	1	Fern centre

Table 43: Potted plants (3): every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Pot	truncated cone	3	(.42, .44, .42)	(-.75/0/.75, .22, 0)	(180, 0, 0)	
Soil	cylinder	3	(.32, .02, .32)	(-.75/0/.75, .43, 0)	–	
Stem	cylinder	1	(.04, .4, .04)	(-.75, .6, 0)	–	
Leaf cards	leaf card	5	(.6, .6, .6)	(-.86. . .-.64, .85/.97, -.097/.037/.12)	(0, -144. . .144, 0)	
Leaf core	sphere (low)	1	(.42, .42, .42)	(-.75, .9, 0)	–	
Fern fronds	quad	6	(.5, .8, 1)	(0, .75, 0)	(35, -180. . .120, 0)	
Fern centre	cross card	1	(.7, .9, .7)	(0, .44, 0)	–	
Snake-plant leaves	cone	7	(.09, .65/.73/.81, .025)	(.68. . ..82, .75/.8/.85, -.064. . ..07)	($\pm$6, -156. . .153, 0)	

Construction. Every pot is a truncated cone turned upside down ($R_x(180°)$) with a soil disc. Variant 0 has a stem, a dark core sphere and five leaf cards turned by $R_y(72° \, i)$; variant 1 has six fern fronds (quads tilted $R_x(35°)$ and turned $R_y(60° \, i)$) and a cross card; variant 2 (snake plant) has seven thin cones turned by $R_y(51° \, i)$ and tilted $\pm 6°$.

Colours, textures and sound. White or terracotta pots, leaf, fern and plain green materials. No sound.

8.2.10 Counter and wall cabinet

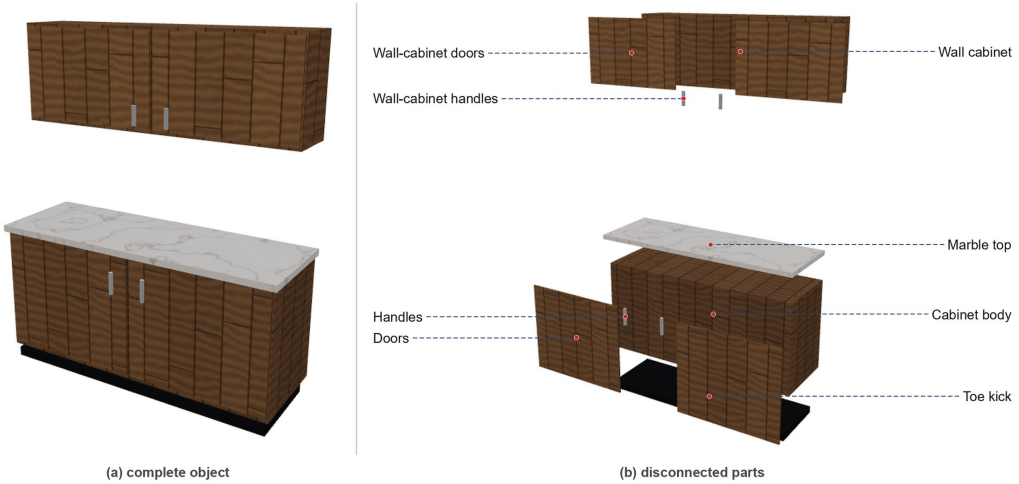


Figure 30: Kitchen counter with a wall cabinet: (a) complete, (b) disconnected parts.

Table 44: Counter + wall cabinet: the basic shapes it uses (12 shapes in total)

Shape	Name	No.	Used for
	cube	12	Cabinet body, Toe kick, Marble top, Doors, Handles, Wall cabinet, Wall-cabinet doors, Wall-cabinet handles

Table 45: Counter + wall cabinet: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Cabinet body	cube	1	(1.8, .82, .6)	(0, .5, 0)	–	
Toe kick	cube	1	(1.76, .06, .54)	(0, .03, .02)	–	
Marble top	cube	1	(1.84, .04, .63)	(0, .9, .015)	–	
Doors	cube	2	(.88, .76, .012)	($\pm$.45, .5, .31)	–	
Handles	cube	2	(.02, .14, .02)	($\pm$.11, .7, .32)	–	
Wall cabinet	cube	1	(1.8, .6, .34)	(0, 1.85, -.13)	–	
Wall-cabinet doors	cube	2	(.88, .56, .012)	($\pm$.45, 1.85, .045)	–	
Wall-cabinet handles	cube	2	(.02, .12, .02)	($\pm$.11, 1.65, .06)	–	

Construction. The counter is a cabinet body, a black toe kick, a marble top and $\max(2, \lfloor w/0.6 \rfloor)$ doors with handles; the handles alternate between the left and right edge of the doors. The wall cabinet uses the same rule. The coffee machine stands on this counter.

Colours, textures and sound. Wood texture, marble texture, polished metal handles. No sound.

8.2.11 Bench

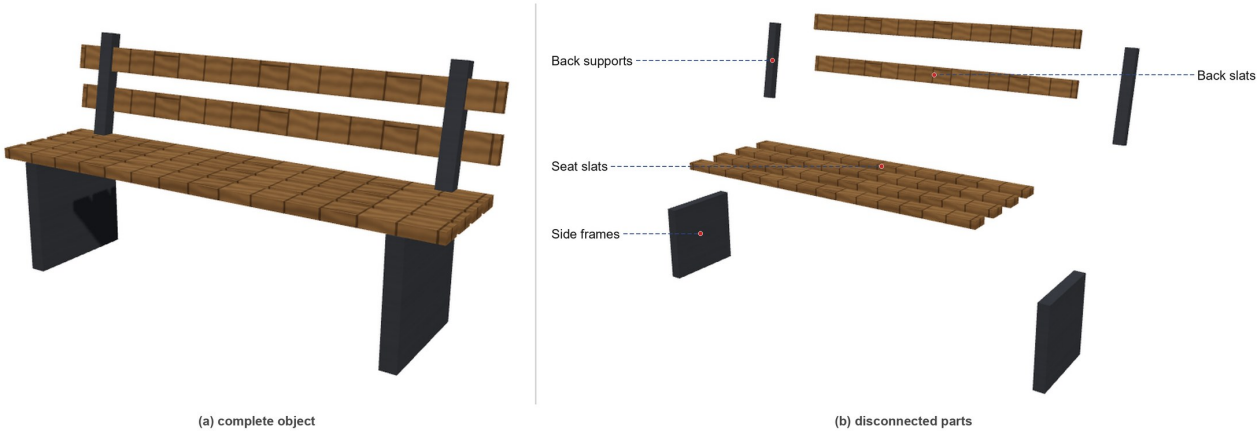


Figure 31: Outdoor bench: (a) complete, (b) disconnected parts.

Table 46: Bench: the basic shapes it uses (10 shapes in total)

Shape	Name	No.	Used for
	cube	10	Side frames, Back supports, Seat slats, Back slats

Table 47: Bench: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Side frames	cube	2	(.06, .44, .45)	($\pm$.75, .22, 0)	–	
Back supports	cube	2	(.06, .5, .05)	($\pm$.75, .65, -.22)	(-10, 0, 0)	
Seat slats	cube	4	(1.8, .04, .09)	(0, .46, -.17. . .17)	–	
Back slats	cube	2	(1.8, .1, .035)	(0, .62/.79, -.25)	(-10, 0, 0)	

Construction. Two metal side frames, two back supports tilted by $R_x(-10°)$, four seat slats and two back slats with the same tilt. Two benches stand on the sidewalk.

Colours, textures and sound. Wood texture slats, dark metal frame. No sound.

8.2.12 Waste bin

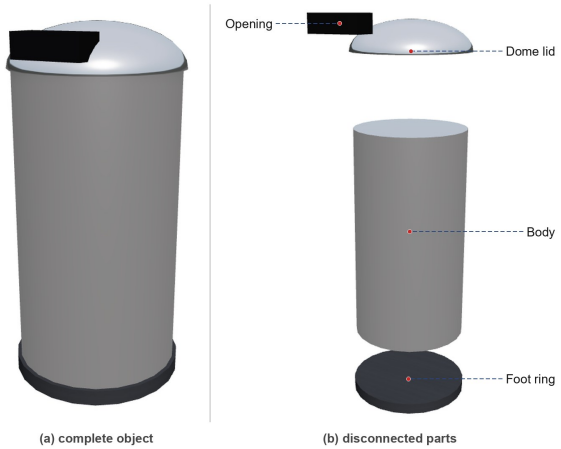


Figure 32: Waste bin: (a) complete, (b) disconnected parts.

Table 48: Waste bin: the basic shapes it uses (4 shapes in total)

Shape	Name	No.	Used for
	cylinder	2	Body, Foot ring
	hemisphere	1	Dome lid
	cube	1	Opening

Table 49: Waste bin: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Body	cylinder	1	(.42, .8, .42)	(0, .4, 0)	–	
Dome lid	hemisphere	1	(.44, .22, .44)	(0, .8, 0)	–	
Opening	cube	1	(.18, .06, .08)	(0, .86, .18)	–	
Foot ring	cylinder	1	(.44, .04, .44)	(0, .02, 0)	–	

Construction. A cylinder body, a hemisphere lid, a box for the opening and a foot ring. Bins stand outside next to the benches and inside the building.

Colours, textures and sound. Polished stainless steel, which shows a bright specular highlight (Section 9). No sound.

8.2.13 Light switch

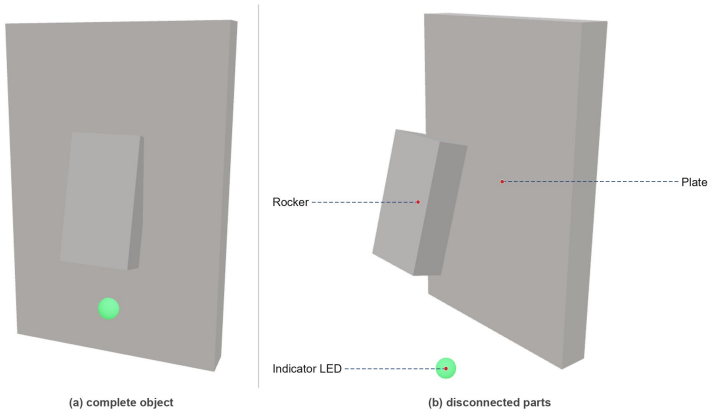


Figure 33: Light switch: (a) complete, (b) disconnected parts.

Table 50: Light switch: the basic shapes it uses (3 shapes in total)

Shape	Name	No.	Used for
	cube	2	Plate, Rocker
	sphere	1	Indicator LED

Table 51: Light switch: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Plate	cube	1	(.08, .12, .012)	(0, 0, .006)	–	
Rocker	cube	1	(.025, .045, .012)	(0, 0, .016)	(-15, 0, 0)	
Indicator LED	sphere	1	(.008, .008, .008)	(0, -.042, .013)	–	glow

Construction. A plate, a rocker tilted by $R_x(\mp 15°)$ depending on the state, and an indicator LED sphere.

Function. Six switches on the walls control the light groups (lounge, spot lights, ATM room, library, outdoor). Looking at a switch highlights it; pressing E toggles its group and the rocker flips (Section 15).

Colours, textures and sound. White plastic, green LED when on and red when off. A click is heard.

8.2.14 Security camera

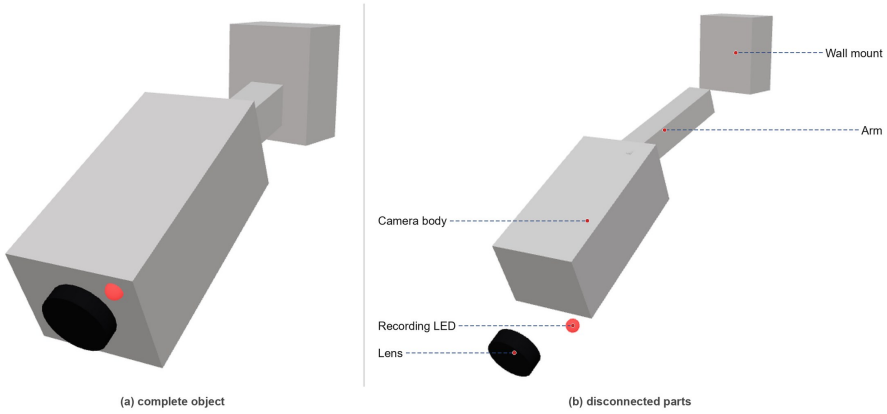


Figure 34: Security camera: (a) complete, (b) disconnected parts.

Table 52: Security camera: the basic shapes it uses (5 shapes in total)

Shape	Name	No.	Used for
	cube	3	Wall mount, Arm, Camera body
	cylinder	1	Lens
	sphere	1	Recording LED

Table 53: Security camera: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Wall mount	cube	1	(.1, .12, .06)	(0, 0, .03)	–	
Arm	cube	1	(.04, .04, .2)	(0, -.05, .14)	(20, 0, 0)	
Camera body	cube	1	(.12, .1, .24)	(0, -.1, .24)	(25, 12, 0)	

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Lens	cylinder	1	(.06, .02, .06)	(.023, -.15, .35)	(65, -168, 180)	
Recording LED	sphere	1	(.015, .015, .015)	(.064, -.12, .35)	(25, 12, 0)	glow

Construction. A wall mount, an arm tilted by $R_x(20°)$ and a camera node $T(0, -0.10, 0.24) \cdot R_x(25°)\, R_y(30° \sin 0.4t)$ that holds the body, the lens and a recording LED.

Function. The camera slowly pans left and right; its LED blinks once per second.

Colours, textures and sound. White plastic, black lens, red glowing LED. No sound.

8.2.15 Paintings

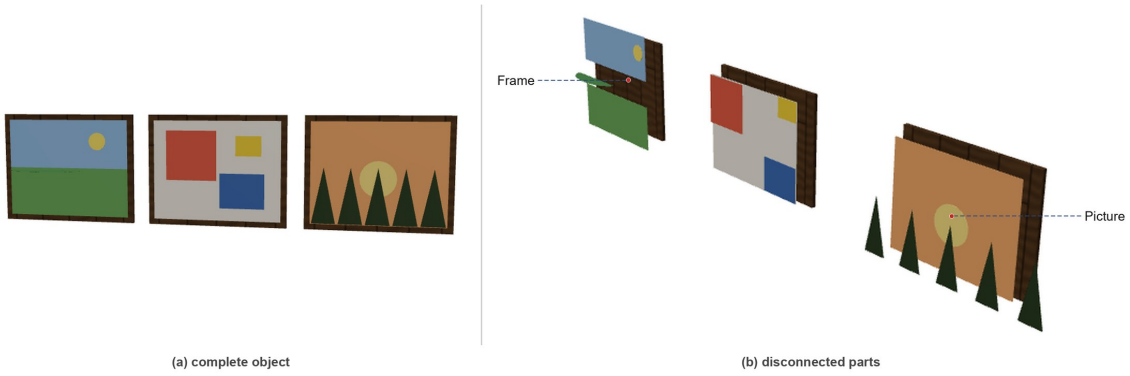


Figure 35: The three wall paintings: (a) complete, (b) disconnected parts.

Table 54: Paintings (3): the basic shapes it uses (18 shapes in total)

Shape	Name	No.	Used for
	cube	10	Frame, Picture
	cone	5	Picture
	cylinder	2	Picture
	hemisphere	1	Picture

Table 55: Paintings (3): every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Frame	cube	3	(.98, .78, .03)	(-1.1/0/1.1, 0, .015)	–	
Picture	cube	7	(.18. . .9, .14. . .7, .003/.004)	(-1.1. . .1.1, -.17. . .17, .032/.035/.036)	–	+3
Picture	hemisphere	1	(.54, .24, .003)	(-1.28, -.02, .034)	(90, 0, 0)	
Picture	cylinder	2	(.13/.24, .003, .13/.24)	(-.88/1.1, -.035/.2, .034/.035)	(90, 0, 0)	
Picture	cone	5	(.16, .39, .003)	(.74. . .1.46, -.15, .036)	–	

Construction. A wooden frame box and a picture made of flat shapes: a landscape (sky, hills as

a hemisphere rotated $R_x(90°)$, a sun disc), abstract blocks, and a jungle sunset with five flattened cones as trees.

Colours, textures and sound. Matte colours, wood frame. No sound.

8.2.16 Sliding entrance doors

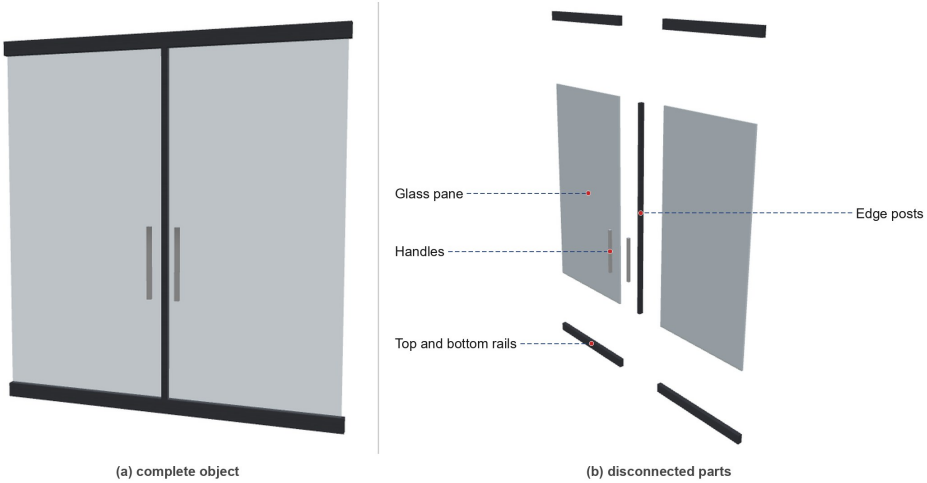


Figure 36: Automatic sliding doors (closed): (a) complete, (b) disconnected parts.

Table 56: Sliding doors: the basic shapes it uses (10 shapes in total)

Shape	Name	No.	Used for
	cube	10	Glass pane, Top and bottom rails, Edge posts, Handles

Table 57: Sliding doors: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Glass pane	cube	2	(1.2, 2.44, .02)	($\pm$.6, 1.3, 0)	–	glass
Top and bottom rails	cube	4	(1.24, .1, .05)	($\pm$.6, .05/2.55, 0)	–	
Edge posts	cube	2	(.04, 2.4, .05)	(0, 1.3, 0)	–	
Handles	cube	2	(.03, .5, .03)	($\pm$.1, 1.05, .045)	–	

Table 58: Sliding doors in the world, placed with $M_{place} = T(0, 0, 5.18)$: centre of each key part and the world position $M_{place}\,M_{part}\,\mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Glass pane (1)	cube	$(-0.60, 1.30, 5.18)$	$(-1.20, 0.08, 5.17)$	$(0.00, 2.52, 5.19)$
Glass pane (2)	cube	$(0.60, 1.30, 5.18)$	$(0.00, 0.08, 5.17)$	$(1.20, 2.52, 5.19)$
Top and bottom rails (1)	cube	$(-0.60, 2.55, 5.18)$	$(-1.22, 2.50, 5.16)$	$(0.02, 2.60, 5.21)$
Top and bottom rails (2)	cube	$(-0.60, 0.05, 5.18)$	$(-1.22, 0.00, 5.16)$	$(0.02, 0.10, 5.21)$
Edge posts (1)	cube	$(0.00, 1.30, 5.18)$	$(-0.02, 0.10, 5.16)$	$(0.02, 2.50, 5.21)$
Edge posts (2)	cube	$(0.00, 1.30, 5.18)$	$(-0.02, 0.10, 5.16)$	$(0.02, 2.50, 5.21)$
Handles (1)	cube	$(-0.10, 1.05, 5.22)$	$(-0.12, 0.80, 5.21)$	$(-0.09, 1.30, 5.24)$
Handles (2)	cube	$(0.10, 1.05, 5.22)$	$(0.09, 0.80, 5.21)$	$(0.12, 1.30, 5.24)$

Construction. Each leaf is a glass pane, a top and a bottom rail, an edge post and a handle. The left leaf is drawn in the node $T(-x_{off}, 0, z)$ and the right leaf in $T(+x_{off}, 0, z)$, with

$$x_{off} = 0.6 + 1.15 \cdot o,$$

where $o \in [0, 1]$ is the opening.

Function. The doors open while a person (the player or a visitor) is within 2.2×2.8 m of the entrance, o moving towards its target at 1.6 per second. While they are closed they block the way.

Colours, textures and sound. Transparent glass, dark metal rails, polished metal handles. A short whoosh is heard when the doors start to move.

8.2.17 Ceiling and roof fixtures

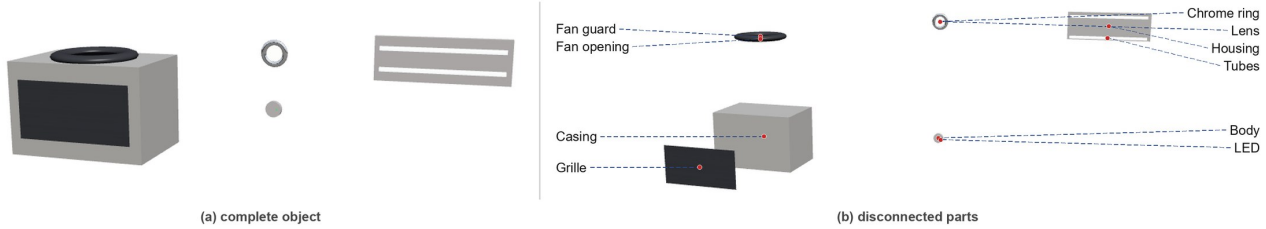


Figure 37: Roof air conditioner, recessed spot light, tube light and smoke detector (the ceiling fixtures are turned to face the viewer): (a) complete, (b) disconnected parts.

Table 59: Ceiling + roof fixtures: the basic shapes it uses (11 shapes in total)

Shape	Name	No.	Used for
	cylinder	5	Fan opening, Lens, Tubes, Body
	cube	3	Casing, Grille, Housing
	torus	2	Fan guard, Chrome ring
	sphere	1	LED

Table 60: Ceiling + roof fixtures: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Casing	cube	1	(1.2, .8, .8)	(-1.6, .4, 0)	–	
Fan opening	cylinder	1	(.6, .02, .6)	(-1.6, .81, 0)	–	
Fan guard	torus	1	(.75, .4, .75)	(-1.6, .82, 0)	–	
Grille	cube	1	(1, .5, .01)	(-1.6, .4, .41)	–	
Chrome ring	torus	1	(.22, .12, .22)	(0, .9, .01)	(-90, 0, 0)	
Lens	cylinder	1	(.13, .01, .13)	(0, .9, .012)	(-90, 0, 0)	glow
Housing	cube	1	(1, .06, .34)	(1.3, .9, .03)	(-90, 0, 0)	
Tubes	cylinder	2	(.035, .92, .035)	(1.3, .82/.98, .07)	(-90, 180, -90)	glow
Body	cylinder	1	(.13, .04, .13)	(0, .45, .02)	(-90, 0, 0)	
LED	sphere	1	(.012, .012, .012)	(.03, .45, .04)	(-90, 0, 0)	glow

Construction. The air conditioner is a casing box, a fan opening (disc), a torus fan guard and a grille. The spot light is a torus ring around a glowing lens; the tube light is a housing box with two glowing tubes (cylinders rotated $R_z(90°)$); the smoke detector is a disc with a small LED.

Function. Spot and tube lights glow when their light group is on and carry the spot and point lights of the rooms.

Colours, textures and sound. Light grey plastic, polished metal rings, glowing lenses. No sound.

8.2.18 Street lamp and bollard

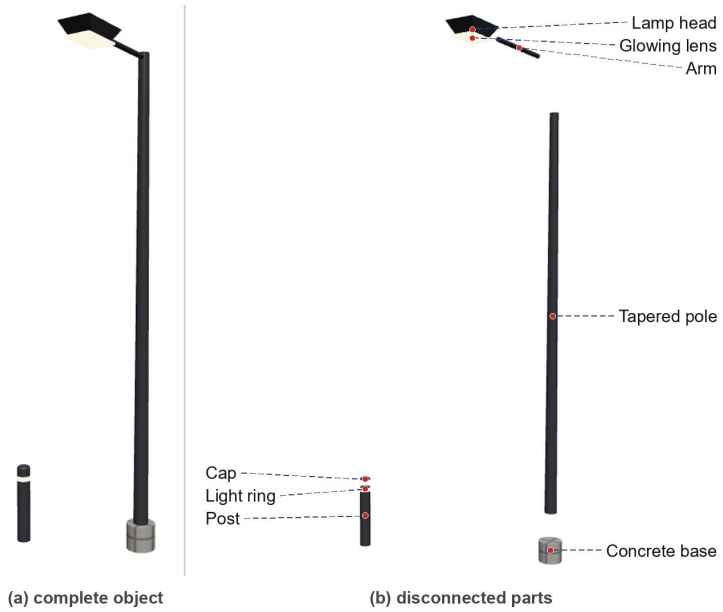


Figure 38: Street lamp and a sidewalk bollard: (a) complete, (b) disconnected parts.

Table 61: Street lamp + bollard: the basic shapes it uses (8 shapes in total)

Shape	Name	No.	Used for
	cylinder	4	Concrete base, Arm, Post, Light ring
	truncated cone	1	Tapered pole
	tapered box	1	Lamp head
	cube	1	Glowing lens
	hemisphere	1	Cap

Table 62: Street lamp + bollard: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Concrete base	cylinder	1	(.34, .3, .34)	(0, .15, 0)	–	
Tapered pole	truncated cone	1	(.18, 5.8, .18)	(0, 2.9, 0)	–	
Arm	cylinder	1	(.06, 1.1, .06)	(0, 5.75, .55)	(90, 0, 0)	
Lamp head	tapered box	1	(.38, .14, .62)	(0, 5.74, 1.15)	(180, 0, 0)	
Glowing lens	cube	1	(.3, .02, .5)	(0, 5.66, 1.15)	–	glow
Post	cylinder	1	(.16, .9, .16)	(-1.3, .45, .6)	–	

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Light ring	cylinder	1	(.17, .06, .17)	(-1.3, .78, .6)	–	glow
Cap	hemisphere	1	(.16, .08, .16)	(-1.3, .9, .6)	–	

Table 63: Street lamp + bollard in the world, placed with $M_{place} = T(-18, 0, 8.25)$: centre of each key part and the world position $M_{place}\, M_{part}\, \mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Concrete base	cylinder	$(-18.00, 0.15, 8.25)$	$(-17.83, 0.00, 8.25)$	$(-17.83, 0.30, 8.25)$
Tapered pole	truncated cone	$(-18.00, 2.90, 8.25)$	$(-17.91, 0.00, 8.25)$	$(-17.94, 5.80, 8.25)$
Arm	cylinder	$(-18.00, 5.75, 8.80)$	$(-17.97, 5.75, 8.25)$	$(-17.97, 5.75, 9.35)$
Lamp head	tapered box	$(-18.00, 5.74, 9.40)$	$(-18.19, 5.81, 9.71)$	$(-17.87, 5.67, 9.21)$
Glowing lens	cube	$(-18.00, 5.66, 9.40)$	$(-18.15, 5.65, 9.15)$	$(-17.85, 5.67, 9.65)$

Construction. The street lamp is a concrete base, a tapered pole (truncated cone) of height $h = 5.8$ m, an arm (cylinder rotated $R_x(90°)$), a lamp head (tapered box rotated $R_x(180°)$ so that it is wider at the top) and a glowing lens. The bollard is a post, a light ring and a hemisphere cap.

Function. The outdoor lamps switch on automatically at dusk. Each street lamp carries a point light that stays on at any distance.

Colours, textures and sound. Concrete texture, dark metal, warm sodium-coloured glow. No sound.

8.2.19 The building

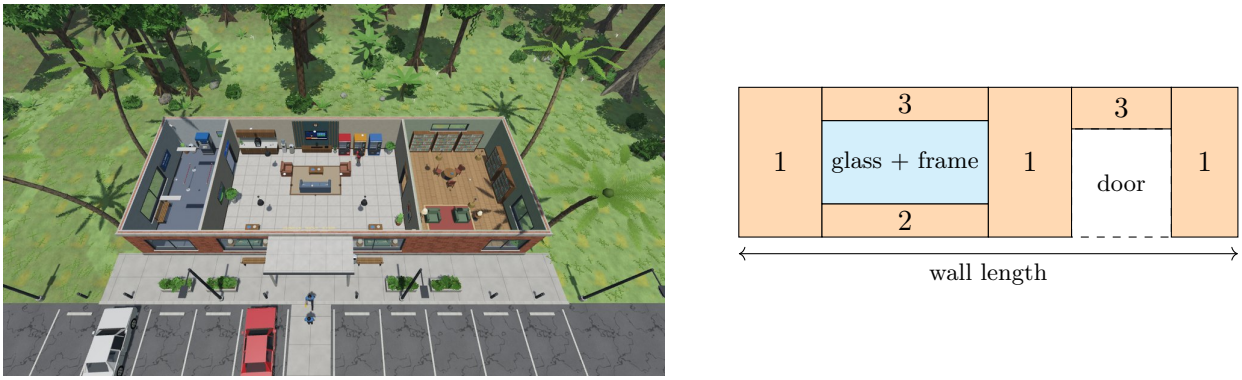


Figure 39: Left: the building with the roof hidden. Right: a wall with one window and one door is split into (1) full pieces, (2) a piece below the window and (3) pieces above the openings.

Construction. All walls are made by one function, `buildWall()`, which takes a straight wall and a list of openings and splits the wall into boxes: full-height pieces between openings, a piece below each window, a lintel above each opening, and for a window a glass pane, an aluminium frame (four bars and a mullion) and a marble sill; for a door, frame trims. Every piece has two half-thickness layers: brick outside and plaster inside, in the colour of the room (beige lounge, light blue ATM room, sage green library). Wooden skirting boards run along the floor. The floors are tiles (lounge), carpet (ATM room) and wood (library). The roof, the parapet, two air conditioners and the sign board are drawn per frame, so that the roof can be hidden in the cut-away view.

Colours, textures and sound. Brick, plaster, floor tile, carpet, wood, ceiling tile, roof and concrete textures.

8.3 People

There are two types of person, a man and a woman, built by the same function with the same hierarchy (Figure 40); only a few sizes, the hair style and the colours differ. The player's own body is the man.

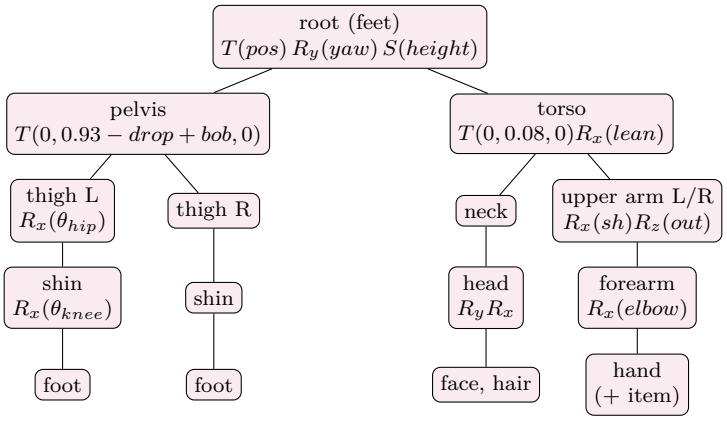


Figure 40: Scene graph of a person. A pose only changes the joint angles; every body part stays attached to its parent.

8.3.1 Man

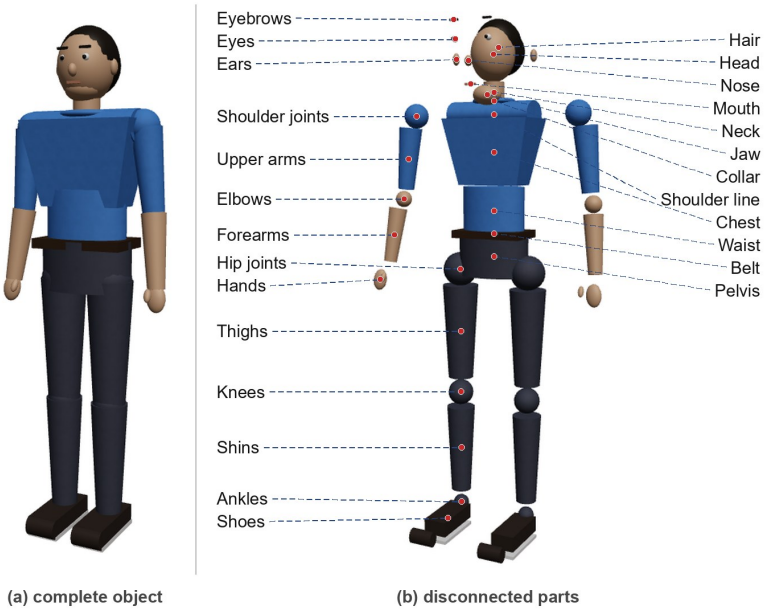


Figure 41: The man (standing): (a) complete, (b) disconnected parts with their names.

Table 64: Man: the basic shapes it uses (49 shapes in total)

Shape	Name	No.	Used for
	sphere	24	Hip joints, Knees, Ankles, Shoulder joints, Jaw, Head, Nose, Eyes, Ears, Hair, Elbows, Hands
	cube	8	Belt, Shoes, Eyebrows, Mouth
	truncated cone	8	Thighs, Shins, Upper arms, Forearms
	cylinder	6	Pelvis, Shoes, Waist, Shoulder line, Neck
	tapered box	1	Chest
	torus	1	Collar
	hemisphere	1	Hair

Table 65: Man: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Pelvis	cylinder	1	(.34, .18, .21)	(0, .95, 0)	–	
Belt	cube	1	(.34, .035, .2)	(0, 1.03, 0)	–	
Hip joints	sphere	2	(.16, .16, .17)	(±.092, .9, 0)	–	
Thighs	truncated cone	2	(.17, .45, .17)	(±.092, .68, 0)	(180, 0, 0)	
Knees	sphere	2	(.12, .12, .12)	(±.092, .45, 0)	–	
Shins	truncated cone	2	(.14, .42, .14)	(±.092, .24, 0)	(180, 0, 0)	
Ankles	sphere	2	(.075, .075, .075)	(±.092, .03, 0)	–	
Shoes	cube	4	(.11/.12, .015/.07, .26/.27)	(±.092, -.028/.005, .045)	–	
Shoes	cylinder	2	(.07, .11, .06)	(±.092, .005, .17)	(0, 0, 90)	
Waist	cylinder	1	(.3, .22, .19)	(0, 1.11, 0)	–	
Chest	tapered box	1	(.42, .3, .23)	(0, 1.31, 0)	(0, 0, -180)	
Shoulder line	cylinder	1	(.12, .4, .17)	(0, 1.44, 0)	(0, 0, 90)	
Shoulder joints	sphere	2	(.12, .12, .12)	(±.21, 1.44, 0)	–	
Collar	torus	1	(.2, .12, .17)	(0, 1.48, 0)	–	
Neck	cylinder	1	(.095, .09, .095)	(0, 1.51, 0)	–	
Jaw	sphere	1	(.13, .09, .12)	(0, 1.56, .03)	–	
Head	sphere	1	(.2, .24, .21)	(0, 1.64, .005)	–	
Nose	sphere	1	(.032, .045, .04)	(0, 1.62, .1)	–	
Eyes	sphere	4	(.014/.026, .014/.026, .014/.026)	(±.042, 1.66, .092/.1)	–	
Eyebrows	cube	2	(.04, .008, .01)	(±.042, 1.69, .098)	–	
Ears	sphere	2	(.03, .055, .035)	(±.1, 1.64, 0)	–	
Mouth	cube	1	(.05, .009, .01)	(0, 1.58, .098)	–	
Hair	hemisphere	1	(.21, .25, .23)	(0, 1.66, -.018)	–	
Hair	sphere	1	(.21, .21, .17)	(0, 1.65, -.045)	–	
Upper arms	truncated cone	2	(.11, .29, .11)	(±.24, 1.29, 0)	(0, ±180, ±174)	
Elbows	sphere	2	(.078, .078, .078)	(±.26, 1.14, 0)	(-8, ±1, ±6)	
Forearms	truncated cone	2	(.088, .26, .088)	(±.27, 1.01, .018)	(8, ±179, ±174)	
Hands	sphere	4	(.025/.068, .05/.1, .025/.045)	(-.28. . .28, .86/.87, .045/.067)	(-28/-8, -3. . .3, -7. . .7)	

Table 66: Man in the world, placed with $M_{place} = T(0,0,6)\,R_y(180°)$: centre of each key part and the world position $M_{place}\,M_{part}\,\mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x,y,z)	Vertex A	Vertex B
Pelvis	cylinder	$(0.00, 0.95, 6.00)$	$(-0.17, 0.86, 6.00)$	$(-0.17, 1.04, 6.00)$
Chest	tapered box	$(0.00, 1.31, 6.00)$	$(-0.21, 1.46, 6.12)$	$(0.15, 1.16, 5.93)$
Head	sphere	$(0.00, 1.64, 6.00)$	$(0.00, 1.52, 6.00)$	$(0.00, 1.76, 6.00)$
Thighs (1)	truncated cone	$(0.09, 0.68, 6.00)$	$(0.01, 0.90, 6.00)$	$(0.03, 0.45, 6.00)$
Thighs (2)	truncated cone	$(-0.09, 0.68, 6.00)$	$(-0.18, 0.90, 6.00)$	$(-0.15, 0.45, 6.00)$
Shins (1)	truncated cone	$(0.09, 0.24, 6.00)$	$(0.02, 0.45, 6.00)$	$(0.04, 0.03, 6.00)$
Shins (2)	truncated cone	$(-0.09, 0.24, 6.00)$	$(-0.16, 0.45, 6.00)$	$(-0.14, 0.03, 6.00)$
Upper arms (1)	truncated cone	$(0.24, 1.29, 6.00)$	$(0.17, 1.42, 6.00)$	$(0.22, 1.14, 6.00)$
Upper arms (2)	truncated cone	$(-0.24, 1.29, 6.00)$	$(-0.28, 1.44, 6.00)$	$(-0.29, 1.15, 6.00)$
Shoes (1)	cube	$(0.09, 0.01, 5.96)$	$(0.15, -0.03, 6.08)$	$(0.04, 0.04, 5.83)$
Shoes (2)	cylinder	$(0.09, 0.01, 5.83)$	$(0.04, 0.04, 5.83)$	$(0.15, 0.04, 5.83)$

Construction. Every joint is a sphere and every limb a truncated cone. The cones are flipped with $R_x(180°)$, so the legs are thicker at the hip and the arms thicker at the shoulder. The chest is a tapered box rotated by $R_z(180°)$ (wide at the shoulders), the waist an elliptic cylinder. The face has sphere eyes, pupils, nose, ears and jaw and box eyebrows and mouth; the short hair is a hemisphere and a sphere.

Worked example: the knee. The knee of the left leg is found by following the chain of nodes:

$$\mathbf{p}_{knee} = \underbrace{T(\mathbf{p})R_y(\psi)}_{\text{person}} \cdot \underbrace{T(0,0.93,0)}_{\text{pelvis}} \cdot \underbrace{T(0.092,-0.03,0)\,R_x(\theta_{thigh})}_{\text{hip}} \cdot \underbrace{T(0,-0.45,0)}_{\text{knee}} \cdot (0,0,0,1)^T,$$

with $R_x(\theta) : (x,y,z) \mapsto (x,\ y\cos\theta - z\sin\theta,\ y\sin\theta + z\cos\theta)$. In person space:

- standing, $\theta_{thigh} = 0$: knee at $(0.092,\ 0.90 - 0.45,\ 0) = (0.092,\ 0.45,\ 0)$;
- walking, leg forward, $\theta_{thigh} = -26°$: $(0.092,\ 0.90 - 0.45\cos 26°,\ 0.45\sin 26°) = (0.092,\ 0.496,\ 0.197)$.

The knee moves 20 cm forward and 4.6 cm up, and the shin and the foot follow because they are its children.

Function (animation). Only the joint angles change. For walking with the phase φ ($\varphi \mathrel{+}= 7\,\Delta t$):

$$\theta^{L,R}_{thigh} = \mp 26° \sin\varphi, \quad \theta^{L}_{knee} = 38° \max(0, \cos\varphi) + 4°, \quad \theta^{L,R}_{shoulder} = \pm 22° \sin\varphi, \quad bob = 0.035\,(1 - |\sin\varphi|).$$

Sitting lowers the pelvis by 0.44 m, turns the thighs by $-88°$ and bends the knees by $88°$. Driving, pressing buttons and bending down are further fixed poses. Every angle is blended linearly over 0.35 s, so changes between poses are smooth.

Colours, textures and sound. Skin colour, blue fabric shirt, dark fabric trousers, dark shoes with white soles, black hair. Footsteps are silent.

8.3.2 Woman

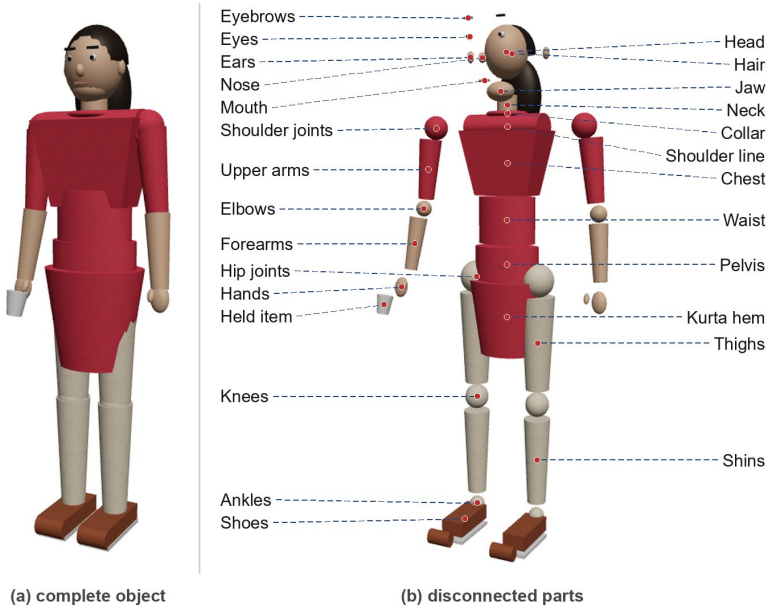


Figure 42: The woman holding a coffee cup: (a) complete, (b) disconnected parts with their names.

Table 67: Woman: the basic shapes it uses (50 shapes in total)

Shape	Name	No.	Used for
	sphere	24	Hip joints, Knees, Ankles, Shoulder joints, Jaw, Head, Nose, Eyes, Ears, Hair, Elbows, Hands
	truncated cone	10	Thighs, Shins, Kurta hem, Upper arms, Forearms, Held item
	cube	7	Shoes, Eyebrows, Mouth
	cylinder	6	Pelvis, Shoes, Waist, Shoulder line, Neck
	tapered box	1	Chest
	torus	1	Collar
	hemisphere	1	Hair

Table 68: Woman: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Pelvis	cylinder	1	(.3, .17, .2)	(0, .9, 0)	–	
Hip joints	sphere	2	(.14, .15, .16)	($\pm$.082, .85, 0)	–	
Thighs	truncated cone	2	(.15, .43, .16)	($\pm$.082, .64, 0)	(180, 0, 0)	
Knees	sphere	2	(.11, .11, .12)	($\pm$.082, .43, 0)	–	
Shins	truncated cone	2	(.12, .4, .13)	($\pm$.082, .23, 0)	(180, 0, 0)	
Ankles	sphere	2	(.071, .071, .071)	($\pm$.082, .029, 0)	–	
Shoes	cube	4	(.1/.11, .014/.067, .25/.25)	($\pm$.082, -.027/.005, .043)	–	

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Shoes	cylinder	2	(.067, .1, .057)	(±.082, .005, .16)	(0, 0, 90)	
Kurta hem	truncated cone	1	(.36, .34, .28)	(0, .72, 0)	(180, 0, 0)	
Waist	cylinder	1	(.27, .21, .18)	(0, 1.05, 0)	–	
Chest	tapered box	1	(.38, .28, .22)	(0, 1.24, 0)	(0, 0, -180)	
Shoulder line	cylinder	1	(.11, .36, .16)	(0, 1.36, 0)	(0, 0, 90)	
Shoulder joints	sphere	2	(.11, .11, .11)	(±.19, 1.36, 0)	–	
Collar	torus	1	(.19, .11, .16)	(0, 1.41, 0)	–	
Neck	cylinder	1	(.09, .086, .09)	(0, 1.44, 0)	–	
Jaw	sphere	1	(.12, .086, .11)	(0, 1.49, .029)	–	
Head	sphere	1	(.19, .23, .2)	(0, 1.56, .005)	–	
Nose	sphere	1	(.03, .043, .038)	(0, 1.54, .1)	–	
Eyes	sphere	4	(.013/.025, .013/.025, .013/.025)	(±.04, 1.58, .087/.098)	–	
Eyebrows	cube	2	(.038, .008, .009)	(±.04, 1.61, .093)	–	
Ears	sphere	2	(.029, .052, .033)	(±.097, 1.55, 0)	–	
Mouth	cube	1	(.048, .009, .009)	(0, 1.5, .093)	–	
Hair	hemisphere	1	(.21, .25, .22)	(0, 1.57, -.019)	–	
Hair	sphere	1	(.2, .34, .12)	(0, 1.48, -.067)	–	
Upper arms	truncated cone	2	(.1, .28, .1)	(±.21, 1.22, 0)	(0, ±180, ±174)	
Elbows	sphere	2	(.074, .074, .074)	(±.23, 1.08, 0)	(-8, ±1, ±6)	
Forearms	truncated cone	2	(.084, .25, .084)	(±.24, .96, .017)	(8, ±179, ±174)	
Hands	sphere	4	(.024/.065, .048/.095, .024/.043)	(-.26. . .26, .81/.82, .043/.063)	(-28/-8, -3. . .3, -7. . .7)	
Held item	truncated cone	1	(.076, .086, .076)	(-.26, .76, .084)	(8, -179, -174)	

Construction. The same hierarchy with a root scale $S(0.95)$ and a width factor $w = 0.94$ (her hip is at $x = 0.092 \cdot 0.94 = 0.086$). The long hair adds a tall flattened sphere behind the head. She wears a kurta: the pelvis uses the shirt colour and a hem (an upside-down truncated cone) covers the upper legs. A held item (cup, can, snack or phone) is a child of the right hand.



(a) Standing with a cup **(b)** Walking pose **(c)** Sitting on a chair

Figure 43: Poses produced by the same hierarchical model.

Colours, textures and sound. Red kurta, light trousers, brown shoes, long black hair.

8.4 Car

There is one car model, a sedan of $4.4 \times 1.78 \times 1.42$ m, painted red, blue or white.

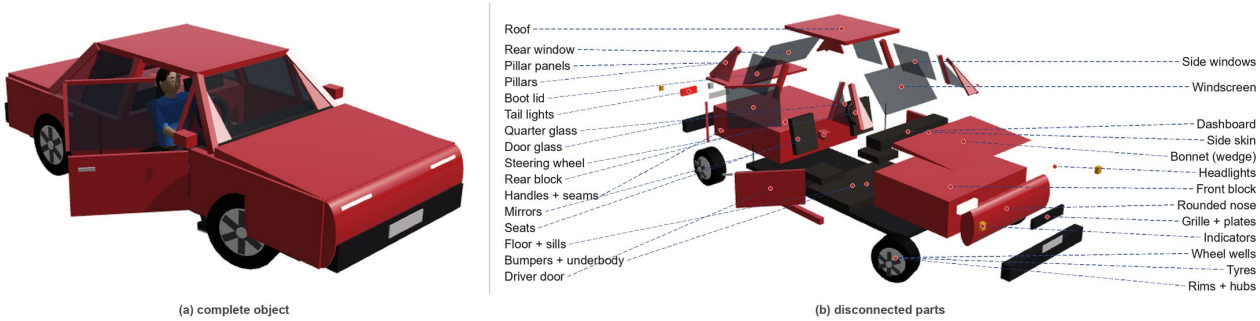


Figure 44: The car (driver door open 55°, front wheels steered 18°): (a) complete with its driver, (b) disconnected parts.

Table 69: Car: the basic shapes it uses (95 shapes in total)

Shape	Name	No.	Used for
	cube	68	Front block, Rear block, Side skin, Floor + sills, Bumpers + underbody, Boot lid, Windscreen, Rear window, Pillars, Side windows, Roof, Dashboard, Seats, Headlights, Indicators, Tail lights, Grille + plates, Mirrors, Handles + seams, Rims + hubs, Driver door, Door glass
	cylinder	17	Rounded nose, Wheel wells, Tyres, Rims + hubs
	wedge	9	Bonnet (wedge), Pillar panels, Quarter glass
	torus	1	Steering wheel

Table 70: Car: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; ± = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Front block	cube	1	(1.78, .56, 1.42)	(0, .58, 1.44)	–	
Rear block	cube	1	(1.78, .56, 1.88)	(0, .58, -1.21)	–	
Side skin	cube	1	(.06, .56, 1)	(.86, .58, .23)	–	
Floor + sills	cube	2	(.08/1.76, .1/.12, 1)	(-.85/0, .35/.36, .23)	–	
Rounded nose	cylinder	1	(.55, 1.74, .24)	(0, .58, 2.15)	(0, 0, 90)	
Bumpers + underbody	cube	3	(1.6/1.71, .08/.22, .14/3.78)	(0, .27/.42, -2.2/0/2.22)	–	
Bonnet (wedge)	wedge	1	(1.73, .08, 1.15)	(0, .9, 1.57)	–	
Boot lid	cube	1	(1.71, .04, .8)	(0, .88, -1.75)	–	
Windscreen	cube	1	(1.15, .69, .012)	(0, 1.12, .72)	(-41, 0, 0)	glass
Rear window	cube	1	(1.15, .69, .012)	(0, 1.12, -1.12)	(-41, 180, 0)	glass
Pillar panels	wedge	6	(.03, .57/.69, .25/.46)	(±.7, 1.12, -1.12/.72)	(0, -90/90/180, -41. . .41)	
Quarter glass	wedge	2	(.012, .57, .46)	(±.7, 1.12, .72)	(0, 0, ±25)	glass
Pillars	cube	8	(.04. . .09, .57/.74, .04. . .09)	(±.7, 1.12, -1.12. . .72)	(-39/0/39, -9/0/9, ±25)	
Side windows	cube	3	(.62/.76, .57, .012)	(±.7, 1.12, -.58/.11)	(±25, 90, 0)	glass
Roof	cube	1	(1.19, .06, 1.42)	(0, 1.39, -.2)	–	
Dashboard	cube	1	(1.51, .16, .3)	(0, .84, .83)	–	
Seats	cube	5	(.46/.48/1.42, .12/.14/.6, .1/.45/.5)	(-.38/0/.38, .38/.41/.71, -.9/-.37/-.1)	(-12/0, 0, 0)	
Steering wheel	torus	1	(.38, .38, .38)	(-.38, .98, .5)	(70, 0, 0)	
Headlights	cube	2	(.36, .12, .06)	(±.59, .74, 2.21)	–	glow
Indicators	cube	4	(.07/.08, .08/.1, .05/.06)	(±.78, .74/.78, -2.17/2.18)	–	glow
Tail lights	cube	4	(.08/.32, .1/.12, .06)	(-.64. . .64, .78, -2.18)	–	glow
Grille + plates	cube	3	(.46/.71, .11/.14, .01/.04)	(0, .44/.62/.66, -2.21/2.23/2.3)	–	
Mirrors	cube	2	(.12, .1, .16)	(±.97, .98, .77)	–	

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Handles + seams	cube	5	(.01/.02, .03/.53, .012/.14)	(-.9. . . .9, .58/.74, -1/-.5/0)	–	■■
Wheel wells	cylinder	4	(.66, .24, .66)	(±.78, .32, ±1.32)	(0, 0/18, 90)	■
Tyres	cylinder	4	(.64, .22, .64)	(±.78, .32, ±1.32)	(0, 0/18, 90)	■
Rims + hubs	cylinder	8	(.096/.42, .02/.23, .096/.42)	(-.91. . . .91, .32, -1.32. . . 1.36)	(0, 0/18, 90)	■
Rims + hubs	cube	12	(.02, .38, .06)	(-.9. . . .9, .32, -1.32/1.29/1.36)	(0/60/120, -162/0/18, 0/180)	■
Driver door	cube	6	(.02. . . .04, .03. . . .76, .035. . . 1)	(-1.62. . . -1.22, .53. . . 1.38, -.015. . . .45)	(0/90, -125/55, -25/0/180)	■■■
Door glass	cube	1	(.76, .57, .012)	(-1.29, 1.12, .2)	(25, 145, 0)	■ glass

Table 71: Car in the world, placed with $M_{place} = T(-3, 0, 11.2)$: centre of each key part and the world position $M_{place}\, M_{part}\, \mathbf{v}$ of two of its real mesh vertices A and B (Table 3)

Part	Shape	Centre (x, y, z)	Vertex A	Vertex B
Front block	cube	$(-3.00, 0.58, 12.64)$	$(-3.89, 0.30, 11.93)$	$(-2.11, 0.86, 13.35)$
Rear block	cube	$(-3.00, 0.58, 9.99)$	$(-3.89, 0.30, 9.05)$	$(-2.11, 0.86, 10.93)$
Roof	cube	$(-3.00, 1.39, 11.00)$	$(-3.59, 1.36, 10.29)$	$(-2.41, 1.42, 11.71)$
Bonnet (wedge)	wedge	$(-3.00, 0.90, 12.78)$	$(-3.86, 0.86, 13.35)$	$(-2.14, 0.94, 12.20)$
Windscreen	cube	$(-3.00, 1.12, 11.92)$	$(-3.57, 0.86, 12.15)$	$(-2.43, 1.38, 11.69)$
Tyres (1)	cylinder	$(-3.78, 0.32, 12.53)$	$(-3.68, 0.64, 12.49)$	$(-3.88, 0.64, 12.56)$
Tyres (2)	cylinder	$(-2.22, 0.32, 12.53)$	$(-2.12, 0.64, 12.49)$	$(-2.32, 0.64, 12.56)$
Driver door (1)	cube	$(-4.32, 0.58, 11.65)$	$(-4.74, 0.32, 11.38)$	$(-3.90, 0.85, 11.92)$
Driver door (2)	cube	$(-4.30, 0.58, 11.62)$	$(-4.70, 0.33, 11.36)$	$(-3.90, 0.83, 11.88)$

Construction. The body is a front block and a rear block (boxes), a passenger-side skin and a floor, with a rounded nose (cylinder rotated $R_z(90°)$), bumpers and an underbody. The bonnet is a wedge sloping to the front. The windscreen, the rear window and the side windows are thin parallelograms and triangles built directly from their corner points (Section 7); pillars are beams between two points. Inside are seats, a rear bench, a dashboard and a steering wheel (torus rotated $R_x(70°)$).

Worked example: a tyre. The tyre is a cylinder with $S = (0.64, 0.22, 0.64)$ rotated by $R_z(90°)$, so its axis points sideways, inside the wheel node $T(-0.78, 0.32, 1.325)$. With $R_z(90°) : (x, y, z) \mapsto (-y, x, z)$:

Table 72: Three vertices of the front tyre (car space)

Vertex	unit $\mathbf{p}$	after S	after $R_z(90°)$	after wheel T
$\mathbf{p}_0^{top}$	$(0.5, 0.5, 0)$	$(0.32, 0.11, 0)$	$(-0.11, 0.32, 0)$	$(-0.89, 0.64, 1.325)$
$\mathbf{p}_0^{bot}$	$(0.5, -0.5, 0)$	$(0.32, -0.11, 0)$	$(0.11, 0.32, 0)$	$(-0.67, 0.64, 1.325)$
$\mathbf{p}_6^{top}$	$(0, 0.5, 0.5)$	$(0, 0.11, 0.32)$	$(-0.11, 0, 0.32)$	$(-0.89, 0.32, 1.645)$

The top of the tyre is at $y = 0.64$, twice the wheel radius, so the tyre touches the ground at $y = 0$. When the car moves, the spin $R_x(\theta_{spin})$, the steering $R_y(\delta)$ (front wheels) and the car's own $T(\mathbf{p}_{car})R_y(\psi)$ are added.

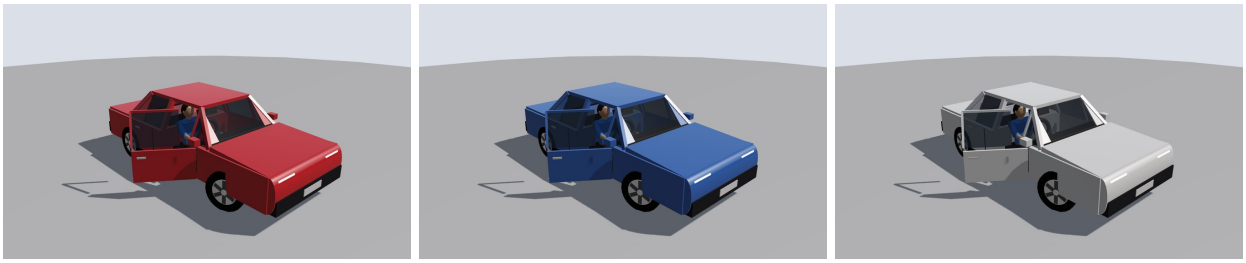
Worked example: the driver door. The door is a child node $T(\mathbf{h}) \cdot R_y(\theta_{door})$ with the hinge $\mathbf{h} = (-0.905, 0, 0.73)$. Its outer panel has $S = (0.04, 0.534, 1.0)$ and $T = (-0.01, 0.582, -0.5)$, so the rear-bottom corner v_0 is at $(-0.03, 0.315, -1.0)$ in door space. With $R_y(\theta) : x' = x\cos\theta + z\sin\theta,\ z' = -x\sin\theta + z\cos\theta$:

- door closed, $\theta = 0°$: $(-0.03, 0.315, -1.0) + \mathbf{h} = (-0.935,\ 0.315,\ -0.27)$;

- door open, $\theta = 70°$: $(-0.950,\ 0.315,\ -0.314) + \mathbf{h} = (-1.855,\ 0.315,\ 0.416)$.

The back edge swings 92 cm out while the points on the hinge line stay fixed. The door glass is a child of the door node, so it swings with the door.

Function. The wheels turn with θ_{spin} += $\Delta s/R$, where Δs is the distance driven, and the front wheels steer with $\delta = \arctan(L_{wb}\,\Delta\psi/\Delta s)$ (L_{wb} = wheel base, $\Delta\psi$ = change of heading). Headlights, tail and brake lights, reverse lights and blinking indicators follow the driving state; at night the headlights are real spot lights. How the cars drive is explained in Section 16.



(a) Red **(b)** Blue **(c)** White

Figure 45: The car model in its three paint colours.

Colours, textures and sound. Glossy car paint, tinted transparent glass ($\alpha = 0.45$), black rubber tyres, polished metal rims and handles, glowing lights. Sounds: the door opening and closing when a visitor gets in or out.

8.5 Jungle

8.5.1 Rainforest tree

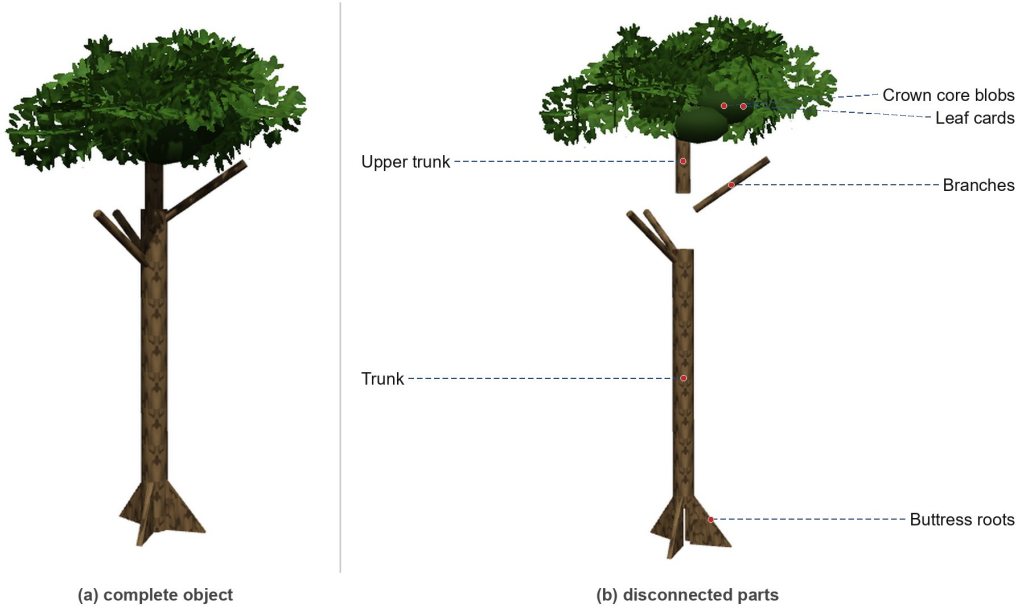


Figure 46: Rainforest tree: (a) complete, (b) disconnected parts.

Table 73: Rainforest tree: the basic shapes it uses (30 shapes in total)

Shape	Name	No.	Used for
	leaf card	18	Leaf cards
	cylinder (low)	5	Trunk, Upper trunk, Branches
	wedge	4	Buttress roots
	sphere (low)	3	Crown core blobs

Table 74: Rainforest tree: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Trunk	cylinder (low)	1	(.62, 7.35, .62)	(0, 3.67, 0)	–	
Upper trunk	cylinder (low)	1	(.42, 1.8, .42)	(0, 8.15, 0)	–	
Buttress roots	wedge	4	(.13, 1.5, 1.2)	(-.58. . . .57, .75, -.6. . . .57)	(0, -175. . . 109, 0)	
Branches	cylinder (low)	3	(.2, 2/2.31/2.49, .2)	(-.74/-.077/.96, 6.6/6.66/7.67, -.37/-.28/.95)	(53/56, -111/-5/111, 0)	
Crown core blobs	sphere (low)	3	(1.74/1.77/1.82, 1.01/1.02/1.05, 1.74/1.77/1.82)	(.28/.63/.87, 8.81/9.22/9.5, -.54/-.023/.58)	(0, 38/40/146, 0)	
Leaf cards	leaf card	18	(1.91. . .2.58, 1.91. . .2.58, 1.91. . .2.58)	(-1.81. . .1.96, 9.1. . .10.33, -2.27. . .2.09)	(-30. . .32, 22. . .170, -29. . .30)	

Construction. A low-poly trunk (6.5–9 m) and an upper trunk, four buttress roots (wedges), three branches in nodes $R_y(yaw)\,R_x(45^\circ \ldots 60^\circ)$, sometimes a hanging vine, and a crown: three dark core blobs inside give the crown volume, and 15–19 alpha-tested leaf cards, each turned randomly about all three axes, are placed on an ellipsoid shell around the crown centre **c**:

$$\mathbf{p} = \mathbf{c} + k\,(\cos\alpha\sqrt{1-y^2}\,r_x,\ y\,r_y,\ \sin\alpha\sqrt{1-y^2}\,r_z), \qquad \alpha \in [0, 2\pi),\ y \in [-0.55, 1],\ k \in [0.55, 1].$$

Worked example: a buttress root. Fin i is a wedge with $S = (0.13, 1.5, 1.2)$ and $T = (0, 0.75, 0.6)$ inside a node turned by $R_y(90^\circ i)$. For $i = 1$: the bottom-front vertex $w_0 = (-.5, -.5, .5)$ becomes $(-0.065, -0.75, 0.6)$ after S, $(-0.065, 0, 1.2)$ after T, and $(1.2,\ 0,\ 0.065)$ after $R_y(90^\circ)$; the top-back vertex $w_2 = (-.5, .5, -.5)$ becomes $(0,\ 1.5,\ 0.065)$. Each fin therefore starts on the trunk 1.5 m above the ground and reaches the ground 1.2 m away from it.

Colours, textures and sound. Bark texture, leaf-cluster cut-out texture, every card with a slightly different green. Birds sing in the background by day (Section 17.4).

8.5.2 Coconut palm

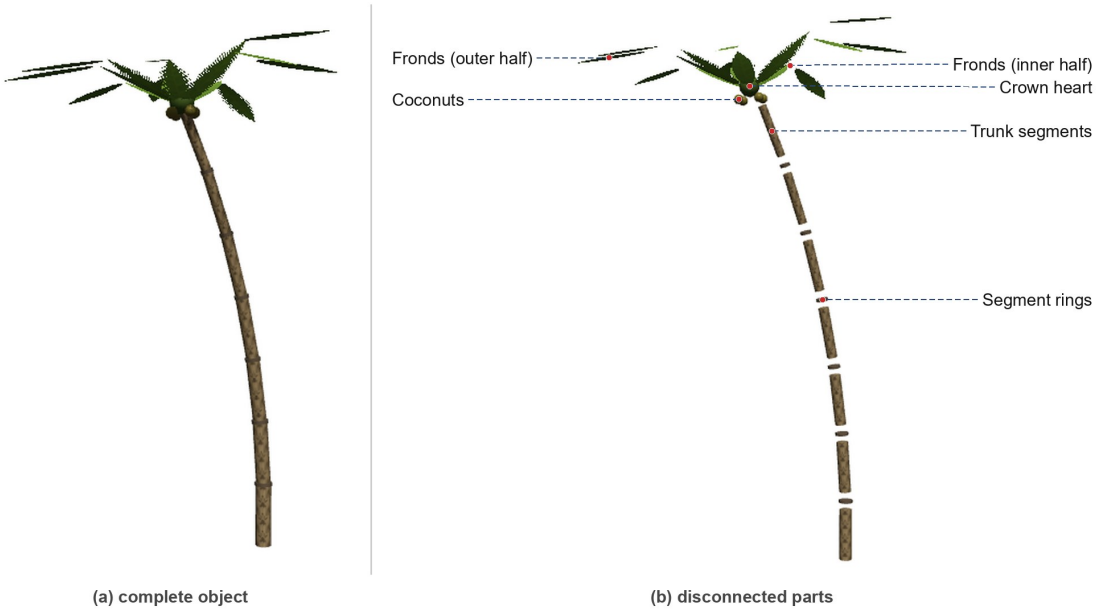


Figure 47: Coconut palm: (a) complete, (b) disconnected parts.

Table 75: Coconut palm: the basic shapes it uses (37 shapes in total)

Shape	Name	No.	Used for
	quad	18	Fronds (inner half), Fronds (outer half)
	cylinder (low)	14	Trunk segments, Segment rings
	sphere (low)	5	Crown heart, Coconuts

Table 76: Coconut palm: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Trunk segments	cylinder (low)	7	(.23. . .36, 1.44, .23. . .36)	(-.74. . .0, .72. . .9.08, 0. . .1.79)	(0. . .26, -23, 0)	
Segment rings	cylinder (low)	7	(.26. . .4, .08, .26. . .4)	(-.86. . .0, 1.41. . .9.7, 0. . .2.07)	(0. . .26, -23, 0)	
Crown heart	sphere (low)	1	(.55, .6, .55)	(-.87, 9.82, 2.1)	(6, -23, 0)	
Fronds (inner half)	quad	9	(1, 1.61. . .2.09, 1)	(-1.65. . .-.068, 10.01. . .10.27, 1.15. . .2.84)	(55. . .69, -143. . .179, -14. . .15)	
Fronds (outer half)	quad	9	(.85, 1.61. . .2.09, 1)	(-3.24. . .1.65, 10.09. . .10.74, -.75. . .4.37)	(70. . .84, -174. . .153, -176. . .163)	
Coconuts	sphere (low)	4	(.26, .26, .26)	(-1.11. . .-.6, 9.54. . .9.61, 1.82. . .2.33)	(6, -23, 0)	

Construction. The trunk is a chain of seven segments. Each segment is a child of the previous one with an extra $R_x(3^\circ \ldots 7^\circ)$, so the rotations add up and the trunk bends smoothly: $M_{k+1} = M_k \cdot T(0, L, 0) \cdot R_x(\beta)$. Each segment has a ring. The crown has a heart (sphere), nine fronds made of two quads (the outer half is a child of the inner half and droops by another $30^\circ \ldots 45^\circ$) and four coconuts.

Colours, textures and sound. Bark texture, palm frond cut-out texture.

8.5.3 Emergent tree

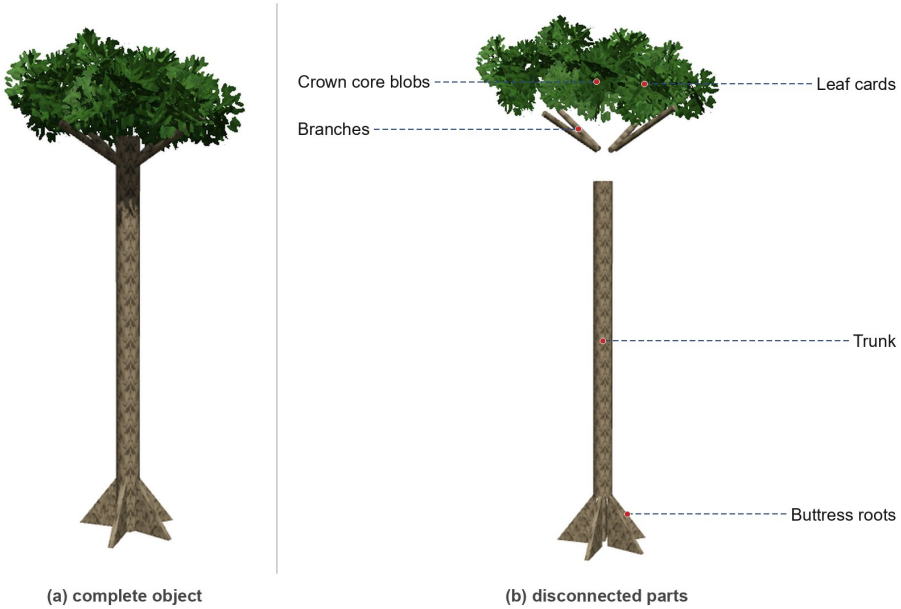


Figure 48: Emergent giant tree: (a) complete, (b) disconnected parts.

Table 77: Emergent tree: the basic shapes it uses (35 shapes in total)

Shape	Name	No.	Used for
	leaf card	22	Leaf cards
	cylinder (low)	5	Trunk, Branches
	wedge	5	Buttress roots
	sphere (low)	3	Crown core blobs

Table 78: Emergent tree: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Trunk	cylinder (low)	1	(.95, 15.36, .95)	(0, 7.68, 0)	–	
Buttress roots	wedge	5	(.2, 2.6, 2)	(-.96. . .1, 1.3, -.9. . .97)	(0, -154. . .147, 0)	
Branches	cylinder (low)	4	(.3, 3.47. . .4.19, .3)	(-1.51. . .1.68, 14.89. . .15.17, -1.89. . .1.49)	(57. . .66, -170. . .83, 0)	
Crown core blobs	sphere (low)	3	(2.72/2.84/2.93, .84/.88/.91, 2.72/2.84/2.93)	(-.88/-.038/.94, 16.64/16.93/17.02, -.85/.38/.91)	(0, 2/101/169, 0)	
Leaf cards	leaf card	22	(2.28. . .2.95, 2.28. . .2.95, 2.28. . .2.95)	(-3.42. . .2.56, 16.12. . .17.87, -2.92. . .3.46)	(-33. . .32, 22. . .177, -30. . .32)	

Construction. A 14–18 m trunk, five large buttress roots turned by $R_y(72° \, i)$, four branches and a wide, flat crown made of 20–24 leaf cards on a flattened ellipsoid ($r_x = r_z = 4.2$, $r_y = 1.3$). These trees rise above the rest of the forest.

Colours, textures and sound. Light bark texture, leaf-cluster texture.

8.5.4 Bamboo clump

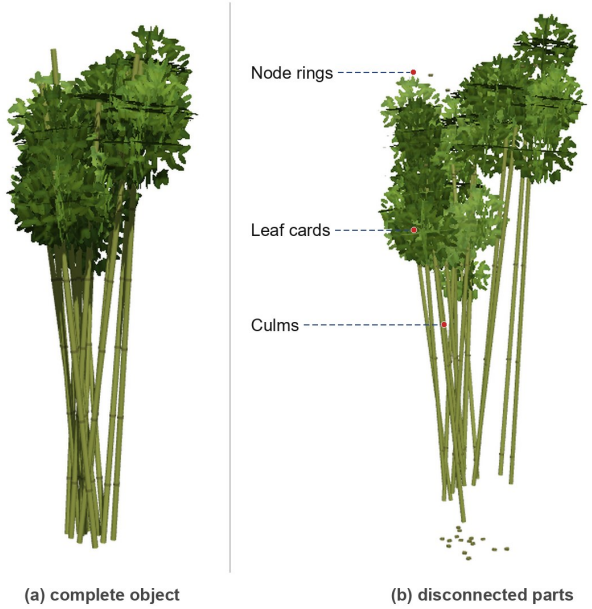


Figure 49: Bamboo clump: (a) complete, (b) disconnected parts.

Table 79: Bamboo clump: the basic shapes it uses (130 shapes in total)

Shape	Name	No.	Used for
	cylinder (low)	100	Culms, Node rings
	leaf card	30	Leaf cards

Table 80: Bamboo clump: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Culms	cylinder (low)	15	(.088, 5.75…7.49, .088)	(-.32….88, 2.85…3.73, -.4….76)	(-8…8, -178…145, -8…6)	
Node rings	cylinder (low)	85	(.1, .04, .1)	(-.52…1.12, .87…6.07, -.53…1.02)	(-8…8, -178…145, -8…6)	
Leaf cards	leaf card	30	(1.36, 1.36, 1.36)	(-.46…1.46, 4.65…7.09, -.61…1.11)	(-9…8, -170…176, -8…7)	

Construction. 9–15 culms (thin low-poly cylinders) in nodes with a random position on a disc and a small random tilt $R(\pm 8°$, random yaw, $\pm 8°)$, node rings every 1.3 m and two leaf cards per culm.

Colours, textures and sound. Plain green culms, leaf-cluster texture.

8.5.5 Banana plant

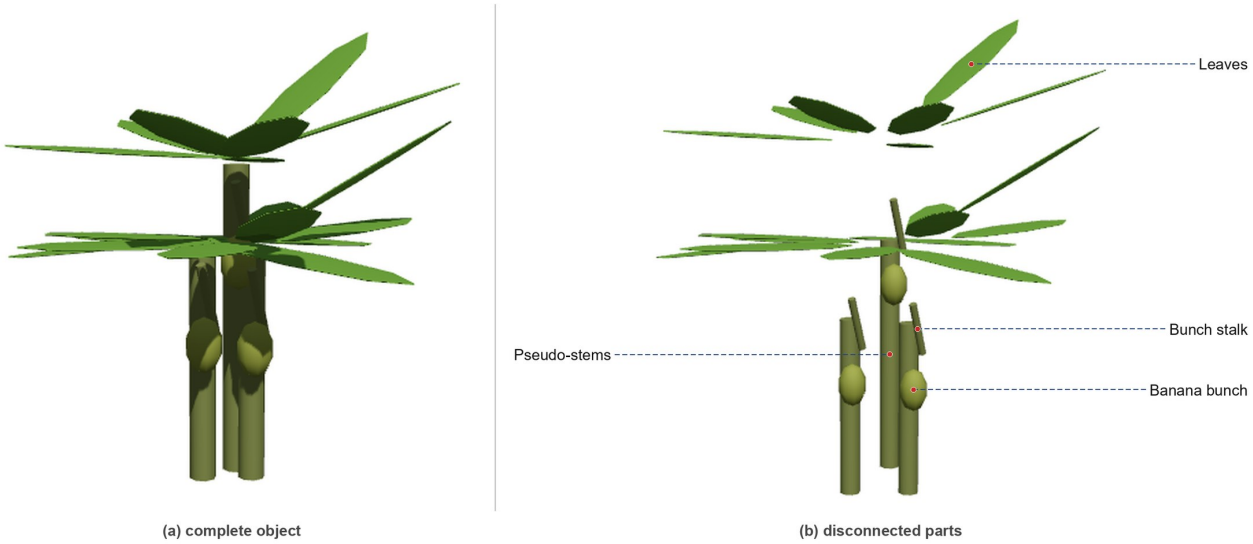


Figure 50: Banana plant: (a) complete, (b) disconnected parts.

Table 81: Banana plant: the basic shapes it uses (28 shapes in total)

Shape	Name	No.	Used for
	sphere (low)	22	Leaves, Banana bunch
	cylinder (low)	6	Pseudo-stems, Bunch stalk

Table 82: Banana plant: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Pseudo-stems	cylinder (low)	3	(.22, 1.89/1.93/2.55, .22)	(.011/.12/.33, .94/.96/1.28, -.22/-.13/.1)	–	
Leaves	sphere (low)	19	(.55, .04, 1.9)	(-.92…1.1, 1.84…3.13, -1.1…1.04)	(-35…8, -160…162, 0)	
Bunch stalk	cylinder (low)	3	(.08, .6, .08)	(.16/.27/.48, 1.49/1.53/2.15, -.072/.022/.25)	(20, 0, 0)	
Banana bunch	sphere (low)	3	(.3, .45, .3)	(.19/.3/.51, 1.14/1.18/1.8, .028/.12/.35)	–	

Construction. One to three pseudo-stems (cylinders); at the top of each, 5–8 long leaves are spheres squashed to $(0.55, 0.04, 1.9)$ and turned by $R_y(360° \, l/n)\, R_x(-35° \ldots 15°)$. Some plants carry a bunch of bananas on a stalk.

Colours, textures and sound. Matte greens; the leaves use two-sided foliage lighting.

8.5.6 Undergrowth

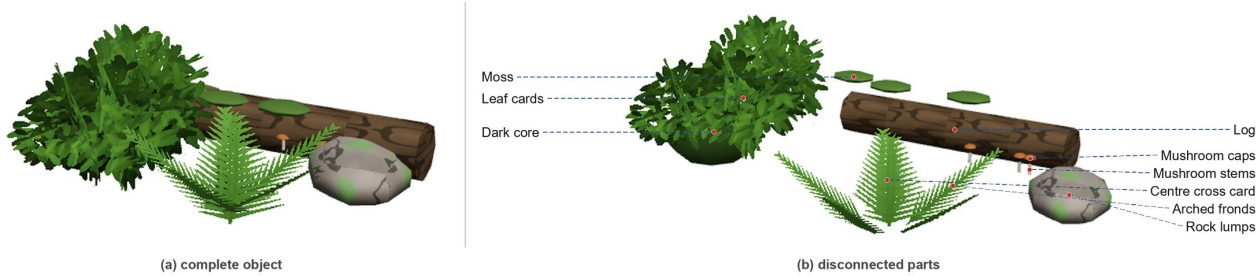


Figure 51: Undergrowth: a bush, a fern, a rock and a fallen log: (a) complete, (b) disconnected parts.

Table 83: Undergrowth: the basic shapes it uses (23 shapes in total)

Shape	Name	No.	Used for
	sphere (low)	5	Dark core, Rock lumps, Moss
	leaf card	5	Leaf cards
	quad	5	Arched fronds
	cylinder (low)	4	Log, Mushroom stems
	hemisphere	3	Mushroom caps
	cross card	1	Centre cross card

Table 84: Undergrowth: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \ldots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Dark core	sphere (low)	1	(1.1, .7, 1.1)	(-1, .38, 0)	(0, 79, 0)	
Leaf cards	leaf card	5	(1.09…1.34, 1.09…1.34, 1.09…1.34)	(-1.06…-.53, .41….75, -.14….5)	(-17…29, 2…145, -20…30)	
Arched fronds	quad	5	(.71, 1.17, 1)	(.2…1, .39, .063/.53/.82)	(55, -144…144, 0)	
Centre cross card	cross card	1	(1.04, .91, 1.04)	(.6, 0, .4)	–	
Rock lumps	sphere (low)	1	(.98, .52, .95)	(1.02, .08, -1.11)	(-4, -77, -9)	
Log	cylinder (low)	1	(.53, 3, .53)	(0, .24, -1.4)	(0, 0, 90)	
Moss	sphere (low)	3	(.6, .12, .43)	(-.91/-.5/.16, .45, -1.42/-1.34/-1.31)	–	
Mushroom stems	cylinder (low)	3	(.035, .12, .035)	(.25/.64/.72, .16, -1.16)	–	
Mushroom caps	hemisphere	3	(.14, .08, .14)	(.25/.64/.72, .22, -1.16)	–	

Construction. A bush is a dark core sphere and 5–7 leaf cards (sometimes with flowers). A fern is five quads tilted by $R_x(55°)$ and turned by $R_y(72° \, i)$ plus a cross card. A rock is 1–3 squashed low spheres with random rotations. A fallen log is a horizontal cylinder ($R_z(90°)$) with moss blobs and mushrooms made of a cylinder stem and a hemisphere cap.

Colours, textures and sound. Leaf, fern, rock and bark textures.

8.5.7 Bird

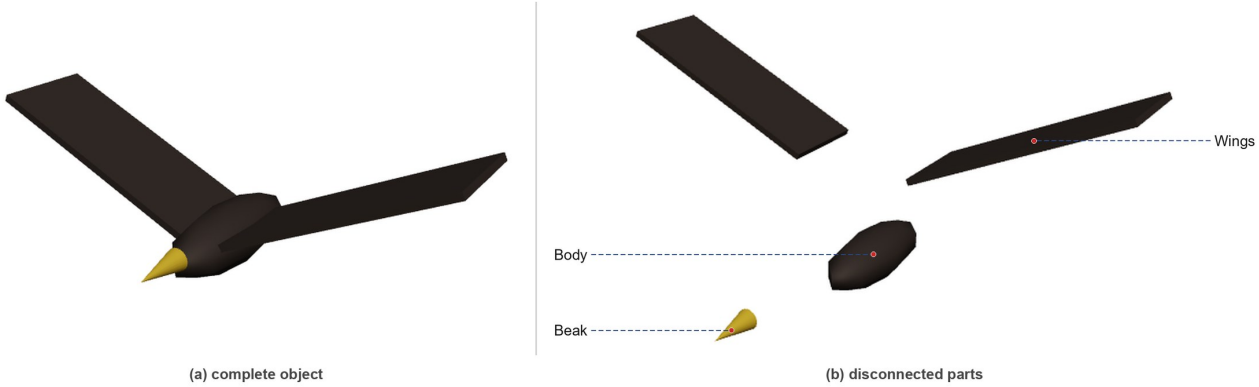


Figure 52: Bird: (a) complete, (b) disconnected parts.

Table 85: Bird: the basic shapes it uses (4 shapes in total)

Shape	Name	No.	Used for
	cube	2	Wings
	sphere (low)	1	Body
	cone	1	Beak

Table 86: Bird: every part with its basic shape and its transformation $M = T\,R\,S$ (object space, metres and degrees; $\pm$ = mirrored pair, $a \dots b$ = range over the copies)

Part	Shape	No.	Scale $S = (s_x, s_y, s_z)$	Translation $T = (t_x, t_y, t_z)$	Rotation $R = (\theta_x, \theta_y, \theta_z)$	Colour / texture
Body	sphere (low)	1	(.18, .16, .5)	(0, 0, 0)	–	■
Beak	cone	1	(.06, .14, .06)	(0, .02, .3)	(90, 0, 0)	■
Wings	cube	2	(.7, .02, .28)	($\pm$.32, .15, 0)	(0, 0, $\pm$25)	■

Construction. A body (squashed sphere), a cone beak rotated $R_x(90°)$ and two wings. Each wing is a box in a child node rotated by $R_z(\pm\phi)$ about the body axis, with $\phi = 35° \sin(\text{flap})$, so the two wings flap symmetrically.

Function. Nine birds fly on circles above the jungle by day (Section 17).

8.5.8 Placement of the jungle, terrain and road

The world (280×280 m) is covered with a *jittered grid* of 6.6 m cells. Each cell gets one random point, and points inside the cleared area (building, lot and road with a margin) are skipped. A random number then chooses the species: 40% rainforest tree, 18% palm, 7% emergent tree, 8% bamboo, 11% banana, 8% bush and 8% nothing. Every cell also gets 1–5 ferns or bushes and 4–12 grass or flower tufts, a rock with 12% and a log with 5% chance. In total the jungle has 795 broad-leaf trees, 279 palms, 132 bamboo clumps, 1,503 bushes, 2,193 ferns, 4,783 grass tufts, 199 rocks and 86 logs. Every trunk gets a circle collider.

The terrain is flat in the clearing and along the road and becomes hilly further away:

$$H(x,z) = 14\,\big(\mathrm{fbm}(0.018x,\,0.018z) - 0.45\big) + 1.2\,\mathrm{fbm}(0.07x,\,0.07z)$$
$$h(x,z) = \max\Big(-1.5,\;\; H(x,z)\cdot \mathrm{smoothstep}(0,30,d)\cdot \mathrm{smoothstep}\big(5,14,|z - z_{road}(x)|\big)\Big)$$

where d is the distance outside the clearing. The road is straight through the lot ($z = 16$) and curves in the jungle. With $t = |x| - 30$,

$$z_{road}(x) = 16 + 7\sin(0.035\,t)\,\mathrm{sign}(x)\cdot \mathrm{smoothstep}(0,25,t).$$

The smoothstep factor gives the bend a slope of zero at $t = 0$, so the straight and the curved part join smoothly. The road is built from 2 m boxes turned to follow the curve; neighbouring boxes overlap by 40 cm.

9 Lighting and Shading

9.1 Illumination model

The light leaving a surface point is computed with the **Phong illumination model** using Blinn's half vector (the *Blinn–Phong* model). It is the sum of three reflection terms:

Table 87: The three terms of the illumination model

Term	Formula	Meaning
Ambient	$I_a\,k_a$	light scattered by the whole scene; it reaches every surface, also in shadow
Diffuse	$I_l\,k_d\,\max(\mathbf{N}\cdot\mathbf{L},0)$	Lambert's cosine law: the brightness depends on the angle between the surface and the light, not on the viewer
Specular	$I_l\,k_s\,\max(\mathbf{N}\cdot\mathbf{H},0)^n$	the highlight; it moves with the viewer. A large shininess n gives a small sharp highlight

Here $\mathbf{N}$ is the unit surface normal, $\mathbf{L}$ the unit vector towards the light, $\mathbf{V}$ the unit vector towards the camera and $\mathbf{H}$ the half vector,

$$\mathbf{H} = \frac{\mathbf{L}+\mathbf{V}}{|\mathbf{L}+\mathbf{V}|}, \qquad \mathbf{R} = 2(\mathbf{N}\cdot\mathbf{L})\,\mathbf{N} - \mathbf{L}.$$

Phong's original specular term $(\mathbf{R}\cdot\mathbf{V})^n$ uses the mirror direction $\mathbf{R}$; the half-vector form $(\mathbf{N}\cdot\mathbf{H})^n$ gives nearly the same highlight and is cheaper to compute. For one light the model is

$$I = I_a k_a + I_l\big(k_d\max(\mathbf{N}\cdot\mathbf{L},0) + k_s\max(\mathbf{N}\cdot\mathbf{H},0)^n\big).$$

The fragment shader evaluates this sum over all active lights:

$$C = \underbrace{\mathrm{mix}\big(A_{ground}, A_{sky}, \tfrac{N_y+1}{2}\big)\,k_d}_{\text{ambient}} + \sum_{lights}\Big(k_d\max(\mathbf{N}\cdot\mathbf{L},0) + k_s\max(\mathbf{N}\cdot\mathbf{H},0)^n\Big)\,I\,a(d)\,c\,s + E.$$

k_d is the surface colour multiplied by the texture, k_s the specular strength, n the shininess, I the light colour times its intensity, $a(d)$ the distance attenuation, c the spot-light cone factor, s the shadow factor (Section 12) and E the emission of glowing surfaces. The ambient term is a *hemisphere* light: surfaces facing up receive the sky colour and surfaces facing down the ground colour. Inside the building it is multiplied by 0.45, because the roof blocks the sky. Point and spot lights fade smoothly to zero at their range r:

$$a(d) = \frac{\big[\mathrm{clamp}(1-(d/r)^4,\,0,\,1)\big]^2}{1+d^2}, \qquad c = \mathrm{smoothstep}\big(\cos\theta_{out},\;\cos\theta_{in},\;-\mathbf{L}\cdot\mathbf{D}\big),$$

where $\mathbf{D}$ is the spot direction and θ_{in}, θ_{out} are its inner and outer cone angles.

Materials. Every material carries its own k_s and n (Table 88). Leaves and grass use *wrap* lighting, $0.6\,\mathbf{N}\cdot\mathbf{L}+0.4$, are lit from both sides, and gain a translucency term $0.45\,k_d\,I\,s\,\max(-\mathbf{V}\cdot\mathbf{L},0)^3$ when the sun is behind them. Polished metal (bins, door handles, wheel rims) uses a weak diffuse part $0.25\,k_d$ and a highlight in the colour of the metal, $3k_d$; it also reflects a simple painted surrounding $E_{env}(\mathbf{R}_v)$ (sky, horizon and ground) in the mirror direction $\mathbf{R}_v = 2(\mathbf{N}\cdot\mathbf{V})\mathbf{N}-\mathbf{V}$. Car paint adds a clear-coat reflection weighted with Schlick's approximation $F=(1-\mathbf{N}\cdot\mathbf{V})^5$:

$$C_{metal} = C(0.25\,k_d,\ 3k_d)+E_{env}(\mathbf{R}_v)\,k_d\,(0.85{+}0.15F), \qquad C_{paint} = C(k_d,\ 2.5)+E_{env}(\mathbf{R}_v)\,(0.08{+}0.75F).$$

Table 88: Typical material parameters

Material	k_s	n	Used for
Fabric, plaster, bark, brick, matte	0.03–0.05	6–8	sofas, walls, trees, soil
Rubber	0.08	8	tyres
Wood, leather	0.2–0.3	24	furniture, shelves
Plastic, painted metal, steel	0.35–0.6	48–64	machines, keys, lamps
Marble	0.6	96	counter top, window sills
Glass	1.0	160	windows, front glass, aquarium
Car paint	1.0	160	cars (with clear-coat reflection)
Polished metal	1.0	200	bins, handles, rims (with reflection)

9.2 Light sources in the scene

Table 89: Light types used in the scene and the figures in which they can be seen

Type	Behaviour	Where in the scene	Figures
Directional	same direction **L** everywhere, no attenuation	the **sun** by day and the **moon** by night	Figure 6
Point	radiates in all directions from a position, attenuated by $a(d)$	lounge pendants (4), library pendants (3), ATM tube lights (2), vending machine interiors (3), ATM screen, aquarium hood, TV glow, street lamps	Figure 53, Figure 54, Figure 69
Spot	a point light limited to a cone	downlights over the vending machines, the TV wall and the coffee counter, the reading lamp, porch lights, car headlights	Figure 62, Figure 54
Emissive	the surface glows by itself and lights nothing else	bulbs, tubes, LEDs, screens, signs, fireflies	Figure 53, Figure 27
Ambient	sky colour from above, ground colour from below	everywhere, changes with the time of day	Figure 53(b)

Every light belongs to a *group*: lounge, spots, ATM room, library, outdoor or machines. Groups are switched with the F1–F4 keys or with the light switches on the walls, and the lamp models check their group, so a bulb only glows when its light is on. The outdoor group switches on automatically at dusk.



(a) Night, lamps on



(b) Night, lamps off (machine glow and moonlight)

Figure 53: The lounge with and without its point and spot lights.



(a) Library at night: pendants and reading lamp



(b) Street lamps and car lights

Figure 54: Point, spot and emissive lights at night.

9.3 Per-frame light selection

The shader accepts up to 32 point lights and 16 spot lights. Every frame the light manager

1. removes lights that are switched off or whose sphere of influence lies outside the view frustum,
2. keeps the nearest ones, and
3. sorts them by **region**: ATM room, lounge, library or outdoors.

Each pixel determines its region from its position and loops only over the lights of that region. Light therefore never passes through a wall into the next room, and each pixel evaluates fewer lights.

9.4 Shading method

The scene uses **Phong shading**: the illumination model is evaluated once per pixel with an interpolated normal (Figure 55).

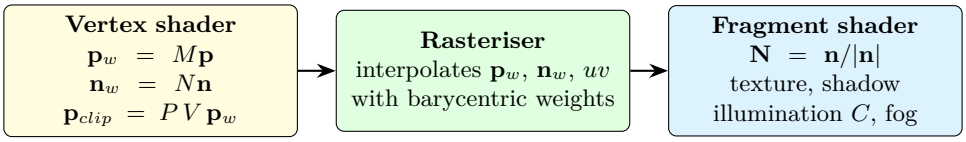


Figure 55: Phong shading in the OpenGL pipeline.

1. The vertex shader transforms the position into world space and the normal with the normal matrix $N = (M^{-1})^T$, and sends both to the rasteriser.

2. For a pixel inside a triangle with vertices $\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2$, the rasteriser computes the barycentric weights (α, β, γ), $\alpha + \beta + \gamma = 1$, and interpolates every value (perspective-correct):

$$\mathbf{n}(\alpha, \beta, \gamma) = \alpha\,\mathbf{n}_0 + \beta\,\mathbf{n}_1 + \gamma\,\mathbf{n}_2, \qquad \mathbf{p}(\alpha, \beta, \gamma) = \alpha\,\mathbf{p}_0 + \beta\,\mathbf{p}_1 + \gamma\,\mathbf{p}_2.$$

3. The interpolated normal is shorter than 1 inside the triangle, so the fragment shader normalises it, $\mathbf{N} = \mathbf{n}/|\mathbf{n}|$, computes $\mathbf{L}$, $\mathbf{V} = \text{normalize}(\mathbf{e} - \mathbf{p})$ and $\mathbf{H}$, and evaluates the illumination model C of the previous section for this pixel.

Whether a surface looks smooth or faceted is decided by its vertex normals. A cube stores separate vertices for every face with the face normal, so all pixels of a face get the same normal and the edges are sharp. A sphere or a cylinder stores one outward normal per vertex, so the interpolated normal changes smoothly across a triangle and the surface looks round even with 20 or 24 sides (compare the shapes in Table 2). Because the specular term is evaluated per pixel, highlights stay sharp in the middle of large triangles, for example on the floor tiles and on the car bonnet. The same shading is used in Basic mode, without textures, shadows and fog.

10 Day and Night

A 24-hour clock runs at one game hour per 20 s (keys change the speed, pause it or jump between day and night). The sun direction is computed from the hour:

$$a = \frac{hour - 6}{12}\pi, \qquad \mathbf{s} = \text{normalize}(0.95\cos a,\ \sin a,\ 0.42), \qquad \mathbf{m} = \text{normalize}(-0.8\,s_x,\ -s_y,\ 0.3).$$

The sun rises in the east ($+x$) at 6:00, is highest at noon and sets at 18:00; the moon $\mathbf{m}$ is on the opposite side. All other values are blended from the sun height s_y: the sky zenith and horizon colours (with an orange glow near sunrise and sunset), the sun colour, the ambient light, the fog colour and density, the cloud colour and the exposure. The night factor

$$\nu = 1 - \text{smoothstep}(-0.14,\ 0.10,\ s_y)$$

shows the stars and the moon, switches on the street lamps and the car headlights and wakes up the fireflies. Figure 6 shows the same view at four times of day.

11 Textures

11.1 Procedural textures

All 25 textures are generated at start-up from *value noise*: random values on a grid are smoothly interpolated, and several octaves are added (fractal Brownian motion):

$$\text{fbm}(\mathbf{p}) = \frac{\sum_{i=0}^{k-1} 0.5^i\,\text{noise}(2^i\mathbf{p})}{\sum_{i=0}^{k-1} 0.5^i}.$$

The noise wraps around at the texture border, so the textures tile without seams. Patterns are built on top of it: rows of bricks with mortar, wooden planks with grain $\sin(6v + \text{noise})$, tiles with grout, marble veins, woven fabric, bark ridges, and *cut-out* textures (leaves, grass, fern, palm frond, flowers, vine) whose alpha channel marks the shape of the leaf.

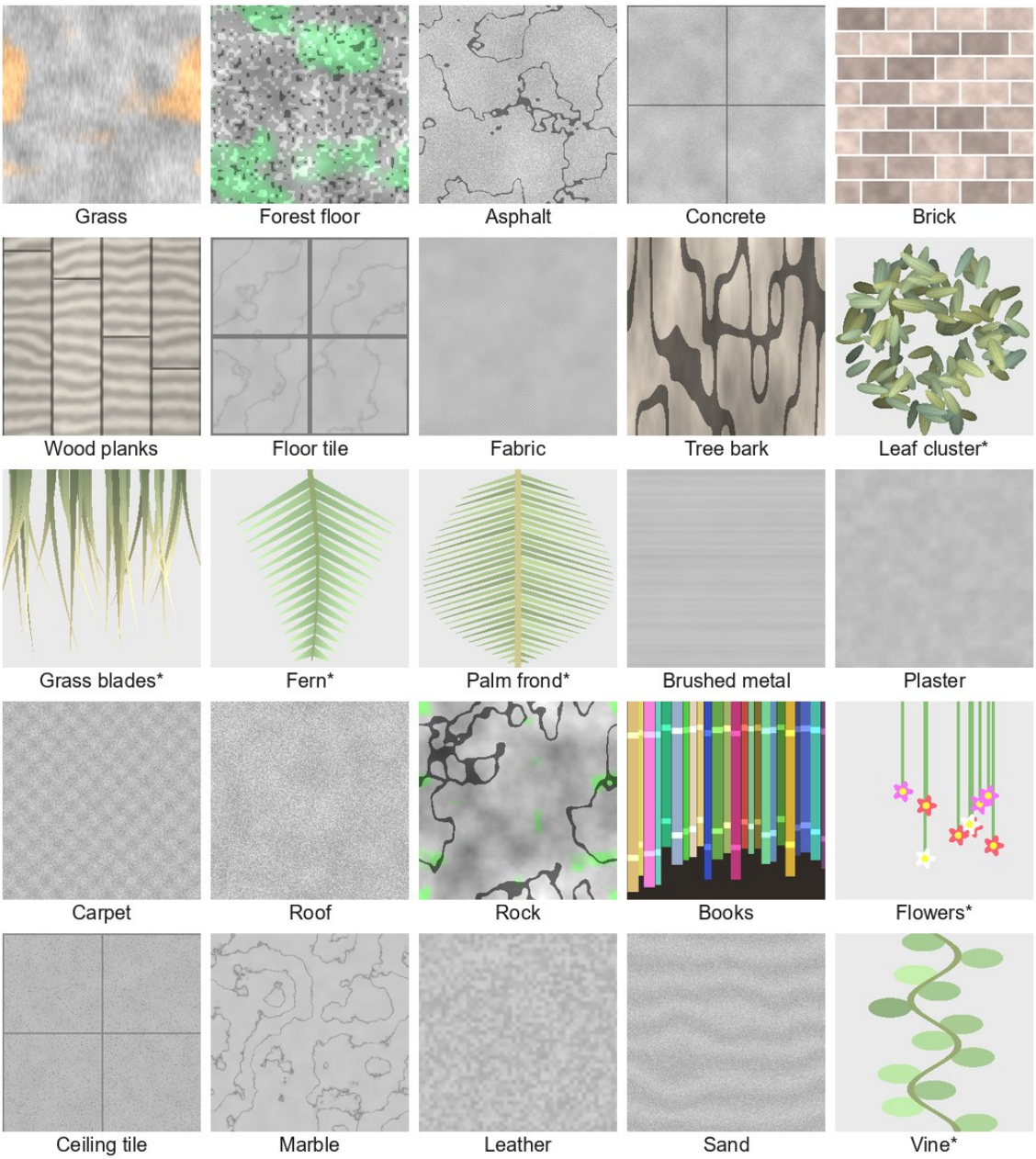


Figure 56: All procedural textures ($*$ = cut-out; light grey marks the transparent part).

11.2 Texture mapping

- All textures are layers of one `GL_TEXTURE_2D_ARRAY` (256×256, mip-maps, $4\times$ anisotropic filtering). An instance only stores a layer number, so a texture change never breaks an instanced draw call.
- A texture stores a *detail* value, and the final colour is $k_d = \text{colour} \times 2 \cdot \text{texel}$, so the same brick texture works for any brick colour.
- **Object-space box mapping**: the shader selects the plane of the largest component of the local normal $\mathbf{n}_\ell$ and multiplies the local position $\mathbf{p}_\ell$ by the object's scale and the number of repeats per metre ρ:

$$(u, v) = \rho \cdot \begin{cases} (x_\ell,\ z_\ell) & |n_y| \geq |n_x|, |n_z| \\ (z_\ell,\ y_\ell) & |n_x| \geq |n_z| \\ (x_\ell,\ y_\ell) & \text{otherwise} \end{cases}$$

A 10 m wall therefore shows ten times more bricks than a 1 m box, without texture coordinates in the mesh.

- Cut-out cards use their mesh texture coordinates, and pixels with $\alpha < 0.5$ are discarded, also in the shadow pass, so leaves cast leaf-shaped shadows.
- The terrain blends lawn grass and forest floor with a large-scale noise and the distance from the clearing.

12 Shadows

Shadows use **shadow mapping** (Figure 57).

1. **Depth pass**: the scene is drawn from the sun with an orthographic camera (± 32 m around the player, a 2048×2048 depth texture, about 3 cm per texel). Only the depth is stored.
2. **Main pass**: every pixel is transformed into the light's space with a small offset along its normal,
$$\mathbf{p}_L = P_L\, V_L\, (\mathbf{p} + 0.05\,\mathbf{N}), \qquad (u, v, z) = \tfrac{1}{2}\,\mathbf{p}_L / w_L + \tfrac{1}{2},$$
and compared with the stored depth $z_{map}(u, v)$: the pixel is lit if $z - b \le z_{map}$, with the slope-scaled bias $b = 0.0006 + 0.0012(1 - \mathbf{N} \cdot \mathbf{L})$.
3. **Percentage-closer filtering**: five hardware-compared samples around (u, v) (the centre counted twice) are averaged, $s = \frac{1}{6} \sum_k w_k\, [z - b \le z_{map}(u_k, v_k)]$, which gives soft shadow edges.

The shadow camera moves only in whole texels along its own axes and the light direction used for shadows changes in steps of 0.5°, so shadow edges do not shimmer when the camera or the sun moves. Glass, grass and other small details do not cast shadows.

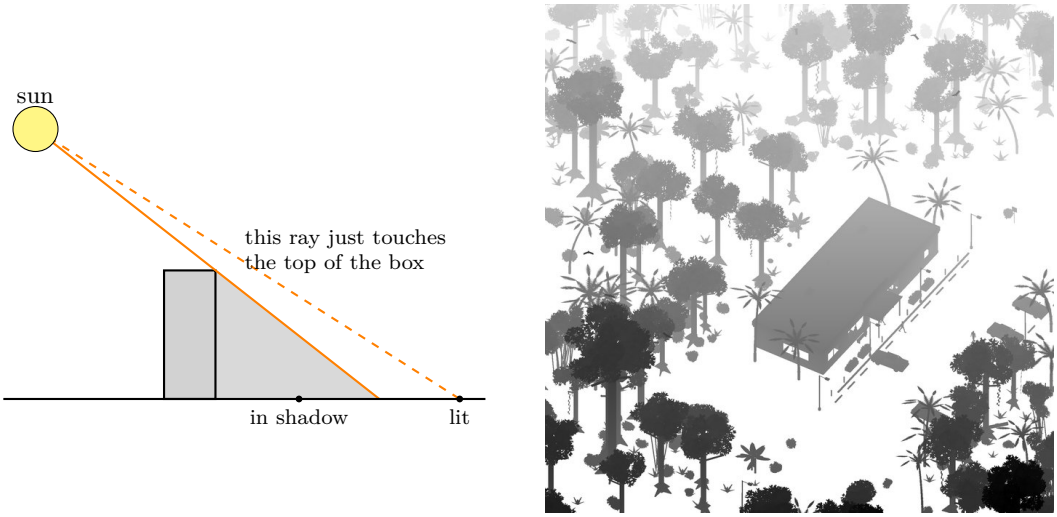


Figure 57: Left: the principle of shadow mapping. Right: the depth image seen by the sun (darker = closer).



(a) Shadows on **(b)** Shadows off

Figure 58: Shadows show where objects touch the ground and give the scene depth.

13 Sky, Fog, Tone Mapping and Wind

- **Sky**: one full-screen triangle. For each pixel the view ray $\mathbf{d}$ is rebuilt with the inverse view-projection matrix. The colour is $\mathrm{mix}(\mathit{horizon}, \mathit{zenith}, h^{0.55})$ (h = height of the ray), plus a sun disc $\max(\mathbf{d}\cdot\mathbf{s}, 0)^{900}$ with a glow, a moon, twinkling stars (a hash on a 3D grid) and moving clouds (4-octave noise projected on a sky plane).
- **Fog**: the colour is blended towards the horizon colour with

$$f = 1 - e^{-(d\rho)^{1.5}}, \qquad C' = (1-f)\,C + f\,C_{fog},$$

  where d is the distance and ρ the density. It hides the end of the world and gives a humid look; it is denser at night and thinner when the camera is high up.
- **Tone mapping**: $C' = 1 - e^{-C\cdot \mathit{exposure}}$ keeps bright lamps from turning flat white. The exposure rises at night.
- **Wind**: leaves and grass sway in the vertex shader. The offset grows with the height inside the shape: $\Delta x = 0.05\,h\,\min(s_y, 2.5)\sin(1.7t + 0.35x + 0.27z)$. The aquarium plants use the same rule as a water current.

14 Basic Mode and Enhanced Mode

B (or the on-screen button) switches between the modes; **T**, **G** and **F** switch textures, shadows and fog one at a time.

Feature	Basic	Enhanced
Basic shapes, transformations, all objects	yes	yes
Blinn–Phong lighting with all lights, Phong shading	yes	yes
Procedural textures	–	yes
Shadow mapping (PCF)	–	yes
Fog	–	yes
Sky with sun, clouds, stars	gradient	yes
Wind animation	–	yes
Tone mapping	clamp	exponential



(a) Basic



(b) Enhanced



(c) Basic (interior)



(d) Enhanced without textures

Figure 59: Basic and Enhanced rendering.

15 The Player and Interaction

The user controls a person (the man model of Section 8.3) who walks through the scene and operates the machines by looking at a part and pressing **E** or the left mouse button. This section explains how the player moves, how the program finds the part that is looked at, and how every object reacts.

15.1 Movement of the player

Walking. The player has a position $\mathbf{p}$ (the feet), a yaw angle ψ (turning left and right) and a pitch angle θ (looking up and down). Moving the mouse by $(\Delta m_x, \Delta m_y)$ pixels changes

$$\psi \leftarrow \psi - 0.12\,\Delta m_x, \qquad \theta \leftarrow \mathrm{clamp}(\theta - 0.12\,\Delta m_y,\ -85^\circ,\ 85^\circ).$$

The keys W, S, A, D give a forward input $w \in \{-1, 0, 1\}$ and a sideways input $a \in \{-1, 0, 1\}$. The forward and the right direction on the ground follow the yaw:

$$\mathbf{f} = (\sin\psi,\ 0,\ \cos\psi), \qquad \mathbf{r} = (-\cos\psi,\ 0,\ \sin\psi), \qquad \mathbf{p} \leftarrow \mathbf{p} + \mathrm{normalize}(w\,\mathbf{f} + a\,\mathbf{r})\ v\,\Delta t,$$

with the speed $v = 2.4$ m/s (walking) or 5.0 m/s (Shift, running). Gravity and jumping act on the height: $v_y \leftarrow v_y - 12\,\Delta t$, $p_y \leftarrow p_y + v_y \Delta t$, and p_y never goes below the terrain height; Space sets $v_y = 4.6$ m/s. While walking, the walk phase grows by $7\,\Delta t$ and drives the body animation of Section 8.3 and a small head bob $0.028\sin(2\varphi)$ of the first-person camera.

Collision. The player is a circle of radius $r = 0.3$ m on the ground. Walls, machines, furniture and cars are oriented boxes with a height range; tree trunks, posts and people are circles. To push the player out of a box, the centre is moved into the box's own frame (rotation by $-yaw_{box}$), the nearest point of the box is found by clamping,

$$\mathbf{q} = \mathrm{clamp}(\mathbf{p}_{local},\ -\mathbf{h},\ \mathbf{h}), \qquad \text{if } |\mathbf{p}_{local} - \mathbf{q}| < r: \quad \mathbf{p}_{local} \leftarrow \mathbf{q} + r\,\frac{\mathbf{p}_{local} - \mathbf{q}}{|\mathbf{p}_{local} - \mathbf{q}|},$$

and the result is turned back. Two circles are separated in the same way along the line between their centres. The test is repeated three times per frame so corners work, and a box only counts when it overlaps the player's height, so the player can walk under a table top. The sliding doors switch their box off while they are open.

Cameras. Key V cycles through three cameras (Figure 60):

- **First person**: the camera is at eye height and looks along the view direction:

$$\mathbf{e} = \mathbf{p} + (0,\ 1.62,\ 0), \qquad \mathbf{d} = (\sin\psi\cos\theta,\ \sin\theta,\ \cos\psi\cos\theta).$$

- **Third person**: the body is visible; the camera is placed 3.4 m behind the head along $-\mathbf{d}$ and 0.35 m higher. A ray from the head to the camera is tested against the collision boxes and the camera moves closer when a wall is in the way.
- **Free camera**: flies without collision (W A S D, Space, Ctrl) to inspect the scene, for example with the roof cut-away view.

The mouse wheel changes the field of view f between 25° and 110°, which changes the $1/\tan(f/2)$ terms of the projection matrix (Section 7).



(a) Third person: walking to the entrance



(b) The sliding doors open in front of the player



(c) First person



(d) Free camera, roof cut-away, night



(e) FOV 35°



(f) FOV 70°



(g) FOV 100°

Figure 60: Camera modes and field of view.

15.2 Object picking

Every interactive part (buttons, slots, the flap, switches, the TV, the aquarium) registers an *oriented box* every frame: its matrix M and its half size $\mathbf{h}$. A ray starts at the camera, $\mathbf{o} = \mathbf{e}$, and goes through the centre of the screen in the view direction $\mathbf{d}$. For each box the ray is moved into the box's own frame, $\mathbf{o}' = M^{-1}\mathbf{o}$, $\mathbf{d}' = M^{-1}\mathbf{d}$, and the *slab test* is applied on each axis a:

$$t_1 = \frac{-h_a - o'_a}{d'_a}, \quad t_2 = \frac{h_a - o'_a}{d'_a}, \qquad t_{near} = \max_a \min(t_1, t_2), \quad t_{far} = \min_a \max(t_1, t_2).$$

The ray hits the box if $t_{near} \le t_{far}$ and $0 \le t_{near} \le 2.4$ m (the reach of an arm). The nearest hit wins: the part is highlighted, its prompt is shown under the crosshair (for example "Insert Tk 50 note"), and E runs its action. The program checks this automatically by aiming the camera at fourteen parts and recording the prompt that is found (Listing 1).

Listing 1: Automatic picking test (part aimed at → prompt found)

```
vending coin slot -> [Insert Tk 10 coin (you have Tk 1070)]
vending note slot -> [Insert Tk 50 note (you have Tk 1070)]
vending key B -> [Press [B]]
vending key 2 -> [Press [2]]
vending OK -> [Press [OK]]
vending flap -> [Collection flap]
ATM card slot -> [Insert bank card]
ATM side key L1 -> [Side key L1]
ATM keypad 5 -> [Keypad [5]]
ATM cash slot -> [Cash dispenser]
coffee button 1 -> [Make Espresso (free)]
TV -> [Change TV channel (now: Nature)]
aquarium -> [Feed the fish]
light switch -> [Turn OFF lounge lights]
```

15.3 How the objects react

Table 90: Interactive parts and their reactions

Object	Part (prompt)	Reaction	Sound
Vending machine	coin slot, note slot	credit +10 / +50 Tk, the cash in the wallet decreases	coin
	keys A–D, 1–3, CLR, OK	the key moves in by 7 mm; the LCD shows the code, the price and the credit	click
	collection flap	swings in by $R_x(50°)$; the item goes into the bag	click
	coin cup, return lever	the change is collected; the credit is returned	coin
ATM	card slot	the card slides in or out	click
	keypad, side keys	PIN stars, menu choice, amount (Figure 63)	beep
	cash dispenser, receipt	notes slide out, cash goes to the wallet; the receipt curls out	cash
Coffee machine	three buttons, cup	cup drops, grinding, pouring, steam; the drink goes into the bag	grinder, pour, beep
Television	screen	next channel with a short static noise	click
Aquarium	glass	food crumbs sink and the fish swim up	click
Light switch	switch	the rocker flips; the lamps of the group and their lights switch	click
Entrance	(no key)	the doors slide open when a person is near	whoosh

15.3.1 Buying from a vending machine

Figure 61 follows one purchase from the player's eyes. The HUD shows the prompt of the part under the crosshair, the cash in the wallet and the items in the bag (top right).



(a) Looking at the note slot: “Insert Tk 50 note”



(b) Credit Tk 50 on the LCD; aiming at key A



(c) Code A1 (Cola, Tk 30) typed; aiming at OK



(d) The spiral turns and pushes the can forward



(e) The can lies in the tray: “Push flap”



(f) The flap swings in; the can is in the bag, Tk 20 change in the cup

Figure 61: The player buys a can of Cola.



(a) Code typed, credit enough



(b) The product falls



(c) “Take your item”

Figure 62: The same steps seen closer, with the downlight (spot light) above the machine.

15.3.2 Using the ATM

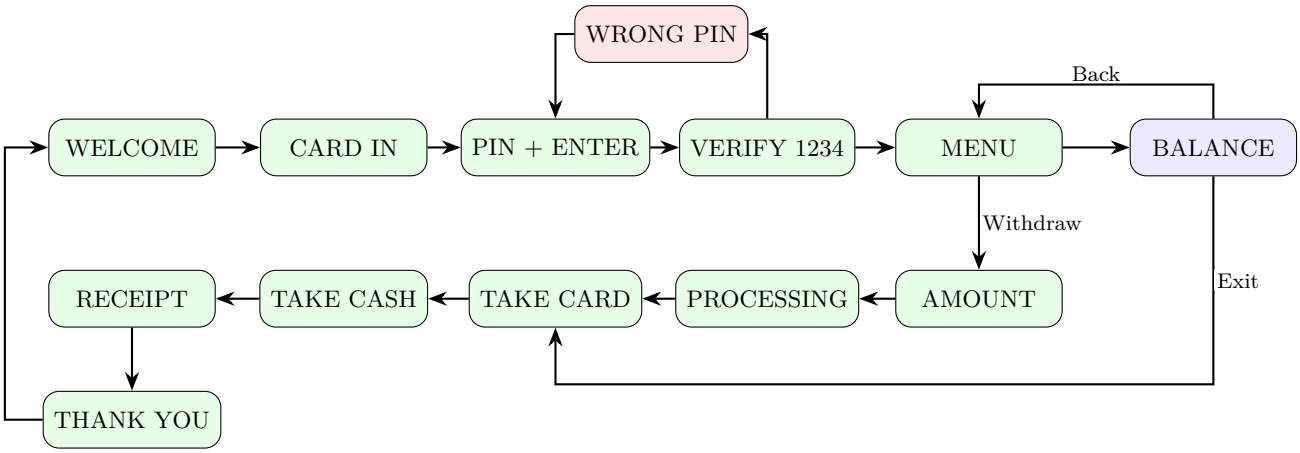


Figure 63: ATM state machine. CANCEL at any step leads to TAKE CARD; the card is always returned before the cash.

The eight steps are listed in a HUD panel and the current step is highlighted. Inside the machine the card slides in and out, the PIN appears as stars, the notes slide out of the dispenser and the receipt curls out of the printer.



(a) 1. Insert card



(b) 2. Enter PIN



(c) 3. Choose a service



(d) 4. Choose amount



(e) 5. Processing



(f) 6. Take card



(g) 7. Take cash



(h) 8. Take receipt



(i) Done

Figure 64: A withdrawal of Tk 1,000 with the step list on the left.

15.3.3 Coffee machine, television, light switch and aquarium



(a) Brewing a cappuccino (stream, rising coffee, steam)



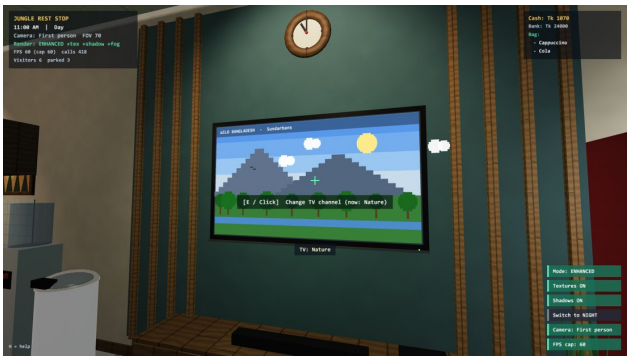
(b) Ready: take the cup



(c) Night: looking at the lounge light switch



(d) After pressing E: the lounge lights are off

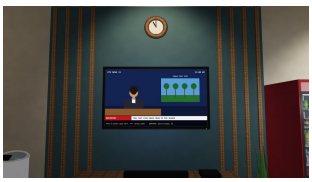


(e) Changing the TV channel

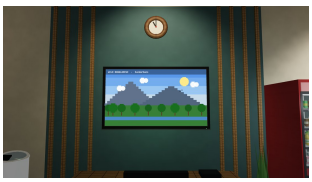


(f) Feeding the fish: they swim up to the food

Figure 65: Reactions of the coffee machine, the light switch, the TV and the aquarium.



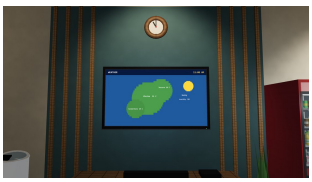
(a) News + ticker



(b) Nature



(c) Live football



(d) Weather (game time)

Figure 66: The four animated TV channels.

16 NPC Visitors

16.1 Visitor life cycle

Visitors appear only inside a car at the far end of the road. A new car comes every 8–18 s when a parking spot is free and the road entrance is clear; at most six visitors are active at once. Every

visitor is a state machine (Figure 67).

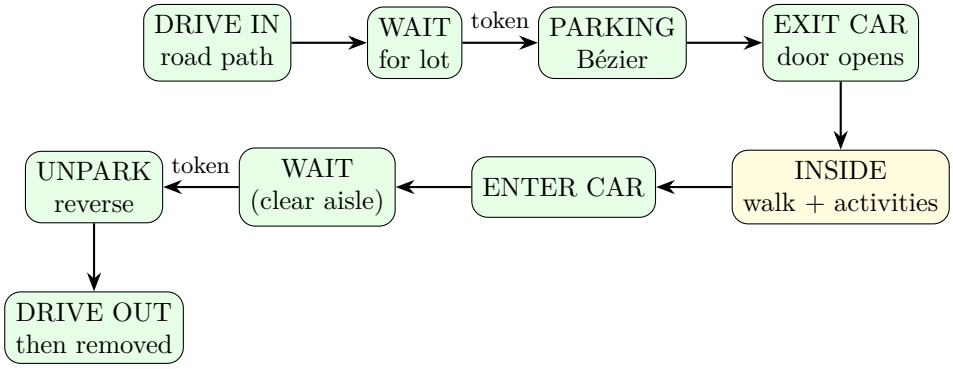


Figure 67: Visitor state machine.

16.2 Driving and parking

A car follows a *path*, a list of points with the running length s. The position is interpolated at s and the heading is the direction of the current segment. The road path samples $z_{road}(x)$ every 3 m. Parking uses a cubic Bézier curve

$$\mathbf{B}(t) = (1-t)^3\mathbf{P}_0 + 3(1-t)^2 t\,\mathbf{P}_1 + 3(1-t)t^2\,\mathbf{P}_2 + t^3\mathbf{P}_3, \qquad t \in [0,1],$$

with $\mathbf{P}_0 = (x_s + 5, 16)$ and $\mathbf{P}_1 = (x_s + 2.2, 16)$ on the aisle and $\mathbf{P}_2 = (x_s, z_s \pm 2.8)$ and $\mathbf{P}_3 = (x_s, z_s)$ in the spot. Since $\mathbf{B}'(0) = 3(\mathbf{P}_1 - \mathbf{P}_0)$ and $\mathbf{B}'(1) = 3(\mathbf{P}_3 - \mathbf{P}_2)$, the curve leaves along the aisle and enters straight into the spot. Leaving uses a second curve driven in *reverse* (heading = direction of motion $+180°$, reverse lights on), then the car drives forward to the west end of the road.

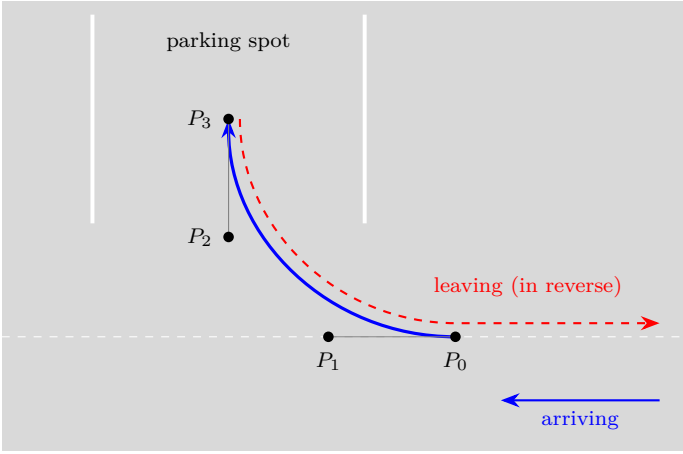


Figure 68: Parking (blue, forward) and leaving (red, reverse) curves for one spot.

The speed is limited by the place (9 m/s on the road, 4 m/s in the lot, 1.8 m/s while parking, 1.3 m/s in reverse) and by the remaining distance s_{left} with a braking deceleration of $2.2\,\mathrm{m/s^2}$:

$$v \le \sqrt{2 \cdot 2.2 \cdot s_{left}} + 0.25.$$

A car stops when another car is less than 7.5 m ahead in its lane or a person is less than 5.5 m ahead. Only the car holding the *lot token* may park or unpark; cars in the aisle wait while it moves, and a car only starts to reverse when the aisle behind it is clear. These two rules prevent deadlocks.

16.3 Indoor navigation and activities

Inside the building visitors walk on a waypoint graph of 18 nodes and 22 edges (doorways, room centres, the queue at the ATM). The shortest route is found with **Dijkstra's algorithm**: starting

from the nearest node, the node u with the smallest known distance is taken and its neighbours are relaxed,

$$d(v) \leftarrow \min\left(d(v),\ d(u) + |\mathbf{x}_u - \mathbf{x}_v|\right),$$

until the goal is reached; the route is then read back through the predecessors. People walk at 1.35 m/s and wait when someone blocks the way. Every visitor has a plan of 1–3 activities, and each activity first *reserves* its resource (a machine, a seat, a reading place), so two people never use the same thing:

- **Vending**: walk to a machine, press the keys (pressing pose), wait for the drop, bend down, take the item and carry it. This runs the machine's own state machine.
- **Coffee**: choose a drink, wait, take the cup and carry it.
- **ATM**: a complete ATM session; the ATM presses its own buttons with short pauses, so the screen shows every step.
- **Watch TV** (7 sofa seats) or **read** (2 library armchairs) for 12–28 s.
- **Browse books** or **watch the fish**.

A machine that the player is using is not available to visitors.



(a) Parked cars and a visitor walking to the entrance



(b) Walking in through the sliding doors



(c) Watching TV (the TV light falls on them)



(d) Buying from a vending machine

Figure 69: NPC visitors.

Listing 2 shows the beginning of a trace recorded by the program.

Listing 2: Start of a visitor trace (simulation time, visitor, spot, activity)

```
t= 4.0s visitor 0 (spot 18): driving in
t= 8.7s visitor 1 (spot 10): driving in
t= 14.2s visitor 2 (spot 13): driving in
t= 19.7s visitor 0 (spot 18): waiting to park
t= 19.7s visitor 0 (spot 18): parking
t= 22.2s visitor 3 (spot 3): driving in
t= 24.5s visitor 0 (spot 18): getting out
t= 26.8s visitor 0 (spot 18): walking in
```

```
t= 27.3s visitor 4 (spot 0): driving in
t= 30.8s visitor 1 (spot 10): waiting to park
t= 30.8s visitor 1 (spot 10): parking
t= 32.9s visitor 5 (spot 14): driving in
t= 35.6s visitor 1 (spot 10): getting out
t= 37.9s visitor 1 (spot 10): walking in
t= 43.5s visitor 2 (spot 13): waiting to park
t= 43.5s visitor 2 (spot 13): parking
t= 47.5s visitor 0 (spot 18): browsing books
t= 48.3s visitor 2 (spot 13): getting out
t= 50.6s visitor 2 (spot 13): walking in
t= 51.3s visitor 3 (spot 3): waiting to park
t= 51.3s visitor 3 (spot 3): parking
t= 51.6s visitor 4 (spot 0): waiting to park
t= 54.1s visitor 1 (spot 10): watching fish
t= 56.1s visitor 3 (spot 3): getting out
t= 56.1s visitor 4 (spot 0): parking
t= 58.4s visitor 3 (spot 3): walking in
t= 60.9s visitor 4 (spot 0): getting out
t= 63.2s visitor 4 (spot 0): walking in
t= 69.6s visitor 5 (spot 14): waiting to park
t= 69.7s visitor 5 (spot 14): parking
```

17 Animation, Particles, Text and Sound

17.1 Autonomous behaviour

Every moving character is a state machine plus a simple movement rule (Table 91).

Table 91: Characters that move on their own

Who	Decision (what to do)	Movement (how)
Visitor	state machine with reserved resources (Figure 67)	Dijkstra route on the waypoint graph, 1.35 m/s, waits when blocked
Car	part of the visitor's state machine; lot token	path following by length s, Bézier parking, braking limit $v \le \sqrt{2 a s_{left}}$
Fish	swim; swim up while there is food	ellipse $x = r_x \cos\varphi$, $z = r_z \sin\varphi$, heading atan2$(\Delta x, \Delta z)$
Birds	fly by day	circles $\mathbf{p} = \mathbf{c} + (r\cos a,\ h + 1.5\sin 2a,\ r\sin a)$, $a \mathrel{+}= \omega \Delta t$
Fireflies	appear at night, outdoors	drift on sine curves around the camera

Fish. The height of a fish moves slowly towards $y_{target} = y_0 + 0.03\sin(0.7t + \varphi)$ at no more than 0.25 m/s; while there is food, y_{target} is the water surface. The tail wags with $R_y(25^\circ \sin(9t + 3\varphi))$.

Birds. The wings are boxes in child nodes rotated by $R_z(\pm 35^\circ \sin(\text{flap}))$ with flap $\mathrel{+}= 9\,\Delta t$.

Fireflies. 110 glowing spheres whose positions are wrapped into a 60×60 m box around the camera, $x' = ((x - x_{cam}) \bmod 60) - 30 + x_{cam}$, so some are always near the viewer. Each blinks with $0.5 + 0.5\sin(3t + 3\varphi)$ and is hidden below 0.25.

17.2 Particles

A particle is a small basic shape with a position $\mathbf{p}$, a velocity $\mathbf{v}$ and a life time. Every frame $\mathbf{v} \mathrel{+}= \mathbf{g}\,\Delta t$ (for falling particles), $\mathbf{p} \mathrel{+}= \mathbf{v}\,\Delta t$ and the life decreases by Δt; at the end of its life the particle is removed.

Table 92: Particle effects

Effect	Shape	Rule
Coffee steam	transparent sphere	a new puff every 0.12 s while brewing; rises 7 cm/s and sways; grows from 2 to 6 cm; $\alpha = 0.25 \cdot \text{life}$; lives 1.6 s
Change coins	thin cylinder	fall with $g = 9.81$ m/s^2 into the coin cup
Falling product	the product	gravity, spin and a bounce $v_y \leftarrow -0.32\, v_y$
Aquarium bubbles	transparent sphere	rise 12–22 cm/s with a wobble; restart at the bottom
Fish food	tiny sphere	sinks 4 cm/s, lives 5–8 s
Fireflies	glowing sphere	see above

17.3 Text rendering

All text (HUD, ATM and vending screens, TV, signs, price tags) uses one **font atlas**:

1. At start-up Windows GDI draws the 95 printable ASCII characters (Consolas) into one bitmap, which becomes an OpenGL texture whose alpha channel is the shape of the letters.
2. Each character is one textured quad (two triangles) whose texture coordinates select its cell in the atlas.
3. **HUD text** uses the orthographic projection $\text{ortho}(0, W, H, 0)$, so one unit is one pixel; depth testing is off, so it is always on top.
4. **Text in the 3D world** is drawn on a *canvas*: a flat rectangle with its own model matrix (for example the ATM screen). A canvas point (x, y) is placed with $\mathbf{p} = M_{canvas}\,(x - \frac{w}{2},\ y - \frac{h}{2},\ 0,\ 1)^T$, so the text sits on the screen and turns with the machine. Depth testing is on, so a person in front hides the text.

Text is blended with $C = \alpha\, C_{text} + (1 - \alpha)\, C_{behind}$, which gives smooth letter edges.

17.4 Sound

The sound uses 14 short recordings that are free to use (CC0, public domain; from Kenney.nl and OpenGameArt, listed in `assets/sounds/CREDITS.txt`), stored as 44.1 kHz mono WAV files. The two background sounds (birds by day, crickets at night) are loops whose end is cross-faded into their start. A mixer thread adds all playing sounds, limits the sum with tanh and sends blocks of 2048 samples to the sound card with the Windows `waveOut` functions. Sounds in the world become quieter with the distance d from the camera,

$$volume = \frac{v_0}{1 + (d/4)^2},$$

so a machine used by a visitor in another room is heard only faintly. The background is quieter inside the building.

Table 93: Sounds

Sound	When
click, beep	any button or switch; ATM keys; coffee ready
coin, thud, motor	vending: money in, change falling, product hitting the tray, spiral turning
cash, grinder, pouring	ATM gives money; coffee machine grinds and pours
door whoosh, car door	entrance doors start to move; a visitor opens or closes the car door
birds / crickets	quiet background loop by day / by night

18 Performance

- **Instanced drawing**: all copies of one basic shape are drawn with one `glDrawElementsInstanced` call; each instance carries its model matrix, normal matrix, colour, emission, texture layer and material values.
- **Static world in chunks**: 84,914 shapes that never move are collected once (in 0.65 s) into 101 chunks of 32×32 m, each with its own GPU buffer.
- **Culling**: a chunk is skipped when its bounding box is outside the view frustum (six planes taken from the view-projection matrix) or farther than 165 m. Grass and flowers end at 45 m, bushes, ferns and leaf cards at 85 m.
- **Occlusion culling**: visible chunks are drawn from near to far, and a chunk that was completely hidden in the previous frame (for example the jungle behind a wall) is skipped; only its bounding box is tested again with an occlusion query.
- **Depth pre-pass**: depth is written first with a minimal shader, then the lighting shader runs with `GL_LEQUAL`, so every pixel is lit exactly once.
- **Per-room light lists**, frustum culling of lights, and a shadow pass that skips grass and glass.
- **Frame cap**: after each frame the program sleeps until $1/cap - 1.5$ ms and then waits the remaining time, which gives a steady 60 or 30 FPS.

Table 94: Measured frame time without the frame cap (1280×720, Intel Iris Xe)

View	Enhanced		Basic		Triangles
	ms	FPS	ms	FPS	(enh.)
Outside, day	13.3	75	11.2	90	2.30M
Main lounge	11.8	85	9.4	107	2.23M
Inside the jungle	10.6	94	7.7	130	1.79M
Parking lot, night	11.7	85	11.6	86	1.69M
Aerial view	15.4	65	13.0	77	2.28M
Aerial view, night	15.8	63	15.3	65	1.76M

19 Conclusion

Jungle Rest Stop is a small living world: three connected rooms with working machines, visitors who arrive by car and act on their own, a dense jungle, and light that changes from morning to night. Every object is a set of basic shapes placed with translation, rotation and scaling and organised in parent and child nodes. The project covers the main topics of the course in one program: transformations and hierarchical modelling, viewing and projection, illumination and shading, several light types, texture mapping, shadow mapping, fog, transparency, animation, picking, collision detection and camera control, and it runs at a steady 60 FPS on integrated graphics.

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